Analog Electronics

Detailed Solutions

GATE INSTRUMENTATION ENGINEERING

(IN)

Himanshu Agarwal & Anish Saha

PrepFusion™

Where Concept Meets Clarity!

Preface

Welcome to PrepFusion™!

This book works, in full, every question GATE has actually asked in Analog Electronics (IN), gathered from the original papers and arranged unit by unit and year by year. Nothing is paraphrased or reconstructed: every question appears as it was set, and every solution is taken step by step rather than asserted.

Each solution carries the same identifier as the question it answers in the companion questions volume, so you can move between practice and explanation without a lookup table. Where the examining authority revised an answer or awarded marks to all (MTA), that is noted with the question it affects.

Used end to end, the book shows you what the paper rewards — which topics repeat, how the framing has shifted over the years, and where your preparation still has gaps.

Meet your Authors

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- 5+ years of teaching experience

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> *"We built this book the way every aspirant deserves one to be built — organized strictly by unit and by year, so that every hour of practice maps directly onto how GATE actually tests you."*
>
> — Himanshu Agarwal & Anish Saha

How to Read the Question Numbers

Every question in this book carries a three-part identifier instead of a plain serial number:

Q3.24.7

3	Unit (chapter) number — Unit III of this book
24	GATE exam year — 2024 (two digits; 98 = 1998)
7	Question number within that unit and year

So **Q3.24.7** is the 7th question that GATE 2024 contributed to Unit III. Numbers are therefore not sequential through the book — they tell you the unit and the exam year at a glance. The same identifier is used in the questions book, the solutions book and the answer keys, so you can jump between them without a lookup table. In the PDF bookmarks panel, expand a unit to see every question ID and click one to go straight to it.

What the Bubbles Mean

Every question ends with three empty circles for you to shade in yourself:

○○○ ●○○ Easy ●●○ Medium ●●● Hard

The companion solutions volume marks the same question with *our* recommended difficulty, using the identical bubbles. If you think a rating should be different, tell us directly — report it under the **Study Hub** section at **pyq.prepfusion.in**.

How to Read the Unit Snapshot

Every unit opens with a **Unit Snapshot** box summarizing that unit's real question data:

Questions	the real total for this unit.
MCQ, MSQ, NAT	that same total, split by question type.
1M, 2M	how many of those questions carry 1 or 2 marks.
Descriptive	questions worth more than 2 marks, or whose marks value isn't recorded.
Avg Weightage	the average marks this unit has contributed per real GATE sitting, not a raw year-by-year total — a year GATE ran multiple real sessions (a genuine forenoon/afternoon split, not the same paper's Set A/B/C/D booklet variants) is divided across those sessions first, so a heavily multi-session year cannot silently outweigh a single-session one. The ~ marks it as a rounded figure, not an exact one.
Range	the lowest and highest that same per-sitting value has been, with the year each occurred.

Avg Weightage and Range are computed from **2010 onward** only: GATE dropped from a 150-mark to a 100-mark paper in 2009, but that year itself was a one-off transition (60 questions, not the 65-question format that became standard from 2010 on) — so 2010 is the first year fully comparable to modern papers, and marks-based figures from earlier years are left out of these two figures specifically (every other count on this page still includes every year). They also depend on knowing which GATE years ran more than one real sitting for a given branch — our own research into this is thorough for some branches and still a work in progress for others. Treat both figures as a **beta**, best-effort estimate: directionally useful, not exact.

MSQ questions specifically weren't introduced until 2021 — an older-year-heavy unit showing **MSQ: 0** reflects that real exam history, not a data gap.

Worked example

Unit Snapshot*

Stream: IN	**Questions:** 45	**MCQ:** 30	**MSQ:** 3	**NAT:** 12
1M: 20	**2M:** 22	**Descriptive:** 3	**Avg Weightage** ~4M	**Range:** 2M(2015)–6M(2021)

45 real questions total; on average this unit has been worth about 4 marks per real GATE sitting since 2010, ranging from as low as 2 (in 2015) to as high as 6 (in 2021).

A quick way to use this: a high **Avg Weightage** points to a consistently high-yield topic worth prioritizing; a wide **Range** usually means the topic's importance has shifted over the years, so it's worth checking *which* years drove that before deciding how much weight to give it today.

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Scan any code with your phone camera, or tap the link beneath it. **pyq.prepfusion.in** is also where you can read this book's worked solutions online — and in its **Study Hub** section, report an error in any question or request a video solution for it.

Contributors

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SOLUTIONS

Diode Circuits and applications

Topics

1. Diode Models with open and short circuit tests
2. Operating Point and Small Signal Analysis of a Diode
3. Zener Diode as a Voltage Regulator
4. Series and Shunt Clipper Circuits
5. Two-Level Clipper Circuits
6. Clamper Circuits and their examples
7. Peak Detector and Average Diode Current
8. Voltage Multiplier Circuits
9. Half Wave Rectifier and Centre Tapped Full Wave Rectifier
10. Full Wave Bridge Rectifier and Capacitive Filters

Unit Snapshot*

Stream: IN	**Questions:** 27	**MCQ:** 18	**MSQ:** 0	**NAT:** 9
1M: 19	**2M:** 8	**Descriptive:** 0	**Avg Weightage** ~2M	**Range:** 1M(2026)–5M(2022)

**See preface for how Avg Weightage and Range are calculated.*

Q1.26.1 NAT 2026 1M Ans: 11 ● ○ ○

At the positive peak of the input, $V_{in} = 15$ V

Since 15 V > 3 V, the Zener diode operates in the breakdown region and maintains a constant voltage of 3 V.

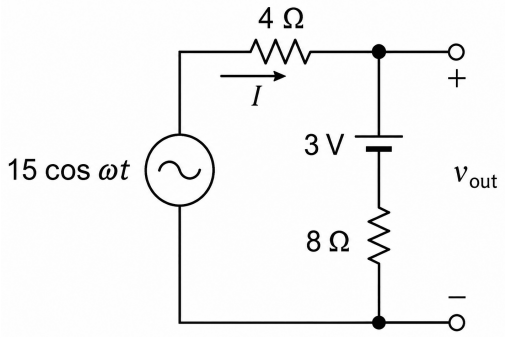

Fig. Solution Q1.26.1

Applying KVL, the current flowing through the circuit is

$$I = \frac{V_{in} - V_Z}{4 + 8} = \frac{15 - 3}{4 + 8} = \frac{12}{12} = 1 \text{ A}$$

The output voltage is the voltage across the Zener branch.

Hence,

$$v_{out} = V_Z + I(8\ \Omega) = 3 + (1 \times 8) = 11 \text{ V}$$

During the negative half-cycle, the Zener diode becomes forward biased and behaves like an ordinary diode.

Therefore, the magnitude of the output voltage is smaller than the positive peak.

Hence, the maximum value of the output voltage is

11

Q1.23.1 MCQ 2023 1M Ans: B ● ○ ○

A **Light Emitting Diode (LED)** emits light only when it is forward biased. In the forward-biased condition, electrons and holes are injected across the *pn*-junction and recombine, releasing energy in the form of photons. A **photodiode** provides maximum sensitivity to light when it is reverse biased. Under reverse bias, the depletion region widens and only a small current flows. When light is incident on the junction, electron-hole pairs are generated, producing a photocurrent proportional to the intensity of the incident light. If the photodiode is forward biased, the large forward current masks the photocurrent, thereby reducing its sensitivity. Hence, the correct answer is

B. Forward, Reverse

Q1.23.2 NAT 2023 1M Ans: 1.00 ● ○ ○

The circuit consists of an ideal diode connected in parallel with the resistor R.

For an ideal diode,

- During the positive half-cycle of the source current, the diode is forward biased and behaves as a short circuit. Hence, the entire current flows through the diode and no current flows through the resistor.
- During the negative half-cycle, the diode is reverse biased and behaves as an open circuit. Hence, the entire source current flows through the resistor.

Here,

$$i_R(t) = \begin{cases} 0, & 0 \le \omega t \le \pi, \\ -\pi \sin(\omega t), & \pi \le \omega t \le 2\pi. \end{cases}$$

Thus, the resistor current is a half-wave rectified sine wave.
The average current over one complete cycle is

$$I_{\text{avg}} = \frac{1}{2\pi} \int_{\pi}^{2\pi} (-\pi \sin\theta)\ d\theta$$

$$= -\frac{1}{2} [-\cos\theta]_{\pi}^{2\pi} = \frac{1}{2} [\cos(2\pi) - \cos(\pi)]$$

$$= \frac{1}{2} [1 - (-1)] = 1 \text{ mA}$$

Hence, the magnitude of the average current through the resistor is

$$\boxed{1.00}$$

Q1.22.1 NAT 2022 1M Ans: 6.0 ●○○

For satisfactory voltage regulation, the Zener diode must conduct at least its minimum current.

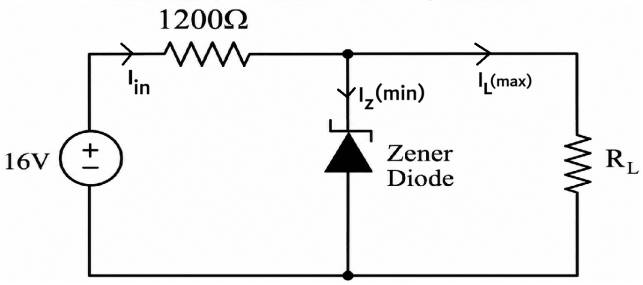

Fig. Solution Q1.22.1

The maximum load current is obtained when the Zener current is at its minimum value.

$$I_{L(\max)} = I_{\text{in}} - I_{Z(\min)}$$

The current through the series resistor is

$$I_{\text{in}} = \frac{V_S - V_Z}{R_S} = \frac{16 - 4}{1200} = 10 \text{ mA}$$

Given, $I_{Z(\min)} = 4$ mA
Therefore,

$$I_{L(\max)} = 10 - 4 = 6 \text{ mA}$$

Hence,

$$\boxed{6.0 \text{ mA}}$$

Q1.22.2 MCQ 2022 2M Ans: D ● ● ○

The given circuit is an ideal positive peak detector.

Initially, the capacitor voltage is $V_C(0) = 0.3$ V

Since the diode is ideal, whenever $V_{in} > V_C$,

the diode becomes forward biased and the capacitor charges to the instantaneous input voltage.

From the given waveform,

- The capacitor initially holds 0.3 V.
- During the first cycle, it charges to 0.4 V.
- At $t = 2$ ms, the input reaches its maximum value of 1.0 V. Hence, the capacitor charges to 1.0 V.

After reaching 1.0 V, the input never exceeds this value again. Therefore, the diode remains reverse biased and the capacitor retains its stored voltage.

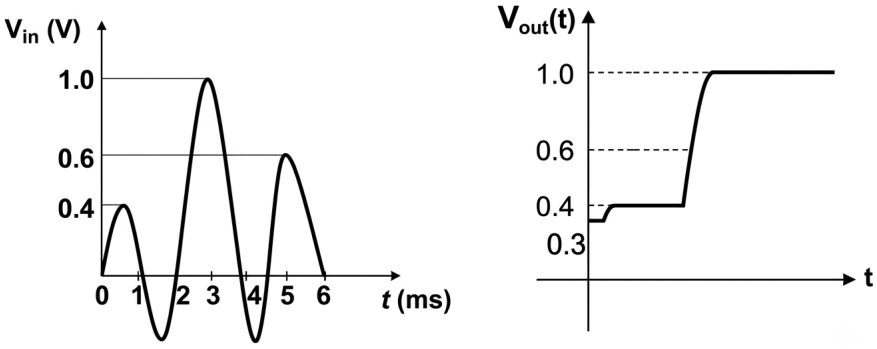

Fig. Solution Q1.22.2

Hence, at $t = 5$ ms, $V_{out} = 1.0$ V

Thus, the correct answer is

D. 1.0 V

Q1.22.3 NAT 2022 2M Ans: -10.0 ● ● ○

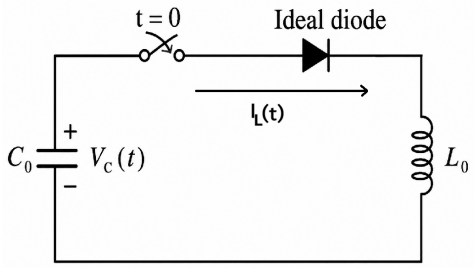

Fig. Solution Q1.22.3

Initially, the capacitor is charged to $V_C(0) = 10$ V

and the inductor current is $i_L(0) = 0$

After the switch is closed, the ideal diode is forward biased and the circuit behaves as a series LC circuit.

The capacitor voltage is

$$V_C(t) = V_0 \cos(\omega_0 t) \qquad \text{where, } \omega_0 = \frac{1}{\sqrt{LC}}$$

The inductor current is

$$i_L(t) = V_0\sqrt{\frac{C}{L}}\sin(\omega_0 t).$$

The diode conducts as long as $i_L(t) \geq 0$.
The current becomes zero for the first time when $\omega_0 t = \pi$.
Hence,

$$t = \frac{\pi}{\omega_0} = \pi\sqrt{LC} = \pi\sqrt{(1\times10^{-3})(10\times10^{-6})} = 0.314 \text{ ms}.$$

At this instant,

$$V_C(t) = V_0\cos(\pi) = -10 \text{ V}.$$

Beyond this instant, the current tends to reverse direction, making the diode reverse biased. Therefore, the diode turns OFF and the capacitor remains isolated, retaining its voltage. And the circuit will reach steady-state.
Since 0.5 s $\gg$ 0.314 ms, the capacitor voltage remains unchanged.
Hence,

$$\boxed{-10.0}$$

Q1.20.1 MCQ | 2020 | 1M | Ans: A

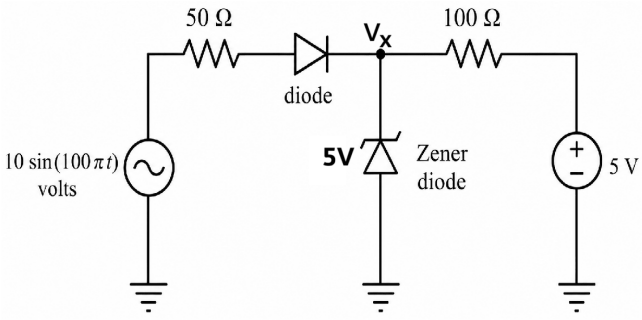

Fig. Solution Q1.20.1

Let the input source be $V_m = 10sin(100\pi t)$ Volts.
For $V_m < 5$ V, the Zener diode is OFF and the series diode is also OFF. Hence, no current flows through the 100 Ω resistor.
For $V_m > 5$ V, the series diode becomes forward biased and the Zener diode operates in the breakdown region, maintaining the node voltage at $V_x = 5$ V.
Since the right terminal of the 100 Ω resistor is also maintained at 5 V by the DC source,

$$V_R = 5 - 5 = 0 \text{ V}.$$

Therefore, the current through the resistor is

$$I_R = \frac{V_R}{100} = \frac{0}{100} = 0 \text{ A}.$$

Hence, the power dissipated in the resistor is

$$P = I_R^2 R = 0.$$

Thus, the correct answer is

$$\boxed{\textbf{A. } 0}$$

Q1.20.2 NAT 2020 2M Ans: 1.8 ●●○

Among the three diodes connected to the right node, the diode connected to the 15 V source alone conducts, since it has the highest voltage.

If that diode gets on then the diodes connected to the 10 V and 5 V sources remain reverse biased.

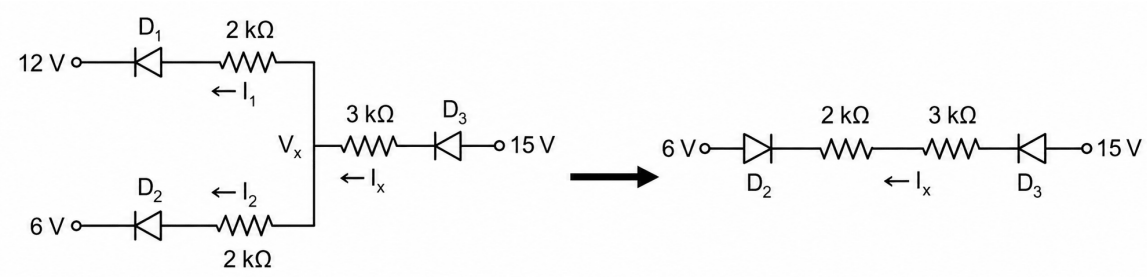

Fig. Solution Q1.20.2

Assume that both left diodes are ON.

Applying KCL at node V_x,

$$I_x = I_1 + I_2$$

$$\frac{15 - V_x}{3} = \frac{V_x - 12}{2} + \frac{V_x - 6}{2}$$

$$\Rightarrow \frac{15 - V_x}{3} = \frac{2V_x - 18}{2}$$

$$\Rightarrow V_x = 10.5 \text{ V}$$

Since, $V_x < 12$ V,

Diode D_1 becomes reverse biased and turns OFF.

Applying Ohm's law,

$$I_x = \frac{15 - 6}{3 + 2} = \frac{9}{5} = 1.8 \text{ mA}$$

Hence,

$$\boxed{1.8 \text{ mA}}$$

Q1.19.1 MCQ 2019 1M Ans: B ●●○

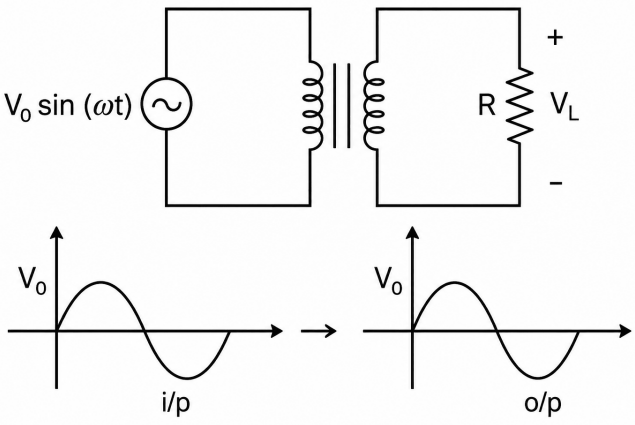

Fig. Solution Q1.19.1

Consider the transformer to have unity turns ratio, the secondary voltage is the same as the input voltage.
Hence, the voltage across the resistor is $V_L(t) = V_0 \sin(\omega t)$
Therefore, the RMS voltage across R is

$$(V_L)_{\text{rms}} = \frac{V_0}{\sqrt{2}}$$

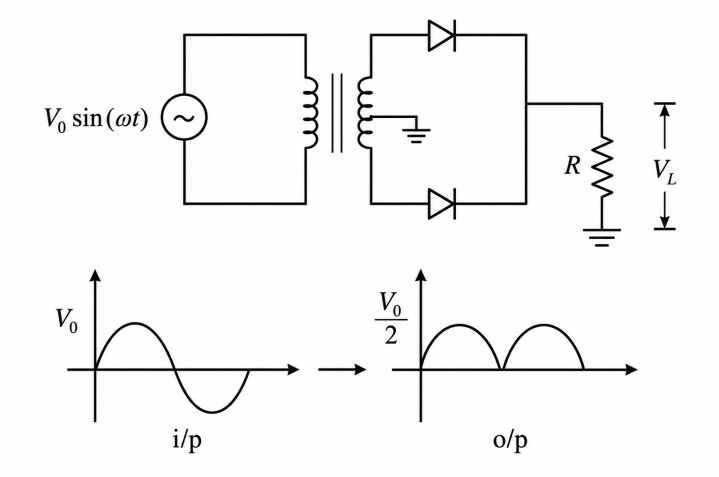

Fig. Solution Q1.19.1

In figure (b), the transformer is centre-tapped.
During each half cycle, only one half of the secondary winding conducts through the corresponding diode.
Hence, the output is a full-wave rectified waveform with peak value $\frac{V_0}{2}$.
The RMS value of a full-wave rectified sine wave is equal to that of the corresponding sine wave having the same peak value.
Therefore,

$$(V_L)_{\text{rms}} = \frac{\frac{V_0}{2}}{\sqrt{2}} = \frac{V_0}{2\sqrt{2}}.$$

Hence, the required ratio is

$$\frac{\frac{V_0}{\sqrt{2}}}{\frac{V_0}{2\sqrt{2}}} = 2.$$

Therefore,

B. 2 : 1

Key Points

- **Root Mean Square (RMS) Value**

  For a periodic signal $x(t)$ with time period T, $x_{\text{rms}} = \sqrt{\frac{1}{T}\int_0^T x^2(t)\,dt}$

  For a discrete-time signal with N samples, $x_{\text{rms}} = \sqrt{\frac{1}{N}\sum_{k=1}^{N} x_k^2}$

- **Common RMS Values**

  Sine Wave: $V_{\text{rms}} = \frac{V_m}{\sqrt{2}}$

Square Wave: $V_{\text{rms}} = V_m$

Triangular Wave: $V_{\text{rms}} = \dfrac{V_m}{\sqrt{3}}$

Sawtooth Wave: $V_{\text{rms}} = \dfrac{V_m}{\sqrt{3}}$

Q1.18.1 NAT | 2018 | 1M | Ans: 200 | ●●○

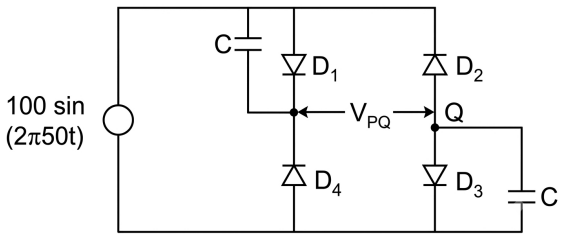

Fig. Solution Q1.18.1

During the positive half cycle, diodes D_1 and D_3 are ON, while diodes D_2 and D_4 are OFF.
Hence, no current flows.
During the negative half cycle, diodes D_1 and D_3 are OFF and thus capacitors gets charged to peak value of input
Since the input is $100\sin(2\pi 50t)$,
The peak voltage is, $V_m = 100$ V.
As there is no resistor in the circuit, the capacitors do not discharge. Therefore, their voltages remain unchanged even at $t = 60$ ms.

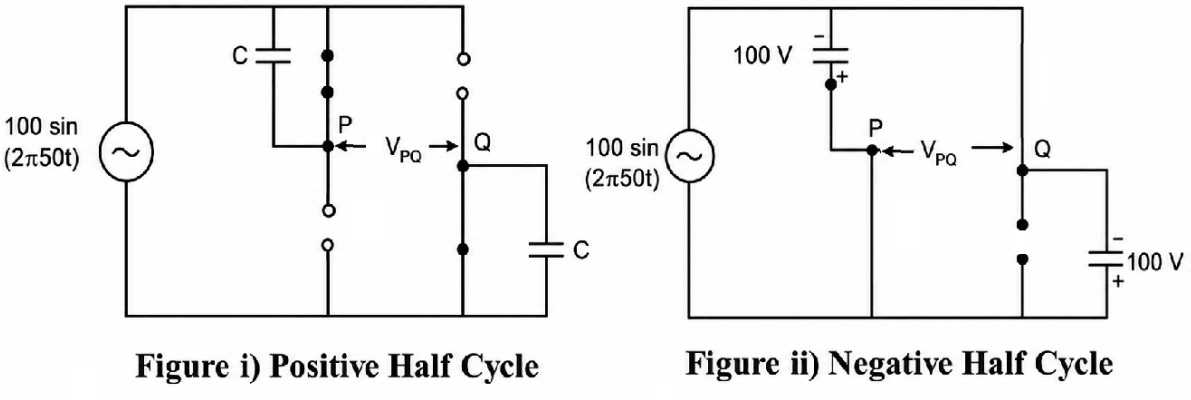

Fig. Solution Q1.18.1

Hence,

$$V_p = 100 \text{ V}, \qquad V_q = -100 \text{ V}.$$

Therefore,

$$V_{pq} = V_p - V_q = 100 - (-100) = 200 \text{ V}.$$

Hence,

$$\boxed{200 \text{ V}}$$

Q1.17.1 MCQ 2017 1M Ans: D ● ○ ○

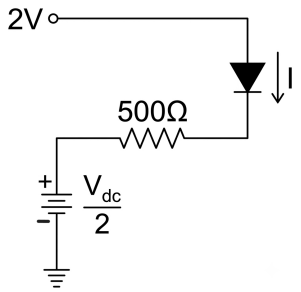

Fig. Solution Q1.17.1

The diode conducts when it is forward biased by at least its barrier potential of 0.7 V.
At the boundary of conduction, the diode current becomes zero.
Applying KVL,

$$I = \frac{2 - 0.7 - \frac{V_{dc}}{2}}{500}$$

For the diode to just turn OFF, $I = 0$.
Hence,

$$2 - 0.7 - \frac{V_{dc}}{2} = 0$$

$$\frac{V_{dc}}{2} = 1.3 \Rightarrow V_{dc} = 2.6 \text{ V}.$$

Therefore, the diode will have no forward current when $V_{dc} > 2.6$ V.
Hence, the correct answer is

D. 2.6 V

Q1.16.1 NAT 2016 1M Ans: 0 ● ○ ○

Assume that the diode is ON.

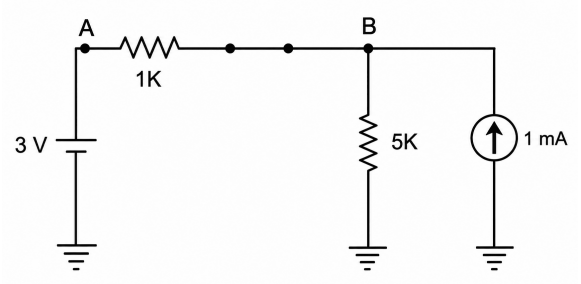

Fig. Solution Q1.16.1

Applying KCL at node B,

$$\frac{V_B - 3}{1\ \text{k}\Omega} + \frac{V_B}{5\ \text{k}\Omega} - 1\ \text{mA} = 0$$

$$5(V_B - 3) + V_B - 5 = 0$$

$$V_B = \frac{10}{3} = 3.33 \text{ V}$$

The current through the 1 kΩ resistor is

$$I = \frac{3 - 3.33}{1\ \text{k}\Omega} = -0.33\ \text{mA}$$

Since the current is negative, the assumption that the diode is ON is incorrect.
Hence, the diode is OFF.
Therefore, no current flows through the 1 kΩ resistor, and the voltage drop across it is

$$V_{ab} = I(1\ \text{k}\Omega) = 0.$$

Hence,

$$\boxed{0\ \text{V}}$$

Q1.15.1 NAT 2015 1M Ans: 0.25 ●○○

Since the diode is ideal, the circuit acts as a half-wave rectifier.
The output voltage across the load is

$$v_o(t) = \begin{cases} 10\sin(100\pi t), & \sin(100\pi t) \geq 0, \\ 0, & \sin(100\pi t) < 0. \end{cases}$$

The RMS value of a **half-wave rectified sine wave** is

$$V_{\text{rms}} = \frac{V_m}{2} = \frac{10}{2} = 5\ \text{V}.$$

Hence, the average power dissipated in the load resistance is

$$P_{\text{avg}} = \frac{V_{\text{rms}}^2}{R_L} = \frac{5^2}{100} = 0.25\ \text{W}.$$

Hence,

$$\boxed{0.25\ \text{W}}$$

Q1.14.1 NAT 2014 1M Ans: 10 ●○○

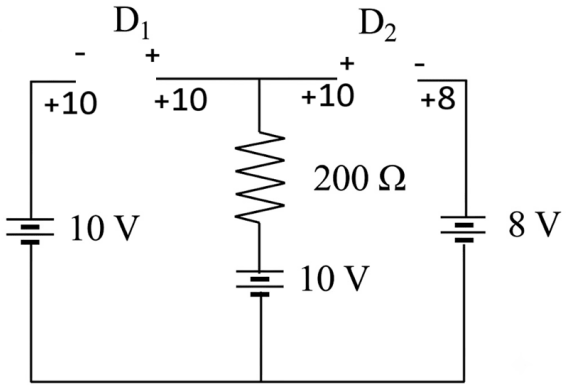

Fig. Solution Q1.14.1

Using the open-circuit test, first assume that both diodes are OFF.
The voltage across diode D_1 is $V_{D_1} = 0$ V.
The voltage across diode D_2 is $V_{D_2} = 10 - 8 = 2$ V.
Since $V_{D_2} > 0$, diode D_2 is ON, whereas diode D_1 remains OFF.

Now, assuming D_2 is ON, the circuit becomes
The voltage across diode D_1 is $V_{D_1} = 8 - 10 = -2$ V.
Since $V_{D_1} < 0$, diode D_1 is reverse biased.
Hence, the assumption that D_2 is ON and D_1 is OFF is correct.
The current through diode D_2 is

$$I_{D_2} = \frac{10 - 8}{200} = \frac{2}{200} = 10 \text{ mA}.$$

Hence,

$$\boxed{10 \text{ mA}}$$

Mistakes to be avoided

- After determining the ON/OFF state of each diode, redraw the circuit with the assumed states and verify that every diode state is consistent with the resulting voltages and currents.

Q1.13.1 MCQ | 2013 | 1M | Ans: D

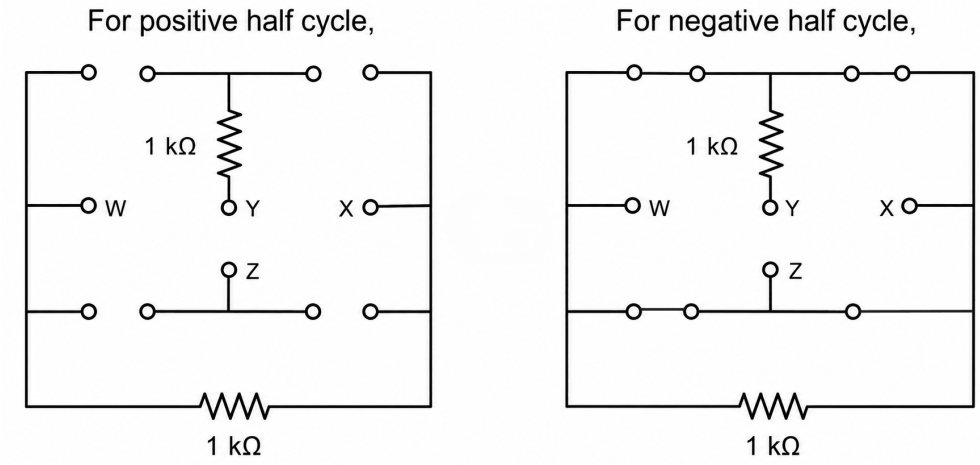

Fig. Solution Q1.13.1

Consider the two half cycles separately. Irrespective of the polarity of the input, $V_{WX} = 0$
Hence, the correct answer is

$$\boxed{\textbf{D. } 0 \text{ for all } t}$$

Q1.13.2 MCQ | 2013 | 2M | Ans: B

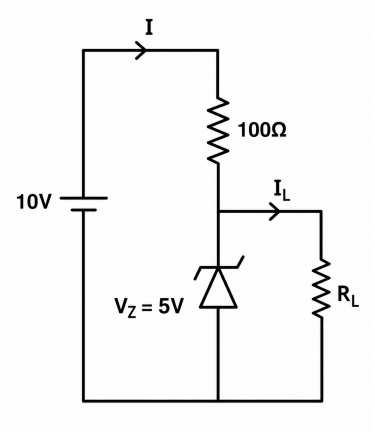

Fig. Solution Q1.13.2

The minimum Zener current required for regulation is

$$I_{Z(\min)} = I_{Z(\text{knee})} = 10 \text{ mA}$$

The current through the 100 Ω resistor is

$$I = \frac{10 - 5}{100} = 50 \text{ mA}$$

Applying KCL, $I = I_Z + I_L$
For the minimum value of R_L, the load current is maximum while the Zener current is at its knee value.
Hence,

$$I_{L(\max)} = 50 - 10 = 40 \text{ mA}$$

Therefore,

$$R_{L(\min)} = \frac{V_Z}{I_{L(\max)}} = \frac{5}{40 \times 10^{-3}} = 125 \ \Omega$$

The maximum Zener current occurs when the load is open-circuited.

$$I_{Z(\max)} = 50 \text{ mA}$$

Hence, the maximum power dissipated in the Zener diode is

$$P_{Z(\max)} = V_Z I_{Z(\max)} = 5 \times 50 \text{ mA} = 250 \text{ mW}$$

Hence, the correct answer is

B. 125 *and* 250

Q1.12.1 MCQ 2012 1M Ans: D ●●○

Method 1

From the given i-v characteristic,

$$i = \frac{v - 0.7}{500}, \qquad v \geq 0.7 \text{ V}$$

Hence, when the diode is ON, it can be modeled as a 0.7 V voltage source in series with a 500 Ω resistance.

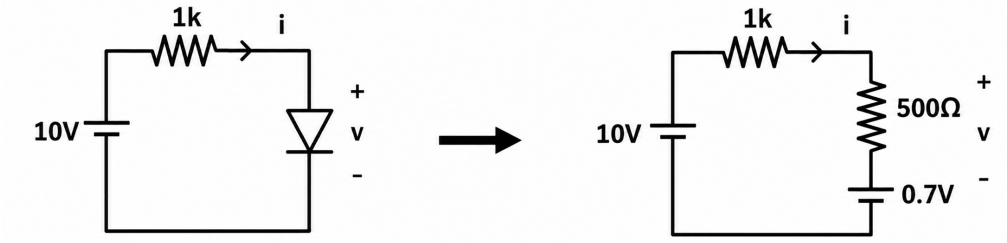

Fig. Solution Q1.12.1

Applying KVL,

$$10 = 0.7 + i(1000 + 500)$$

$$i = \frac{10 - 0.7}{1500} = \frac{9.3}{1500} = 6.2 \text{ mA}$$

Hence, the correct answer is

D. 6.2 mA

Method 2

Applying KVL to the given circuit, $i = \dfrac{10 - v}{1000}$

From the diode characteristic, $i = \dfrac{v - 0.7}{500}$

Equating the two expressions,

$$\frac{10 - v}{1000} = \frac{v - 0.7}{500}$$

$$10 - v = 2v - 1.4 \implies v = 3.8 \text{ V}$$

Therefore,

$$i = \frac{3.8 - 0.7}{500} = 6.2 \text{ mA}$$

Hence, the correct answer is

D. 6.2 mA

PrepFusion

Q1.11.1 MCQ 2011 2M Ans: C ●●○

Assume the Zener D_1 is off and the forward diode D_2 is on. The off Zener branch is an open circuit.

The D_2 branch has its existing 1 kOhm resistor in series with a 0.7 V diode drop. Applying KCL at the output gives

$$\frac{3.7 - V_0}{1\ \text{k}\Omega} = \frac{V_0 - 0.7}{1\ \text{k}\Omega}.$$

Therefore $V_0 =$ 2.2 V, option (C). The diode current is 1.5 mA, consistent with D_2 on, and 2.2 < 2.7 V, consistent with the Zener remaining below reverse breakdown.

If both diodes were off, the unloaded output would be 3.7 V and the assumed off state of D_2 would fail. Thus the checked state is self-consistent.

Q1.08.1 MCQ 2008 1M Ans: B ●○○

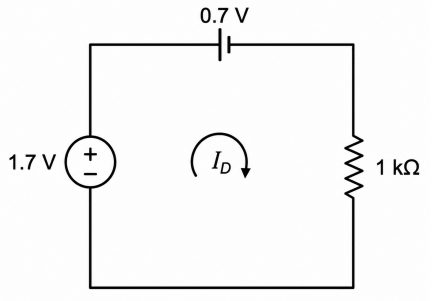

Fig. Solution Q1.08.1

Since the diode is forward biased, it is ON with a constant voltage drop of $V_D = 0.7$ V.
Applying KVL,

$$1.7 - 0.7 - 1\ \text{k}\Omega \times I_D = 0 \Rightarrow I_D = \frac{1}{1000} = 1\ \text{mA}.$$

The dynamic resistance of the diode is given by;

$$r_d = \frac{\eta V_T}{I_D}.$$

At room temperature,

$$V_T \approx 26\ \text{mV}, \qquad \eta = 1.$$

Hence,

$$r_d = \frac{1 \times 26\ \text{mV}}{1\ \text{mA}} = 26\ \Omega \approx 25\ \Omega.$$

Hence, the correct answer is

B. 25 Ω

Key Points

- The **thermal voltage** is given by $V_T = \dfrac{kT}{q}$

  where

  - $k = 1.38 \times 10^{-23}$ J/K is the **Boltzmann constant**,
  - T is the **absolute temperature** in Kelvin (K),
  - $q = 1.602 \times 10^{-19}$ C is the **electronic charge**.

- At room temperature ($T \approx 300$ K),

$$V_T \approx 25.9\ \text{mV} \approx 26\ \text{mV}.$$

Q1.05.1 MCQ | 2005 | 1M | Ans: B | ● ● ○

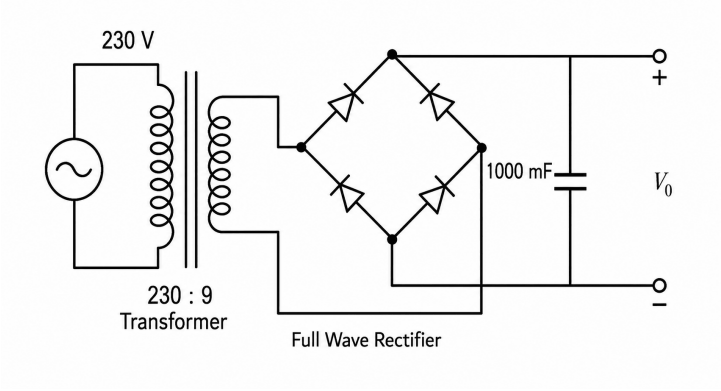

Fig. Solution Q1.05.1

The diode bridge forms a full-wave rectifier.
For the ideal transformer,

$$\frac{V_1}{V_2} = \frac{N_1}{N_2}.$$

Given,

$$\frac{230}{V_2} = \frac{230}{9}$$

$$\Rightarrow V_2 = 9 \text{ V (rms)}.$$

The peak secondary voltage is

$$V_m = \sqrt{2}\, V_{\text{rms}} = \sqrt{2} \times 9 = 12.73 \text{ V}.$$

In a bridge rectifier, two diodes conduct during each half cycle.

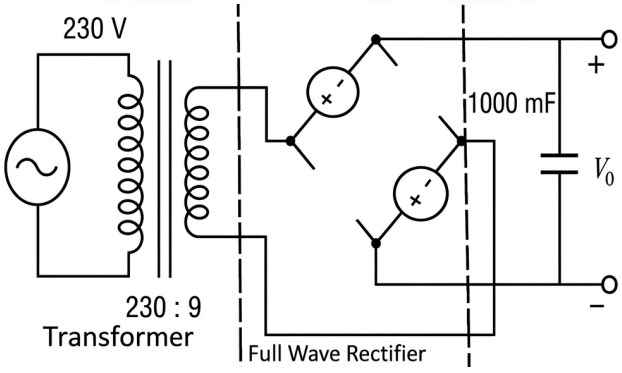

Fig. Solution Q1.05.1

Hence, the total diode drop is $2 \times 0.7 = 1.4$ V.
Therefore, the peak voltage across the capacitor is

$$V_0 = 12.73 - 1.4 = 11.33 \text{ V}.$$

Hence, the correct answer is

B. 11.33 V

Q1.04.1 MCQ | 2004 | 2M | Ans: C | ● ○ ○

Since the diode is ideal, it conducts only during the positive half cycle.
The total resistance in the circuit is

$$R_{eq} = 100 + 200 + 300 = 600\ \Omega.$$

The source voltage is $v(t) = 60\cos(314t)$.
Hence, during the positive half cycle, the current is

$$i(t) = \frac{60\cos(314t)}{600} = \frac{\cos(314t)}{10} \quad \text{A.}$$

The average power dissipated in the 300 Ω resistor is

$$P = \frac{1}{T}\int_0^{T/2} i^2(t)\,(300)\,dt.$$

Substituting $i(t)$,

$$P = \frac{300}{T}\int_0^{T/2}\left(\frac{\cos(314t)}{10}\right)^2 dt.$$

Using the identity

$$\cos^2\theta = \frac{1+\cos 2\theta}{2},$$

we get

$$\frac{1}{T}\int_0^{T/2}\cos^2(314t)\,dt = \frac{1}{2T}\int_0^{T/2}(1+\cos(628t))\,dt$$

$$= \frac{1}{2T}\left[t + \frac{\sin(628t)}{628}\right]_0^{T/2}$$

Since

$$314 = \omega, \qquad T = \frac{2\pi}{314},$$

we have

$$628\left(\frac{T}{2}\right) = 2\pi,$$

hence $\sin(2\pi) = \sin(0) = 0$
Therefore,

$$\frac{1}{T}\int_0^{T/2}\cos^2(314t)\,dt = \frac{1}{2T}\left(\frac{T}{2}\right) = \frac{1}{4}.$$

we get

$$P = 300\left(\frac{1}{100}\right)\left(\frac{1}{4}\right) = 0.75\ \text{W}.$$

Hence, the correct answer is

C. 0.75 W

Q1.04.2 MCQ 2004 1M Ans: D ●○○

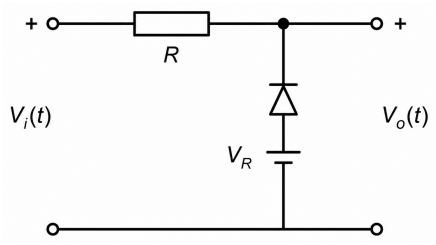

Fig. Solution Q1.04.2

Consider the two possible operating regions.
When

$$V_i(t) \le V_R,$$

the diode is forward biased and turns ON.
Hence, the output is clamped to the reference voltage,

$$V_o(t) = V_R.$$

When

$$V_i(t) > V_R,$$

the diode is reverse biased and turns OFF.
Since no current flows through the resistor, there is no voltage drop across it. Therefore,

$$V_o(t) = V_i(t).$$

Thus, the input-output characteristic is

$$V_o(t) = \begin{cases} V_R, & V_i(t) \le V_R, \\ V_i(t), & V_i(t) > V_R. \end{cases}$$

Hence, the correct answer is

D. $V_o(t) = V_i(t)$, for $V_i(t) > V_R$; $V_o(t) = V_R$, otherwise.

Q1.03.1 MCQ 2003 2M Ans: C ●●○

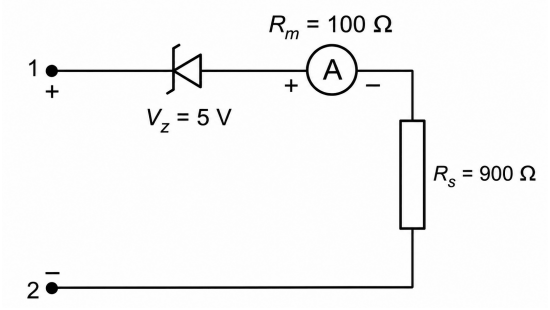

Fig. Solution Q1.03.1

The Zener diode is ideal with a breakdown voltage of $V_Z = 5$ V.

Hence, the Zener remains OFF for

$$V_{in} < 5 \text{ V},$$

and starts conducting when

$$V_{in} \geq 5 \text{ V}.$$

The ammeter has a full-scale current of $I_{FS} = 1$ mA,
With an internal resistance $R_m = 100\ \Omega$.
Therefore, the maximum input voltage corresponding to full-scale deflection is

$$V_{max} = V_Z + I_{FS}(R_m + R_s)$$

$$= 5 + 1 \times 10^{-3}(100 + 900) = 5 + 1 = 6 \text{ V}$$

Thus, the instrument measures voltages from 0 V to 6 V.
Hence, it functions as a

C. 0–6 V DC voltmeter

Q1.03.2 MCQ 2003 2M Ans: A ●●○

The piecewise-linear model of a forward-biased silicon diode is

$$V_D = V_0 + I_D R_{ON} \qquad [\text{where } V_0 \text{ is the cut-in voltage}]$$

When the diode carries negligible current, $V_0 = 0.64$ V.
When the diode current is $I_D = 1$ A,
The diode dissipates

$$P = V_D I_D.$$

Hence,

$$V_D = \frac{P}{I_D} = \frac{1}{1} = 1 \text{ V}.$$

Substituting into the piecewise-linear model,

$$1 = 0.64 + (1)R_{ON}.$$

$$\Rightarrow R_{ON} = 1 - 0.64 = 0.36\ \Omega.$$

Hence, the correct answer is

A. 0.36 Ω

Q1.00.1 MCQ 2000 1M Ans: B ●○○

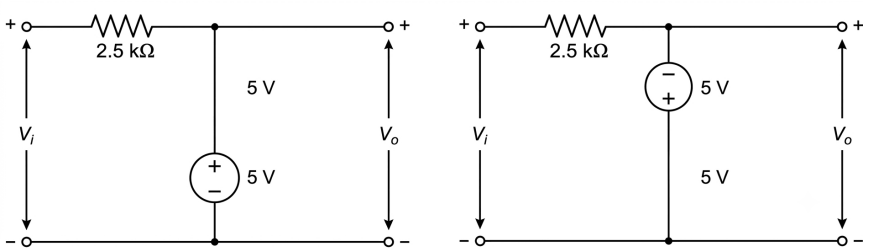

Fig. Solution Q1.00.1

The two Zener diodes are connected back-to-back, forming a symmetrical voltage limiter.
When $V_i > 5$ V, one Zener diode operates in breakdown while the other is forward biased.

Hence, $V_o \approx 5$ V.
Similarly, when $V_i < -5$ V, the other Zener diode goes into breakdown and

$$\Rightarrow V_o \approx -5 \text{ V}.$$

Thus, the output is clipped at ±5 V.

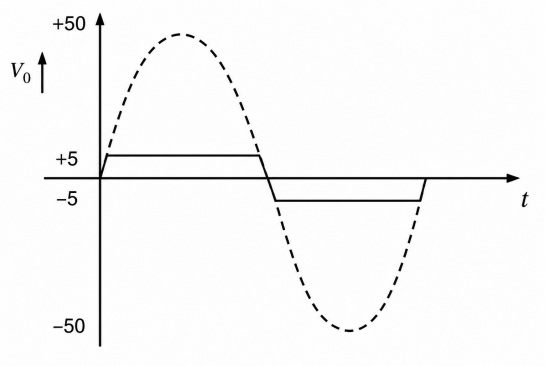

Fig. Solution Q1.00.1

For a sinusoidal input of amplitude 50 V, the output waveform is clipped at ±5 V, resulting in a waveform that closely resembles a square wave.
Hence, the circuit acts as a

B. Square Wave Generator

Q1.99.1 MCQ 1999 1M Ans: A ●●○

The input voltage is

$$V_i = 4 + 3 \sin \omega t.$$

The maximum value of the input is

$$V_{i,\max} = 4 + 3 = 7 \text{ V}.$$

During the positive cycle, the diode is forward biased and the capacitor charges to

$$V_C = 7 \text{ V}.$$

After the capacitor is fully charged, the diode becomes reverse biased and remains OFF.
The equivalent circuit is

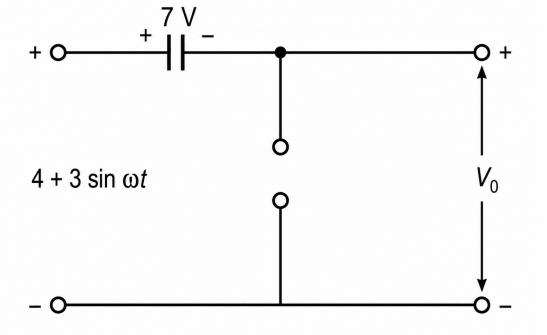

Fig. Solution Q1.99.1

Therefore,

$$V_o = V_i - V_C = (4 + 3 \sin \omega t) - 7 = -3 + 3 \sin \omega t.$$

Since the average value of $\sin \omega t$ over one complete cycle is zero,

$$V_{o,\text{avg}} = -3 + 3\,(0) = -3 \text{ V}.$$

Hence, the correct answer is

A. -3 V

Q1.99.2 MCQ 1999 1M Ans: B

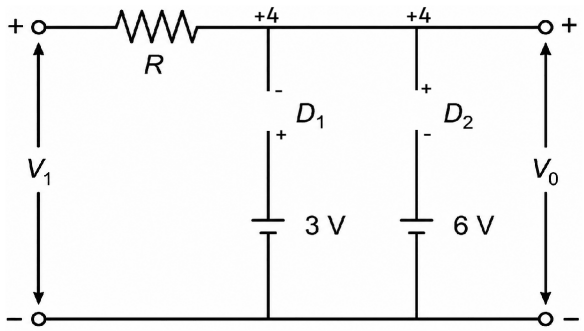

Fig. Solution Q1.99.2

Using the open-circuit test, assume both diodes are OFF.
Since no current flows through the resistor,

$$V_o = V_i = 4 \text{ V}.$$

The voltage across diode D_1 is

$$V_{D_1} = 3 - 4 = -1 \text{ V}.$$

Similarly, the voltage across diode D_2 is

$$V_{D_2} = 4 - 6 = -2 \text{ V}.$$

Since

$$V_{D_1} < 0 \quad \text{and} \quad V_{D_2} < 0,$$

both diodes are reverse biased and remain OFF.
Hence, the assumption is correct, and

$$V_o = 4 \text{ V}.$$

Therefore, the correct answer is

B. 4 V

SOLUTIONS

II

MOSFETs

Topics

1. Introduction to MOS - Depletion and Enhancement Type MOSFETs
2. Important Concepts of MOS Physics
3. Biasing a MOSFET - Transconductance and Output Impedance
4. Biasing a MOSFET - Amplification & Small Signal Analysis
5. Small Signal Model of MOSFET and Port Impedance
6. Common Source Amplifier - With and Without Degeneration
7. Common Source Amplifier with Different Biasing
8. Miller's Theorem
9. Common Gate and Common Drain Amplifiers
10. GmRout Method for Gain
11. PMOS Amplifiers
12. Cascode Current Source and Amplifiers
13. Current Mirror Circuits
14. Multistage Amplifiers and Pass Transistor Logic
15. Frequency Response of MOSFET Amplifiers
16. Frequency Response of Common Gate Amplifiers
17. Introduction to Differential Amplifiers
18. Current Source, Voltage Source as Tail
19. Large Signal Analysis of Differential Amplifiers
20. Concept of Half Circuit
21. Building a Two-Stage Op Amp
22. Types and Effects of Feedback
23. Feedback Topologies - Voltage-Voltage, Voltage-Current, Current-Voltage, Current-Current
24. MOS Amplifiers with Body Effect

Unit Snapshot*

Stream: IN	**Questions:** 13	**MCQ:** 7	**MSQ:** 0	**NAT:** 6
1M: 3	**2M:** 10	**Descriptive:** 0	**Avg Weightage** ~3M	**Range:** 1M(2026)–4M(2022)

**See preface for how Avg Weightage and Range are calculated.*

PrepFusion

Q2.26.1 MCQ 2026 1M Ans: B

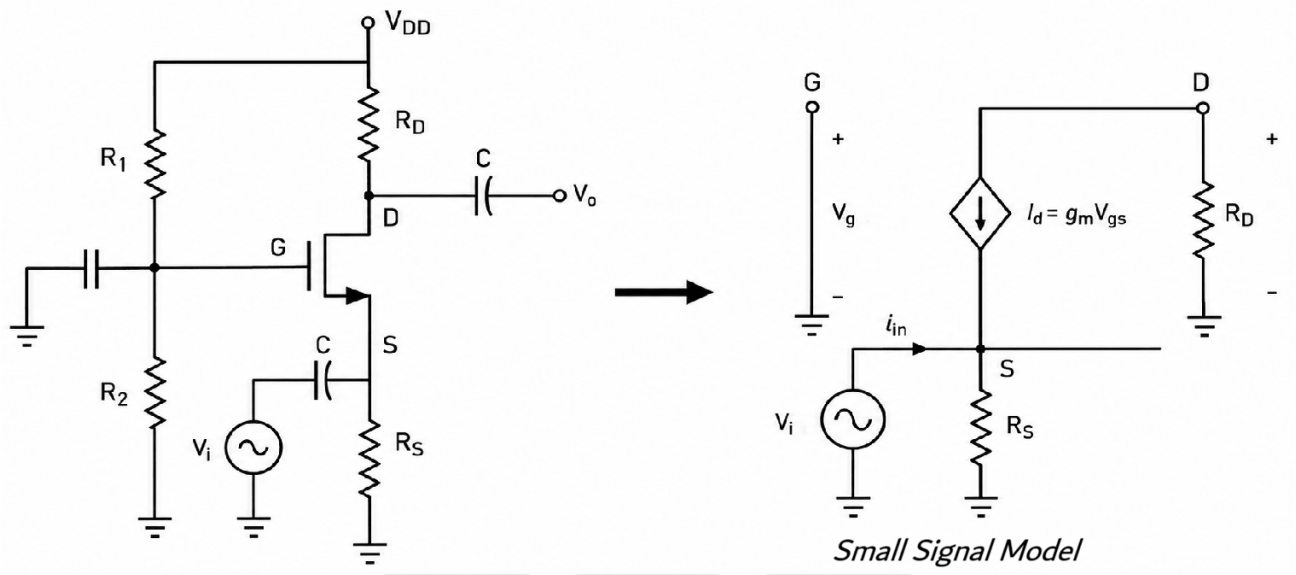

Fig. Solution Q2.26.1

A common-gate amplifier has Low input impedance ($R_{in} \approx \frac{1}{g_m}$) and High output impedance ($R_{out} \approx R_D$)
Hence, it senses an input current without significantly loading the source and delivers an output current to the load.
The current gain is

$$A_i = \frac{I_{\text{out}}}{I_{\text{in}}} \approx 1.$$

Thus,

$$I_{\text{out}} \approx I_{\text{in}}.$$

A Current Controlled Current Source (CCCS) is a dependent source in which the output current is controlled by the input current.
A CCCS has low input impedance so that it can easily sense the input current without causing a significant voltage drop at the input. It has high output impedance so that the output current remains nearly constant and is almost independent of the load resistance, thereby behaving as an ideal current source.
Hence, the common-gate amplifier is modeled as a Current Controlled Current Source (CCCS).
Hence, the correct answer is

B. Current Controlled Current Source

Q2.24.1 NAT 2024 1M Ans: 500

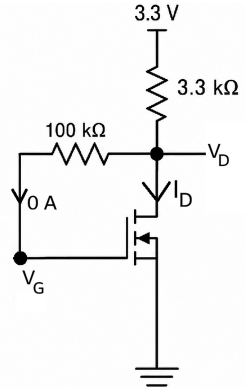

Fig. Solution Q2.24.1

For DC analysis, the coupling capacitors act as open circuits.
Since the gate current of a MOSFET is zero, $I_G = 0$.
Hence, there is no voltage drop across the 100 kΩ resistor, and $\therefore V_G = V_D$.
Since the source is grounded,

$$V_{GS} = V_G = V_D = V_{DS}.$$

Thus,

$$V_{DS} = V_{GS} > V_{GS} - V_T,$$

which confirms that the MOSFET operates in saturation.
Applying KVL,

$$V_D = 3.3 - I_D R_D$$

$$\Rightarrow V_{GS} = 3.3 - 3.3 I_D$$

where I_D is in mA.
The saturation current is

$$I_D = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS} - V_T)^2.$$

Given,

$$\mu_n C_{ox} = 250\ \mu\text{A/V}^2 = 0.25\ \text{mA/V}^2, \quad \frac{W}{L} = 4, \quad V_T = 0.65\ \text{V}.$$

Hence,

$$I_D = \frac{1}{2}(0.25)(4)(V_{GS} - 0.65)^2$$

$$I_D = 0.5(V_{GS} - 0.65)^2.$$

Substituting $V_{GS} = 3.3 - 3.3 I_D$,

$$I_D = 0.5(2.65 - 3.3 I_D)^2.$$

Expanding,

$$I_D = 0.5\left(7.0225 - 17.49 I_D + 10.89 I_D^2\right)$$

$$5.445 I_D^2 - 9.745 I_D + 3.51125 = 0.$$

Using the quadratic formula,

$$I_D = \frac{9.745 \pm \sqrt{9.745^2 - 4(5.445)(3.51125)}}{2(5.445)} = \frac{9.745 \pm 4.30}{10.89}$$

Thus,

$$I_D = 1.29\ \text{mA} \quad \textit{or} \quad I_D = 0.50\ \text{mA}$$

For $I_D = 1.29$ mA,

$$V_{GS} = 3.3 - 3.3(1.29) = -0.957 \text{ V},$$

which is less than V_T. Hence, this root is invalid.
For $I_D = 0.50$ mA,

$$V_{GS} = 3.3 - 3.3(0.50) = 1.65 \text{ V},$$

which satisfies $V_{GS} > V_T$. Hence, this root is valid.
Therefore,

$$I_D = 0.50 \text{ mA} = 500\ \mu\text{A}.$$

Hence,

$$\boxed{500\ \mu\text{A}}$$

Key Points

- **MOSFET Drain Current Equations (Long-Channel Model)**

Region	Condition	Drain Current
Cutoff	$V_{GS} < V_T$	$I_D = 0$
Linear (Triode)	$V_{GS} > V_T$ $V_{DS} < V_{GS} - V_T$	$I_D = \mu_n C_{ox} \frac{W}{L} \left[(V_{GS} - V_T)V_{DS} - \frac{V_{DS}^2}{2} \right]$
Saturation	$V_{GS} > V_T$ $V_{DS} \geq V_{GS} - V_T$	$I_D = \frac{1}{2}\mu_n C_{ox} \frac{W}{L}(V_{GS} - V_T)^2$

Q2.23.1 NAT 2023 2M Ans: 0.67

DC Analysis

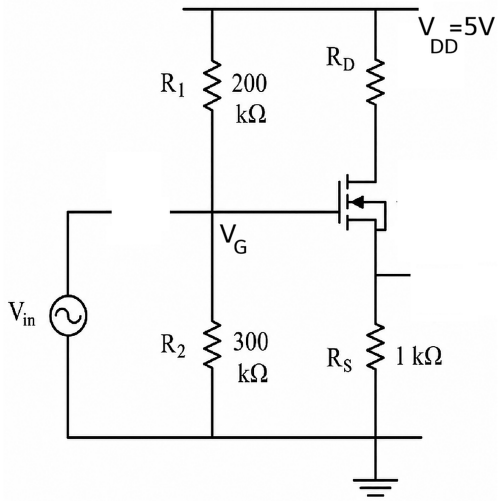

Fig. Solution Q2.23.1

For DC analysis, the coupling capacitors act as open circuits.
The gate voltage is obtained using the voltage divider:

$$V_G = V_{DD}\left(\frac{R_2}{R_1 + R_2}\right) = 5\left(\frac{300}{200 + 300}\right) = 3 \text{ V}.$$

The source voltage is

$$V_S = I_D R_S.$$

$$\Rightarrow V_{GS} = V_G - V_S = 3 - I_D(1\ \text{k}\Omega).$$

Since

$$\mu_n C_{ox} = 1\ \text{mA/V}^2, \qquad \frac{W}{L} = 2, \qquad V_T = 1\ \text{V},$$

the saturation current is

$$I_D = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS} - V_T)^2$$

$$I_D = 1 \times 10^{-3}(V_{GS} - 1)^2.$$

Also,

$$I_D = \frac{3 - V_{GS}}{1000} = 10^{-3}(3 - V_{GS}).$$

Equating the two expressions,

$$3 - V_{GS} = (V_{GS} - 1)^2.$$

$$3 - V_{GS} = V_{GS}^2 - 2V_{GS} + 1.$$

$$V_{GS}^2 - V_{GS} - 2 = 0.$$

$$(V_{GS} - 2)(V_{GS} + 1) = 0.$$

The valid solution is

$$V_{GS} = 2\ \text{V}.$$

Hence,

$$I_D = 1 \times 10^{-3}(2 - 1)^2 = 1\ \text{mA}.$$

The transconductance is

$$g_m = \mu_n C_{ox}\frac{W}{L}(V_{GS} - V_T)$$

$$= (1 \times 10^{-3})(2)(2 - 1) = 2\ \text{mS}.$$

AC Analysis

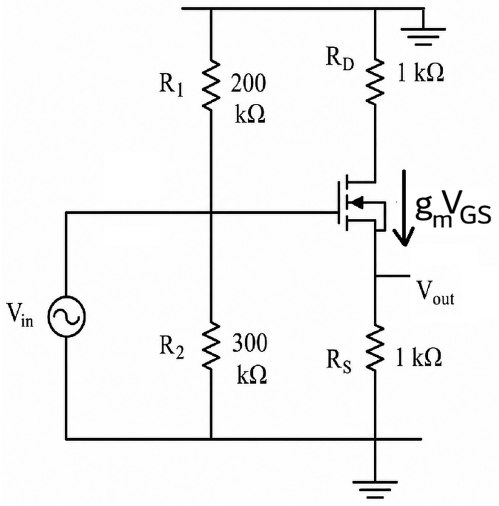

Fig. Solution Q2.23.1

For small-signal analysis, the coupling capacitors are short circuits and V_{DD} is AC ground.

The small-signal drain current is

$$i_d = g_m v_{gs}.$$

Since

$$v_{gs} = v_{in} - v_{out},$$

we get

$$i_d = g_m(v_{in} - v_{out}).$$

The output is taken from the source terminal $\Rightarrow v_{out} = i_d R_S$.
Substituting,

$$v_{out} = g_m R_S(v_{in} - v_{out}).$$

We have,

$$g_m R_S = (2\text{ mS})(1\text{ k}\Omega) = 2,$$

$$\Rightarrow v_{out} = 2(v_{in} - v_{out}) \Rightarrow 3v_{out} = 2v_{in}$$

Hence,

$$\frac{V_{out}}{V_{in}} = \frac{2}{3} = 0.67.$$

Hence,

$$\boxed{0.67}$$

Key Points

- **MOSFET Transconductance (g_m) in Saturation (neglecting CLM)**

$$g_m = \frac{\partial I_D}{\partial V_{GS}} = \mu_n C_{ox} \frac{W}{L}(V_{GS} - V_T)$$

$$g_m = \sqrt{2\mu_n C_{ox} \frac{W}{L} I_D}$$

$$g_m = \frac{2I_D}{V_{GS} - V_T} = \frac{2I_D}{V_{OV}}$$

Q2.22.1 MCQ | 2022 | 2M | Ans: B

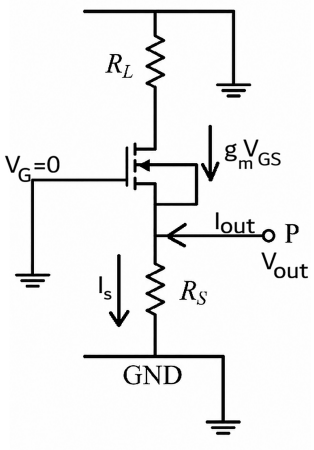

Fig. Solution Q2.22.1

To find the output impedance, all independent sources are set to zero.
Apply a test voltage V_{out} at node P and let the resulting current be I_{out}.
Applying KCL at node P,

$$V_{out} = R_S\left(g_m V_{GS} + I_{out}\right).$$

Since the gate is grounded,

$$V_{GS} = 0 - V_{out} = -V_{out}.$$

Substituting,

$$V_{out} = R_S\left(-g_m V_{out} + I_{out}\right).$$

$$V_{out}\left(1 + g_m R_S\right) = R_S I_{out}.$$

Hence,

$$R_{out} = \frac{V_{out}}{I_{out}} = \frac{R_S}{1 + g_m R_S}.$$

Dividing the numerator and denominator by g_m,

$$R_{out} = \frac{R_S/g_m}{R_S + 1/g_m} = R_S \parallel g_m^{-1}.$$

Hence, the correct answer is

$$\boxed{\textbf{B. } R_S \parallel g_m^{-1}}$$

Q2.22.2 MCQ 2022 2M Ans: D ●●○

The source is at small-signal ground and the gate draws no current. The two equal gate-divider resistors give

$$v_{gs} = v_g = v_o/2.$$

At the output node, the current entering from the supply resistor is $(v_d - v_o)/R_0$. It equals the currents leaving through the two-resistor divider and the transistor:

$$\frac{v_d - v_o}{R_0} = \frac{v_o}{2R_0} + \frac{g_m v_o}{2}.$$

Multiplying by R_0 and collecting terms gives

$$v_d = v_o\left(\frac{3}{2} + \frac{g_m R_0}{2}\right).$$

Hence

$$\frac{v_o}{v_d} = \boxed{\left(\frac{3}{2} + \frac{g_m R_0}{2}\right)^{-1}},$$

option (D).

Mistakes to be avoided

- The current cannot enter Gate terminal but can go to ground via R_0 so do not forget to include $\frac{v_0}{2R_0}$ in KCL

Q2.19.1 MCQ 2019 2M Ans: C ●●○

Method 1

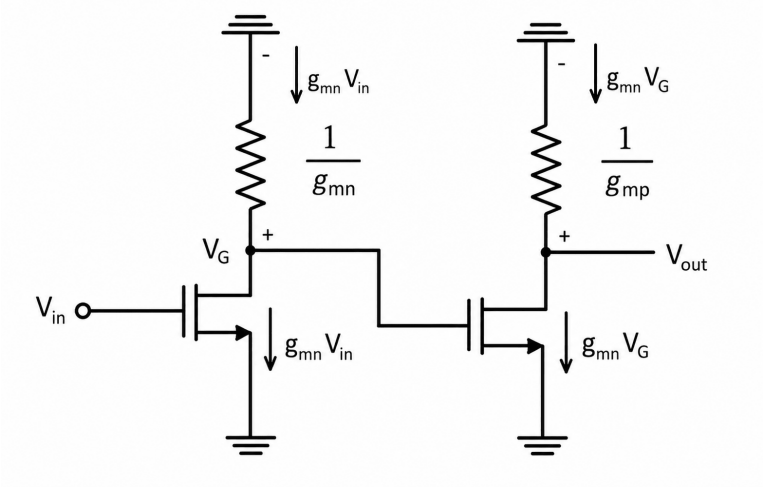

Fig. Solution Q2.19.1

Since M_1 and M_3 are diode-connected MOSFETs, their drain and gate terminals are shorted. Hence, each diode-connected MOSFET can be replaced by its small-signal resistance,

$$\frac{1}{g_{mn}} \quad \text{and} \quad \frac{1}{g_{mp}}, \quad \text{respectively}$$

$$\Rightarrow V_G = -\frac{1}{g_{mn}} g_{mn} v_{in} = -v_{in}$$

The output voltage is

$$v_{out} = \frac{-g_{mn} V_G}{g_{mp}}$$

Hence,

$$\frac{v_{out}}{v_{in}} = \frac{g_{mn}}{g_{mp}}.$$

Method 2

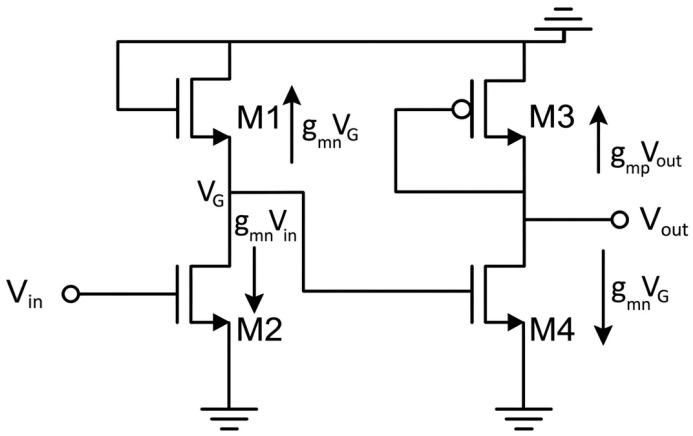

Fig. Solution Q2.19.1

With reference to the above small signal equivalent circuit,

$$g_{mn} v_{in} = -g_{mp} V_G, \qquad \text{Since no current enters gate terminal}$$

Also,

$$g_{mn}V_G = -g_{mp}v_{out}$$

$$\Rightarrow \frac{v_{out}}{v_{in}} = \frac{g_{mn}}{g_{mp}}$$

Hence, the correct answer is

$$\boxed{\textbf{C. } \frac{g_{mn}}{g_{mp}}}$$

Key Points

- Since a MOSFET is a **structurally symmetrical device**, the source and drain terminals are determined by their terminal voltages during operation.
- **NMOS:** The terminal at the **higher potential** acts as the **Drain**, and the terminal at the **lower potential** acts as the **Source**.
- **PMOS:** The terminal at the **lower potential** acts as the **Drain**, and the terminal at the **higher potential** acts as the **Source**.
- For normal operation, the body is usually connected to the source to keep the body-source junction reverse biased.

Mistakes to be avoided

- M3 is a p-channel MOSFET so be careful with source and drain terminal and current direction.

Q2.19.2 NAT | 2019 | 2M | Ans: 4 | ● ● ○

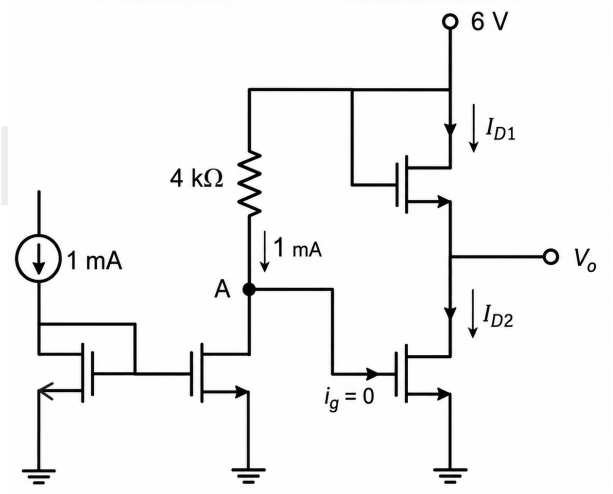

Fig. Solution Q2.19.2

By using concept of current mirror;

Given that all MOSFETs are identical, $\Rightarrow \frac{W}{L}$ is same for all

Hence, the current through $4k\Omega$ is $I_D = 1$ mA.

Applying KVL across the 4 kΩ resistor,

$$6 - 4\ \text{k}\Omega(1\ \text{mA}) - V_A = 0$$

$$V_A = 2\ \text{V}.$$

Since all the transistors are identical and operate in saturation,

$$I_{D1} = I_{D2}.$$

Using the saturation current equation,

$$\frac{k}{2}(V_{GS1} - V_T)^2 = \frac{k}{2}(V_{GS2} - V_T)^2.$$

Hence,

$$V_{GS1} = V_{GS2}.$$

For the upper transistor,

$$V_{GS1} = 6 - V_o.$$

For the lower transistor,

$$V_{GS2} = V_A = 2 \text{ V}.$$

Therefore,

$$6 - V_o = 2$$

$$V_o = 4 \text{ V}.$$

Hence,

$$\boxed{4 \text{ V}}$$

Q2.17.1 NAT 2017 2M Ans: 10

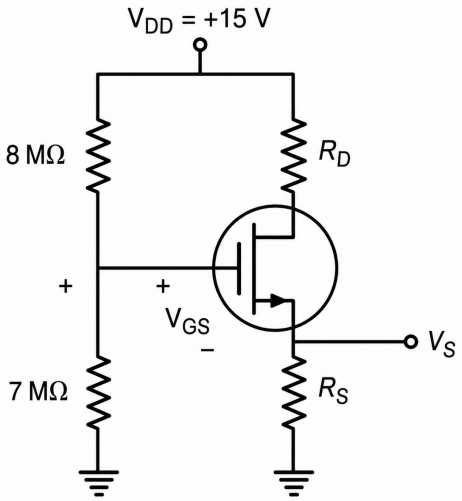

Fig. Solution Q2.17.1

Since the gate current of the MOSFET is zero,
Hence, the gate voltage is obtained using the voltage divider,

$$V_G = 15\left(\frac{7}{8+7}\right) = 7 \text{ V}.$$

Given,

$$V_S = 5 \text{ V}.$$

Therefore,

$$V_{GS} = V_G - V_S = 7 - 5 = 2 \text{ V}.$$

The MOSFET is operating in saturation, and its transfer characteristic is

$$I_D = \frac{1}{2}(V_{GS} - 1)^2 \text{ mA}.$$

Substituting $V_{GS} = 2$ V,

$$I_D = \frac{1}{2}(2 - 1)^2 = 0.5 \text{ mA}.$$

Applying Ohm's law,

$$V_S = I_D R_S.$$

Hence,

$$R_S = \frac{V_S}{I_D} = \frac{5}{0.5 \times 10^{-3}} = 10 \text{ k}\Omega.$$

Hence,

$$\boxed{10}$$

Q2.15.1 NAT 2015 2M Ans: 20

Since the two NMOS transistors are identical and are operating in saturation, both will have same saturation drain current,

$$\frac{k}{2}(V_{GS1} - V_T)^2 = \frac{k}{2}(V_{GS2} - V_T)^2$$

$$\Rightarrow V_{GS1} = V_{GS2}$$

For the upper NMOS,

$$V_{GS1} = 30 - V_{\text{out}}$$

For the lower NMOS,

$$V_{GS2} = 10 - 0 = 10 \text{ V}$$

Equating the two gate-to-source voltages,

$$30 - V_{\text{out}} = 10$$

$$V_{\text{out}} = 20 \text{ V}$$

Therefore,

$$\boxed{20}$$

Q2.15.2 NAT 2015 2M Ans: 1

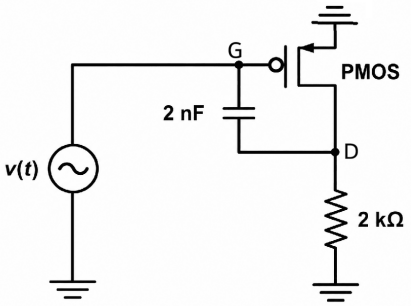

Fig. Solution Q2.15.2

For small-signal analysis, the DC supply is treated as AC ground.

Given,

$$g_m = 1 \text{ mA/V}, \qquad R_D = 2 \text{ k}\Omega,$$
$$C_{gs} = 4 \text{ nF}, \qquad C_{gd} = 2 \text{ nF}, \qquad \omega = 100 \text{ rad/s}$$

The voltage gain of the PMOS stage is

$$A = -g_m R_D = -(1 \times 10^{-3})(2 \times 10^3) = -2.$$

Using Miller's theorem,

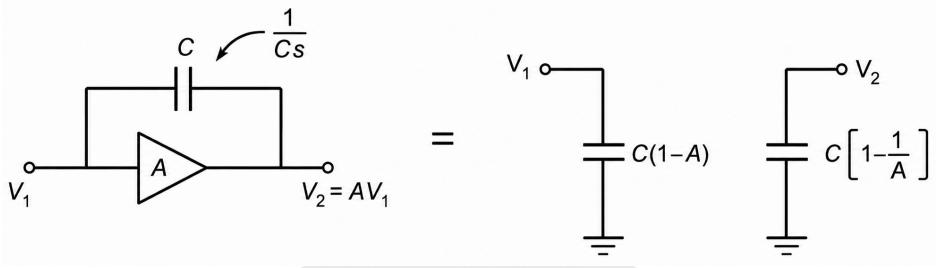

Fig. Solution Q2.15.2

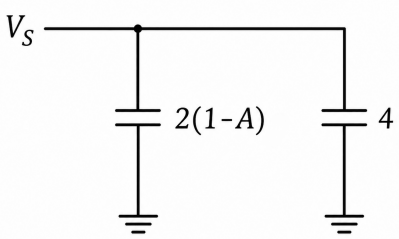

Fig. Solution Q2.15.2

The equivalent input capacitance is

$$C_{\text{in}} = C_{gs} + (1 - A)C_{gd}.$$

Substituting the given values,

$$C_{\text{in}} = 4 + (1 - (-2)) \times 2 = 10 \text{ nF}.$$

Hence, the input impedance is

$$Z_{\text{in}} = \frac{1}{j\omega C_{\text{in}}}.$$

Therefore,

$$|Z_{\text{in}}| = \frac{1}{100 \times 10 \times 10^{-9}} = 10^6 \ \Omega = 1 \ \text{M}\Omega.$$

Hence,

$$\boxed{1}$$

Q2.11.1 MCQ | 2011 | 2M | Ans: B | ●●○

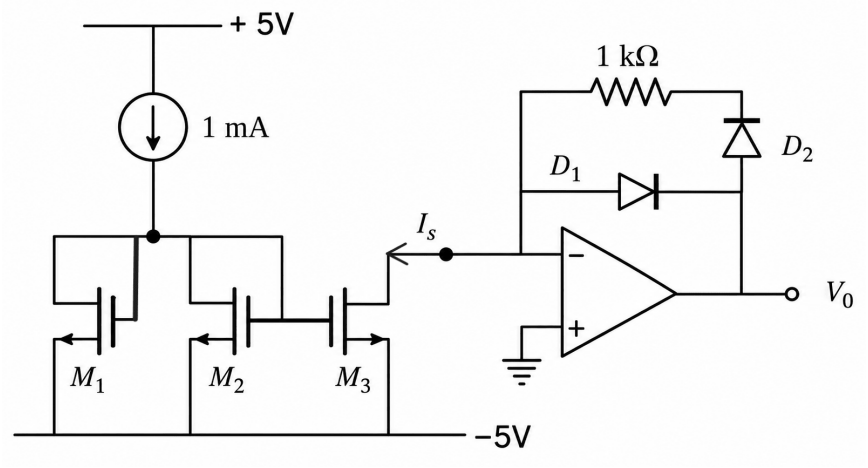

Fig. Solution Q2.11.1

Since M_1 and M_2 are identical, the 1 mA reference current divides equally.
Hence,

$$I_{D1} = I_{D2} = \frac{1}{2} = 0.5 \text{ mA}.$$

Also, M_2 and M_3 form a current mirror.
Therefore,

$$I_{D2} = I_{D3} = 0.5 \text{ mA}.$$

The op-amp input current is zero. Hence,

$$I_S = I_{D3} = 0.5 \text{ mA}.$$

Hence, the correct answer is

B. 0.5 mA

Q2.11.2 MCQ 2011 2M Ans: A ●●○

From Q2.11.1,

$$I_S = 0.5 \text{ mA}.$$

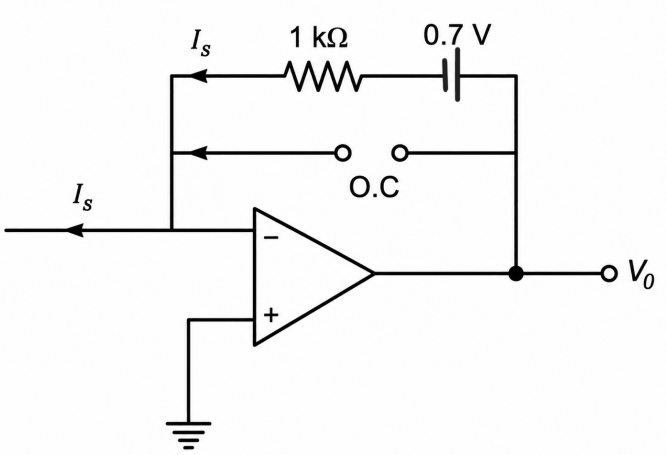

Fig. Solution Q2.11.2

Since the current flows in the direction shown, diode D_1 is reverse biased and diode D_2 is forward biased.

Applying KVL,

$$V_o = 0.7 + I_S(1\ \text{k}\Omega).$$

Substituting $I_S = 0.5$ mA,

$$V_o = 0.7 + 0.5 \times 10^{-3} \times 10^3 = 1.2\ \text{V}.$$

Hence, the correct answer is

A. 1.2 V

Q2.98.1 MCQ | 1998 | 1M | Ans: A

A → P-channel JFET,

B → N-channel JFET,

C → Enhancement-mode MOSFET,

D → Depletion-mode MOSFET.

Hence, the correct answer is

A

PrepFusion

SOLUTIONS

III

Operational Amplifier

Topics

1. Basics of Op-Amp, Comparator and Virtual Short
2. Inverting and Non-Inverting Amplifiers
3. Adder, Differential and Instrumentation Amplifiers
4. Log, Anti-Log and Rectifier Using Op-Amp
5. Filters Using Op-Amp
6. Offsets in Op-Amp
7. CMRR, Slew Rate and Frequency Response of Op-Amp
8. Integrator Using Op-Amp
9. Differentiator Using Op-Amp
10. Active Clipper, Clamper and Peak Detector
11. Multivibrators Using Op-Amp
12. Oscillators Using Op-Amp
13. Schmitt Triggers

Unit Snapshot*

Stream: IN	**Questions:** 127	**MCQ:** 86	**MSQ:** 1	**NAT:** 40
1M: 49	**2M:** 78	**Descriptive:** 0	**Avg Weightage** ~7M	**Range:** 2M(2025)–14M(2024)

**See preface for how Avg Weightage and Range are calculated.*

Q3.26.1 MCQ 2026 2M Ans: C ● ○ ○

The op-amps are ideal and operate with negative feedback. Hence, the inverting and non-inverting terminals are at the same potential.

$$V_{-1} = V_{+1} = 100\,\text{mV} = 0.1\,\text{V}, \qquad V_{-2} = V_{+2} = 0\,\text{V}$$

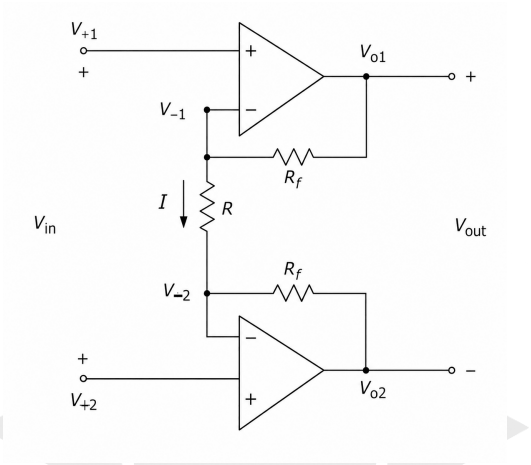

Fig. Solution Q3.26.1

Current through the resistor R is

$$I = \frac{0.1 - 0}{1\,\text{k}\Omega} = 0.1\,\text{mA}$$

For the upper op-amp,

$$\frac{V_{o1} - 0.1}{10\,\text{k}\Omega} = 0.1\,\text{mA}$$

$$V_{o1} = 0.1 + 1 = 1.1\,\text{V}$$

For the lower op-amp,

$$\frac{0 - V_{o2}}{10\,\text{k}\Omega} = 0.1\,\text{mA}$$

$$V_{o2} = -1\,\text{V}$$

Hence,

$$V_{out} = V_{o1} - V_{o2} = 1.1 - (-1) = 2.1\,\text{V}$$

(c) 2.1 V

Q3.26.2 MCQ 2026 2M Ans: C ● ○ ○

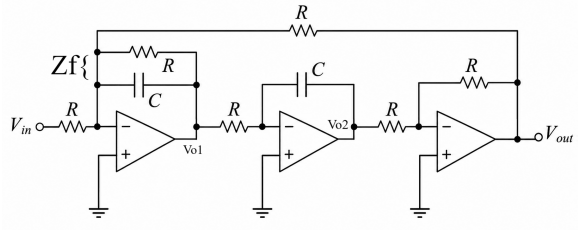

Fig. Solution Q3.26.2

The circuit consists of three cascaded inverting stages with an additional feedback path from V_{out} to the first summing node. For $R = 1\,\text{k}\Omega$ and $C = 1\,\mu\text{F}$,

$$RC = 10^{-3}\,\text{s}, \qquad \omega RC = 1000 \times 10^{-3} = 1$$

The first stage has feedback impedance

$$Z_f = R \parallel \frac{1}{sC} = \frac{R}{1 + sRC}$$

Hence,

$$V_{o1} = -\frac{1}{1 + sRC}(V_{in} + V_{out})$$

The second stage is an inverting integrator,

$$V_{o2} = -\frac{1}{sRC}V_{o1}$$

The third stage is an inverting amplifier of gain -1,

$$V_{out} = -V_{o2} = \frac{V_{o1}}{sRC}$$

Combining,

$$V_{out}\left[(sRC)^2 + sRC + 1\right] = -V_{in}$$

At $\omega RC = 1$, we have $sRC = j$,

$$\frac{V_{out}}{V_{in}} = -\frac{1}{j^2 + j + 1} = -\frac{1}{j} = j$$

Therefore,

$$\left|\frac{V_{out}}{V_{in}}\right| = 1$$

Since the amplitude of V_{in} is 1 V,

$$|V_{out}| = 1\,\text{V}$$

(c) 1.0 V

Key Points

- An inverting integrator has gain $-\frac{1}{sRC}$.
- At $\omega RC = 1$, substitute $sRC = j$.

Q3.25.1 MCQ | 2025 | 1M | Ans: A

For an ideal op-amp operating with negative feedback,

$$V_+ = V_-$$

The non-inverting input voltage is obtained using the voltage divider formed by R_3 and R_4:

$$V_+ = \frac{R_4}{R_3 + R_4} V$$

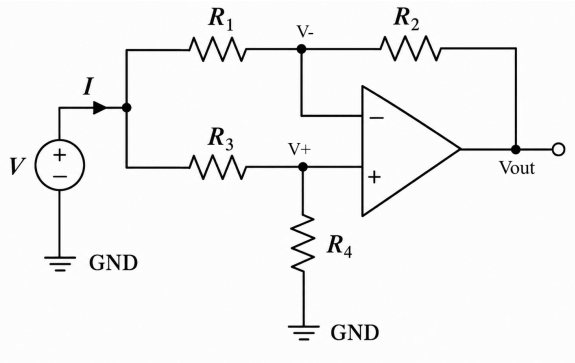

Fig. Solution Q3.25.1

Hence,

$$V_- = \frac{R_4}{R_3 + R_4} V$$

Applying KCL at the inverting node,

$$\frac{V - V_-}{R_1} = \frac{V_- - V_{out}}{R_2}$$

The source current is

$$I = \frac{V - V_-}{R_1} + \frac{V - V_+}{R_3}$$

Since $V_+ = V_- = \frac{R_4}{R_3 + R_4} V$,

$$I = \frac{V}{R_3 + R_4} + \frac{R_3}{R_1(R_3 + R_4)} V$$

$$I = \frac{V(R_1 + R_3)}{R_1(R_3 + R_4)}$$

Therefore,

$$\frac{V}{I} = \frac{R_1(R_3 + R_4)}{R_1 + R_3}$$

$$\boxed{\text{(a)}\ \frac{(R_3 + R_4)}{R_1 + R_3} R_1}$$

Q3.25.2 MSQ 2025 1M Ans: B,C,D ● ○ ○

Check each option based on whether the circuit performs a **linear** or **non-linear** operation.

A. Instrumentation amplifier → Linear amplification (×)

B. Schmitt trigger → Comparator with hysteresis (Non-linear) (✓)

C. Logarithmic amplifier → Produces logarithm of input (Non-linear) (✓)

D. Precision rectifier → Rectifies input signal (Non-linear) (✓)

(b), (c) and (d)

Q3.24.1 NAT 2024 1M Ans: 1.6 ● ○ ○

The CMOS logic has infinite input impedance. Hence, all the current through R_Z flows through the Zener diode.

$$V_o = 5\,\text{V}, \qquad V_Z = 1.8\,\text{V}, \qquad I_{Z(\text{min})} = 2\,\text{mA}$$

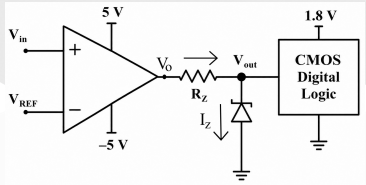

Fig. Solution Q3.24.1

Using Ohm's law,

$$I_Z = \frac{V_o - V_Z}{R_Z}$$

For the maximum allowable value of R_Z,

$$I_Z = I_{Z(\text{min})} = 2\,\text{mA}$$

$$R_{Z(\text{max})} = \frac{5 - 1.8}{2 \times 10^{-3}} = \frac{3.2}{0.002} = 1600\,\Omega = 1.6\,\text{k}\Omega$$

1.6

Key Points

- Infinite CMOS input impedance implies zero input current.
- The Zener diode maintains the output at 1.8 V.
- Maximum R_Z occurs when the Zener current is just equal to its minimum required value.

Q3.24.2 NAT 2024 1M Ans: -3 ● ○ ○

For an ideal op-amp with negative feedback,

$$V_- = V_+ = 0$$

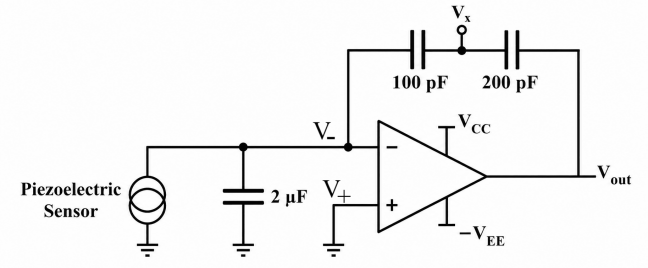

Fig. Solution Q3.24.2

Hence, the left terminal of the 100 pF capacitor is at virtual ground.
Since the change in the midpoint voltage is

$$V_x = -2\,\text{V},$$

the voltage across the 100 pF capacitor is

$$0 - (-2) = 2\,\text{V}$$

Therefore, the charge on the series capacitor is

$$Q = 100\,\text{pF} \times 2\,\text{V} = 200\,\text{pC}$$

The same charge flows through the 200 pF capacitor.

$$V_x - V_{out} = \frac{Q}{200\,\text{pF}} = \frac{200\,\text{pC}}{200\,\text{pF}} = 1\,\text{V}$$

Hence,

$$-2 - V_{out} = 1$$

$$V_{out} = -3\,\text{V}$$

$$\boxed{-3}$$

Q3.24.3 MCQ 2024 2M Ans: A

Since $V_{in} = 0$, only the offset voltage of op-amp A_1 affects the circuit.

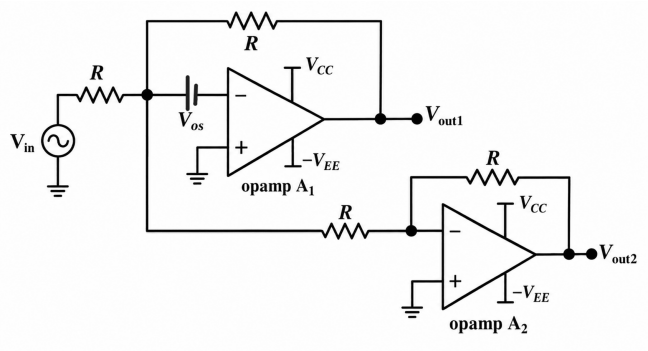

Fig. Solution Q3.24.3

Let the common summing node voltage be V_x.

For op-amp A_1,

$$V_{+1} = 0, \qquad V_{-1} = -V_{OS} = -1\,\text{mV}$$

Hence,

$$V_x = -1\,\text{mV}.$$

The inverting input of op-amp A_2 is at virtual ground,

$$V_{-2} = 0.$$

Applying KCL at node V_x,

$$\frac{V_x - 0}{R} + \frac{V_x - V_{out1}}{R} + \frac{V_x - 0}{R} = 0$$

Substituting $V_x = -1\,\text{mV}$,

$$\frac{-1}{R} + \frac{-1 - V_{out1}}{R} + \frac{-1}{R} = 0$$

$$V_{out1} = -3\,\text{mV}$$

Considering the polarity of the offset source shown in the figure,

$$V_{out1} = 3\,\text{mV}$$

For op-amp A_2, applying KCL at its inverting node,

$$\frac{V_x - 0}{R} + \frac{V_{out2} - 0}{R} = 0$$

$$V_{out2} = V_x = -1\,\text{mV}$$

(a) $V_{out1} = 3\,\text{mV}$, $V_{out2} = -1\,\text{mV}$

Mistakes to be avoided

- Do not neglect the resistor connecting the first summing node to op-amp A_2.
- Do not assume the input of A_2 is driven by V_{out1}; it is driven by the common summing node.
- Use the offset polarity exactly as shown in the figure.

Q3.24.4 MCQ 2024 2M Ans: A ●○○

The circuit is an inverting first-order active low-pass filter. The feedback impedance is

$$Z_f = R_f \parallel \frac{1}{sC} = \frac{R_f}{1 + sR_fC}$$

Hence, the transfer function is

$$\frac{V_{out}}{V_{in}} = -\frac{Z_f}{R} = -\frac{R_f}{R} \cdot \frac{1}{1 + sR_fC}$$

The dc gain is given as

$$-\frac{R_f}{R} = -10$$

Since $R = 1\,\text{k}\Omega$,

$$R_f = 10R = 10\,\text{k}\Omega$$

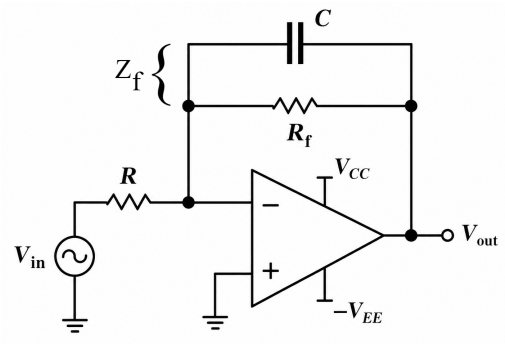

Fig. Solution Q3.24.4

The cut-off frequency is

$$f_c = \frac{1}{2\pi R_f C}$$

Substituting $R_f = 10\,\text{k}\Omega$ and $C = 0.1\,\mu\text{F}$,

$$f_c = \frac{1}{2\pi(10\times10^3)(0.1\times10^{-6})} = \frac{1}{2\pi\times10^{-3}} = 159.15\,\text{Hz}$$

$$f_c \approx 159.1\,\text{Hz}$$

(a) 159.1 Hz

Q3.24.5 NAT 2024 2M Ans: 4

To find the equivalent resistance between terminals A and B, deactivate the independent voltage source V_{in}.

$V_{in} = 0 \Rightarrow$ replace the voltage source by a short circuit.

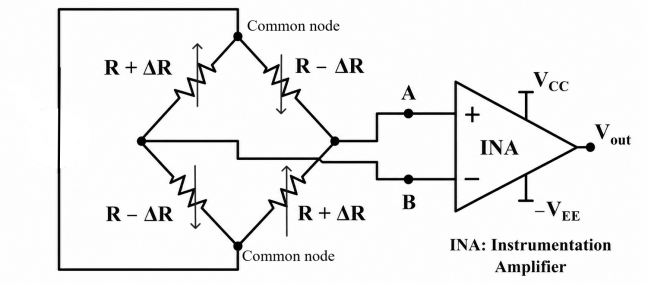

Fig. Solution Q3.24.5

Now,

$$R + \Delta R = 4.5 + 1.5 = 6\,\text{k}\Omega,$$

$$R - \Delta R = 4.5 - 1.5 = 3\,\text{k}\Omega.$$

The resistance from terminal A to the common node is

$$R_A = (6\,\text{k}\Omega) \parallel (3\,\text{k}\Omega) = \frac{6 \times 3}{6 + 3} = 2\,\text{k}\Omega.$$

Similarly, the resistance from terminal B to the common node is

$$R_B = (3\,\text{k}\Omega) \parallel (6\,\text{k}\Omega) = 2\,\text{k}\Omega.$$

Since the only path between A and B is through the common node,

$$R_{AB} = R_A + R_B = 2 + 2 = 4\,\text{k}\Omega.$$

$$\boxed{4}$$

Q3.24.6 NAT 2024 2M Ans: 6.17

The circuit is an op-amp relaxation oscillator. The non-inverting input forms a Schmitt trigger, while the capacitor charges and discharges through R.

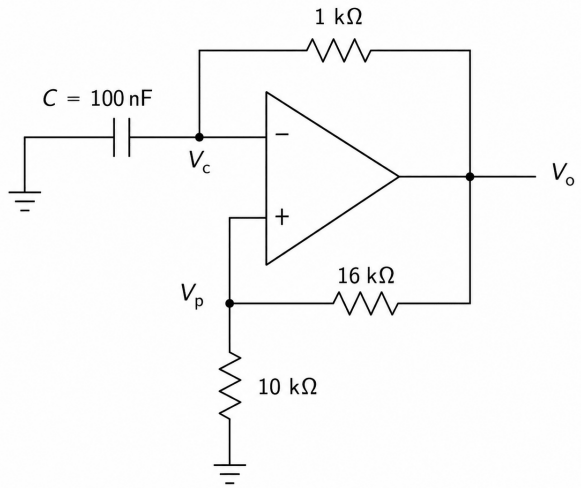

Fig. Solution Q3.24.6

The switching threshold is

$$V_P = \beta V_{sat}, \qquad \beta = \frac{10}{10 + 16} = \frac{5}{13}$$

Since $V_{sat} = 13\,\text{V}$,

$$V_P = \frac{5}{13} \times 13 = 5\,\text{V}$$

The capacitor charges from $-V_P$ to $+V_P$ when the output is at $+V_{sat}$.

$$t = RC \ln\left(\frac{1 + \beta}{1 - \beta}\right)$$

Substituting

$$R = 1\,\text{k}\Omega, \quad C = 100\,\text{nF},$$

$$t = (10^{-4})\ln\left(\frac{18}{8}\right) = 81.09\,\mu\text{s}$$

Hence,

$$T = 2t = 162.19\,\mu\text{s}$$

and

$$f = \frac{1}{T} = \frac{1}{162.19 \times 10^{-6}} = 6.17 \times 10^3\,\text{Hz} = 6.17\,\text{kHz}$$

6.17

Key Points

- Schmitt trigger threshold: $V_P = \beta V_{sat}$, where $\beta = \dfrac{R_1}{R_1 + R_2}$.

Q3.24.7 NAT | 2024 | 2M | Ans: 10 ●○○

The op-amp is configured as a non-inverting amplifier with

$$A_v = 1 + \frac{5R}{R} = 6.$$

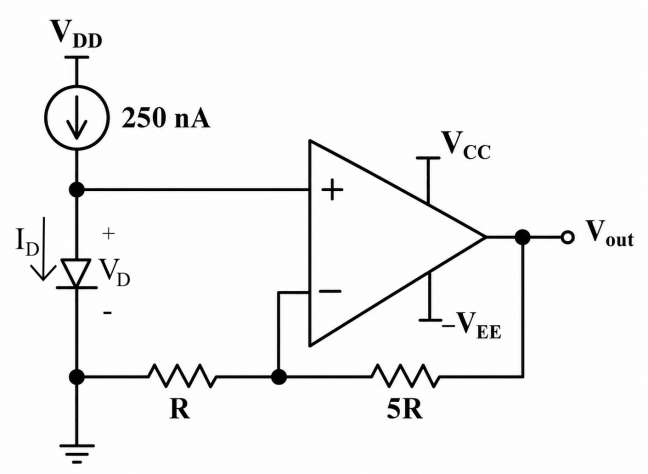

Fig. Solution Q3.24.7

Hence,

$$V_{out} = 6V_D.$$

The current source is ideal, therefore the diode current is constant:

$$I_D = 250\,\text{nA}.$$

Using the given diode equation,

$$I_D = I_S e^{\frac{\alpha V_D}{T}},$$

taking natural logarithm,

$$\ln\left(\frac{I_D}{I_S}\right) = \frac{\alpha V_D}{T}$$

or

$$V_D = \frac{T}{\alpha}\ln\left(\frac{I_D}{I_S}\right).$$

Differentiating with respect to temperature,

$$\frac{dV_D}{dT} = \frac{1}{\alpha}\ln\left(\frac{I_D}{I_S}\right).$$

Substituting

$$I_D = 250\,\text{nA}, \qquad I_S = 10^{-15}\,\text{A}, \qquad \alpha = 1.16 \times 10^4\,\text{K/V},$$

$$\frac{dV_D}{dT} = \frac{\ln\left(\frac{250 \times 10^{-9}}{10^{-15}}\right)}{1.16 \times 10^4} = 1.667\,\text{mV/K}.$$

Therefore,

$$\frac{dV_{out}}{dT} = 6\frac{dV_D}{dT} = 6 \times 1.667 = 10.0\,\text{mV/K}.$$

$$\boxed{10.0}$$

Mistakes to be avoided

- Do not differentiate I_D; it is fixed by the ideal current source.
- Remember to multiply $\frac{dV_D}{dT}$ by the closed-loop gain of the op-amp.

Q3.24.8 NAT 2024 2M Ans: -2.5 ●○○

The given circuit in question is an ideal inverting integrator.
The output voltage is

$$V_{out}(t) = -\frac{1}{RC}\int V_{in}(t)\,dt + V_{out}(0).$$

Since the capacitor is initially uncharged,

$$V_{out}(0) = 0.$$

Given

$$R = 1\,\text{k}\Omega, \qquad C = 2\,\mu\text{F},$$

$$RC = (1 \times 10^3)(2 \times 10^{-6}) = 2\,\text{ms}.$$

From the input waveform,

$$V_{in} = 5\,\text{V} \quad (0 \le t \le 1\,\text{ms}),$$

and

$$V_{in} = 0 \quad (1 < t < 2\,\text{ms}).$$

During the first interval,

$$V_{out}(1\,\text{ms}) = -\frac{1}{2\,\text{ms}}(5 \times 1\,\text{ms}) = -2.5\,\text{V}.$$

From 1 ms to 2 ms,

$$V_{in} = 0$$

so the capacitor neither charges nor discharges. Hence,

$$V_{out}(2\,\text{ms}) = V_{out}(1\,\text{ms}) = -2.5\,\text{V}.$$

$$\boxed{-2.5}$$

Q3.23.1 NAT | 2023 | 1M | Ans: 1110 | ● ○ ○

The circuit is an inverting amplifier with

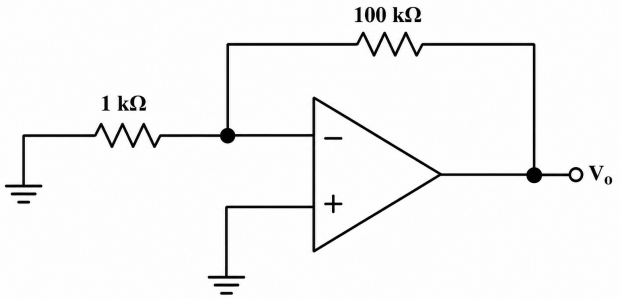

Fig. Solution Q3.23.1

$$R_1 = 1\,\text{k}\Omega, \qquad R_f = 100\,\text{k}\Omega.$$

Hence, the noise gain is

$$A_N = 1 + \frac{R_f}{R_1} = 1 + \frac{100}{1} = 101.$$

The equivalent resistance seen at the inverting input is

$$R_{eq} = R_1 \parallel R_f = \frac{1 \times 100}{1 + 100}\,\text{k}\Omega \approx 0.99\,\text{k}\Omega.$$

The output error due to the input offset voltage is

$$V_{o,OS} = A_N V_{OS} = 101 \times 10\,\mu\text{V} = 1010\,\mu\text{V}.$$

The output error due to the input bias current is

$$V_{o,IB} = I_B R_f = (1\,\text{nA})(100\,\text{k}\Omega) = 100\,\mu\text{V}.$$

For the worst case, both errors add in the same direction.

$$V_{o,\max} = 1010 + 100 = 1110\,\mu\text{V}.$$

$$\boxed{1110}$$

Key Points

- Output offset due to input offset voltage is $A_N V_{OS}$.
- Input bias current flowing through the feedback resistor produces an output error of $I_B R_f$.

Q3.23.2 NAT 2023 2M Ans: 0.50

Since the op-amps are ideal,

$$V_{-1} = V_{+1} = V_{S1}, \qquad V_{-2} = V_{+2} = V_{S2}.$$

The three resistors between V_{o1} and V_{o2} are equal. Hence,

$$V_{S1} = \frac{2V_{o1} + V_{o2}}{3},$$

$$V_{S2} = \frac{V_{o1} + 2V_{o2}}{3}.$$

Solving,

$$V_{o1} = 2V_{S1} - V_{S2},$$

$$V_{o2} = 2V_{S2} - V_{S1}.$$

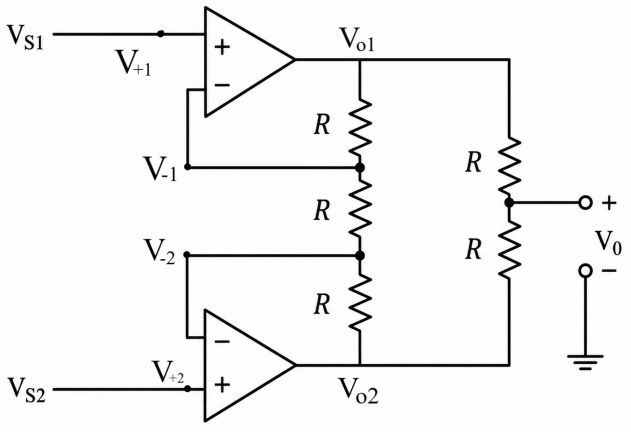

Fig. Solution Q3.23.2

The output V_o is taken from the midpoint of two equal resistors connected between V_{o1} and V_{o2}. Therefore,

$$V_o = \frac{V_{o1} + V_{o2}}{2} = \frac{V_{S1} + V_{S2}}{2}.$$

Substituting

$$V_{S1} = 3 + 0.1\sin(300t),$$

$$V_{S2} = -2 + 0.11\sin(300t),$$

gives

$$V_o = \frac{1 + 0.21\sin(300t)}{2} = 0.5 + 0.105\sin(300t).$$

Hence, the average value of V_o is

$$\boxed{0.50}$$

Q3.23.3 NAT | 2023 | 2M | Ans: 100 | ●○○

Initially, the 10 nF feedback capacitor stores

$$Q_0 = 1\,\text{nC}.$$

Hence, the initial output voltage is

$$V_o(0) = \frac{Q_0}{C_f} = \frac{1\,\text{nC}}{10\,\text{nF}} = 100\,\text{mV}.$$

When V_s is High, the 1 nF capacitor is connected to

$$V_{in} = 100\,\text{mV},$$

so it stores

$$Q_s = C_s V_{in} = (1\,\text{nF})(100\,\text{mV}) = 0.1\,\text{nC}.$$

When V_s is Low, this charge is transferred to the feedback capacitor. Therefore, the output changes by

$$\Delta V_o = -\frac{Q_s}{C_f} = -\frac{0.1\,\text{nC}}{10\,\text{nF}} = -10\,\text{mV}$$

per switching cycle.
The switching frequency is 1 kHz. Thus, in

$$t = 20\,\text{ms},$$

the number of cycles is

$$N = 1000 \times 0.02 = 20.$$

Hence,

$$V_o(20\,\text{ms}) = 100 - 20(10) = -100\,\text{mV}.$$

Therefore,

$$|V_o| = 100\,\text{mV}.$$

$$\boxed{100}$$

Q3.23.4 NAT | 2023 | 2M | Ans: 2 | ●○○

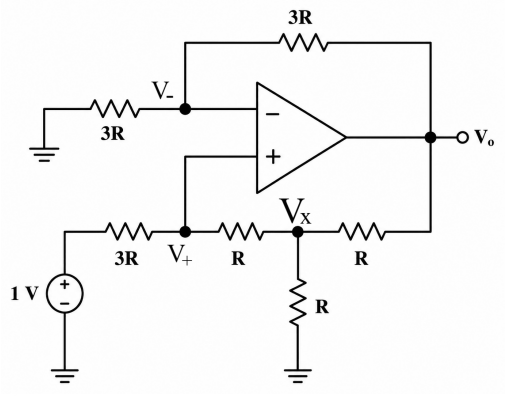

Fig. Solution Q3.23.4

Let the voltage at the inverting and non-inverting terminals be V_- and V_+, respectively.
Since the op-amp is ideal,

$$V_- = V_+.$$

At the inverting input, the two resistors are equal $(3R)$. Hence,

$$V_- = \frac{V_o}{2}.$$

Therefore,

$$V_+ = \frac{V_o}{2}.$$

Let the voltage at the middle node of the lower network be V_x.
Applying KCL at V_+,

$$\frac{V_+ - 1}{3R} + \frac{V_+ - V_x}{R} = 0$$

$$4V_+ - 3V_x = 1.$$

Applying KCL at node V_x,

$$\frac{V_x - V_+}{R} + \frac{V_x - V_o}{R} + \frac{V_x}{R} = 0$$

$$3V_x - V_+ - V_o = 0.$$

Using

$$V_+ = \frac{V_o}{2},$$

the above equation becomes

$$3V_x - \frac{3V_o}{2} = 0$$

$$V_x = \frac{V_o}{2} = V_+.$$

Substituting into

$$4V_+ - 3V_x = 1,$$

gives

$$V_+ = 1\,\text{V}.$$

Hence,

$$V_o = 2V_+ = 2\,\text{V}.$$

$$\boxed{2.0}$$

Q3.22.1 NAT 2022 1M Ans: 10

For a voltage follower, the feedback factor is

$$\beta = 1.$$

The closed-loop output impedance is given by

$$Z_{out(CL)} = \frac{Z_{out}}{1 + A\beta}.$$

Given,

$$A = 10^4, \qquad Z_{out} = 100\,\Omega.$$

Hence,

$$Z_{out(CL)} = \frac{100}{1 + 10^4} = \frac{100}{10001} \approx 0.009999\,\Omega.$$

Converting into milliohms,

$$0.009999\,\Omega = 9.999\,\text{m}\Omega \approx 10.0\,\text{m}\Omega.$$

$$\boxed{10.0}$$

Q3.22.2 NAT 2022 2M Ans: 15.84

The circuit is an inverting amplifier with a parallel RC feedback network.

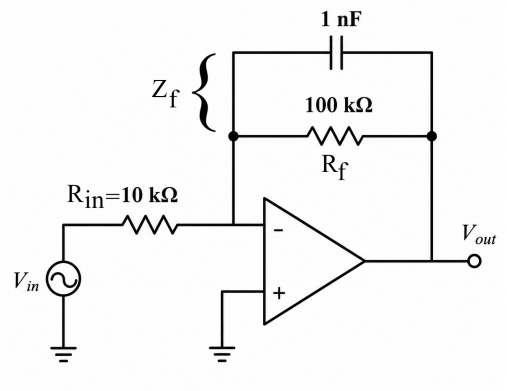

Fig. Solution Q3.22.2

The feedback impedance is

$$Z_f = R_f \parallel \frac{1}{j\omega C} = \frac{R_f}{1 + j\omega R_f C}.$$

Hence, the closed-loop gain is

$$\frac{V_{out}}{V_{in}} = -\frac{Z_f}{R_{in}} = -\frac{R_f}{R_{in}\left(1 + j\omega R_f C\right)}.$$

The magnitude of the gain is

$$\left|\frac{V_{out}}{V_{in}}\right| = \frac{R_f/R_{in}}{\sqrt{1 + \left(\omega R_f C\right)^2}}.$$

For a gain of 0 dB,

$$\left|\frac{V_{out}}{V_{in}}\right| = 1.$$

Substituting $R_f = 100\,\text{k}\Omega$, $R_{in} = 10\,\text{k}\Omega$, and $C = 1\,\text{nF}$,

$$\frac{10}{\sqrt{1 + \left(\omega \times 100 \times 10^3 \times 10^{-9}\right)^2}} = 1.$$

Therefore,

$$1 + \left(\omega \times 10^{-4}\right)^2 = 100,$$

$$\omega = \frac{\sqrt{99}}{10^{-4}} = 9.9499 \times 10^4\ \text{rad/s}.$$

Hence,

$$f = \frac{\omega}{2\pi} = \frac{9.9499 \times 10^4}{2\pi} \approx 15.84\,\text{kHz}.$$

$$\boxed{15.84}$$

Q3.22.3 NAT 2022 2M Ans: 0.367

The comparator switches LOW when the non-inverting input voltage becomes equal to the reference voltage at the inverting input.

The reference voltage is obtained using the divider:

$$V_{ref} = 5\left(\frac{400}{600 + 400}\right) = 2\,\text{V}.$$

Initially, the switch is closed. Hence, the capacitor is charged to

$$V_0 = 5\left(\frac{5}{1 + 5}\right) = 4.167\,\text{V}.$$

At $t = 0$, the switch is opened and the capacitor discharges through the $5\,\text{k}\Omega$ resistor.

The time constant is

$$\tau = RC = (5\,\text{k}\Omega)(0.1\,\mu\text{F}) = 0.5\,\text{ms}.$$

The capacitor voltage is

$$V(t) = 4.167\,e^{-t/\tau}.$$

At the switching instant,

$$V(t) = V_{ref} = 2\,\text{V}.$$

Hence,

$$2 = 4.167\,e^{-t/0.5},$$

$$t = 0.5\ln\left(\frac{4.167}{2}\right) = 0.5\ln(2.0833) \approx 0.367\,\text{ms}.$$

$$\boxed{0.367}$$

Key Points

- The comparator changes state when $V_+ = V_-$.
- After the switch opens, the capacitor discharges as $V(t) = V_0 e^{-t/RC}$.

Q3.21.1 NAT | 2021 | 1M | Ans: 1 | ● ○ ○

At the positive peak of the input,

$$V_s = 0.4\,\text{V}.$$

Since the op-amp is ideal and the diode is forward biased, negative feedback is active. Hence,

$$V_- = V_+ = 0.4\,\text{V}.$$

The diode has a fixed forward drop of 0.6 V. Therefore,

$$V_{oA} = V_- + 0.6 = 0.4 + 0.6 = 1.0\,\text{V}.$$

1

Q3.21.2 MCQ | 2021 | 2M | Ans: C | ● ○ ○

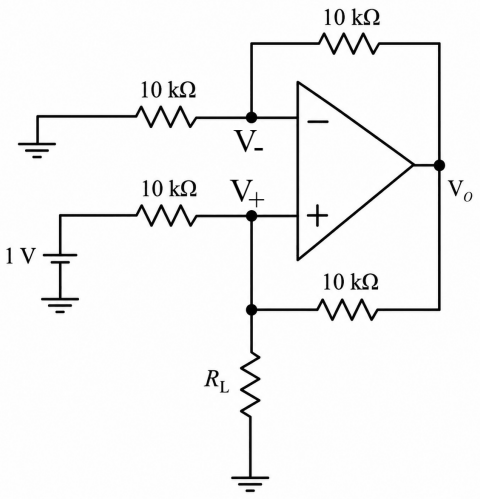

Fig. Solution Q3.21.2

At the inverting terminal, the two 10 kΩ resistors form a voltage divider between V_o and ground.

$$V_- = \frac{10}{10+10} \times 5 = 2.5\,\text{V}.$$

Since the op-amp is ideal and operates with negative feedback,

$$V_+ = V_- = 2.5\,\text{V}.$$

Applying KCL at the non-inverting terminal,

$$\frac{2.5-1}{10\,\text{k}\Omega} + \frac{2.5-5}{10\,\text{k}\Omega} + \frac{2.5}{R_L} = 0.$$

Simplifying,

$$\frac{1.5 - 2.5}{10\,\text{k}\Omega} + \frac{2.5}{R_L} = 0,$$

$$-\frac{1}{10\,\text{k}\Omega} + \frac{2.5}{R_L} = 0.$$

Hence,

$$R_L = 2.5 \times 10\,\text{k}\Omega = 25\,\text{k}\Omega.$$

(c) 25

Q3.21.3 NAT 2021 2M Ans: 1.1 ●○○

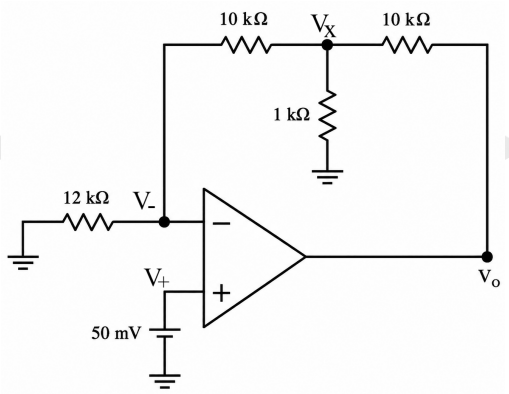

Fig. Solution Q3.21.3

Since the op-amp is ideal and operates with negative feedback,

$$V_- = V_+ = 50\,\text{mV} = 0.05\,\text{V}.$$

Applying KCL at the inverting input node,

$$\frac{V_-}{12\,\text{k}\Omega} + \frac{V_- - V_x}{10\,\text{k}\Omega} = 0,$$

Substituting $V_- = 0.05\,\text{V}$,

$$\frac{0.05}{12\,\text{k}\Omega} + \frac{0.05 - V_x}{10\,\text{k}\Omega} = 0,$$

$$V_x = \frac{11}{6}(0.05) = 0.0917\,\text{V}.$$

Now apply KCL at node V_x:

$$\frac{V_x - V_-}{10\,\text{k}\Omega} + \frac{V_x - V_o}{10\,\text{k}\Omega} + \frac{V_x}{1\,\text{k}\Omega} = 0.$$

Hence,

$$V_o = 12V_x - V_- = 12(0.0917) - 0.05 = 1.05\,\text{V}.$$

Rounded to one decimal place,

$$V_o = 1.1\,\text{V}.$$

1.1

Q3.20.1 NAT | 2020 | 1M | Ans: 7 | ● ○ ○

Since both op-amps are ideal and operate with negative feedback,

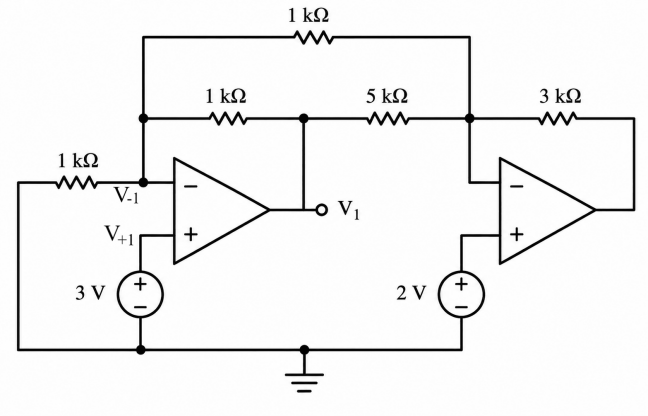

Fig. Solution Q3.20.1

$$V_{-1} = V_{+1} = 3\,\text{V}, \qquad V_{-2} = V_{+2} = 2\,\text{V}.$$

Let the output of the first op-amp be V_1.
Applying KCL at the inverting input node of the first op-amp (3 V),

$$\frac{3-0}{1\,\text{k}\Omega} + \frac{3-V_1}{1\,\text{k}\Omega} + \frac{3-2}{1\,\text{k}\Omega} = 0.$$

Simplifying,

$$3 + (3 - V_1) + 1 = 0,$$

$$V_1 = 7\,\text{V}.$$

$\boxed{7}$

Q3.20.2 NAT | 2020 | 1M | Ans: 4 | ● ○ ○

Since the op-amp is ideal and operates with negative feedback,

$$V_+ = V_- = V_x.$$

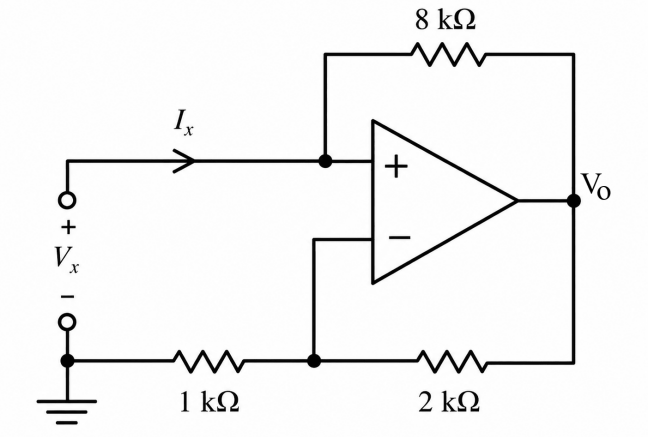

Fig. Solution Q3.20.2

The inverting input is connected to the output through a voltage divider of 2 kΩ and 1 kΩ. Hence,

$$V_- = \frac{1}{1+2}V_o = \frac{V_o}{3}.$$

Therefore,

$$V_o = 3V_x.$$

The op-amp input draws zero current. Hence, the input current I_x flows only through the 8 kΩ resistor:

$$I_x = \frac{V_x - V_o}{8\,\text{k}\Omega} = \frac{V_x - 3V_x}{8\,\text{k}\Omega} = -\frac{V_x}{4\,\text{k}\Omega}.$$

Thus,

$$\frac{V_x}{I_x} = -4\,\text{k}\Omega.$$

$\boxed{-4}$

Q3.20.3 NAT 2020 2M Ans: 1 ●○○

The given circuit is a precision half-wave rectifier using an ideal op-amp.
The inverting amplifier gain is

$$A_v = -\frac{R_f}{R_{in}} = -\frac{10\,\text{k}\Omega}{20\,\text{k}\Omega} = -\frac{1}{2}.$$

The input is

$$v_{in} = 2\pi \sin(5000\pi t)\ \text{V},$$

whose peak value is

$$V_m = 2\pi\ \text{V}.$$

Hence, the peak output of the rectifier is

$$V_{o,m} = \frac{1}{2}(2\pi) = \pi\ \text{V}.$$

A PMMC voltmeter with 1 Hz bandwidth measures the average (DC) value of the rectified waveform. For a half-wave rectified sine wave,

$$V_{\text{avg}} = \frac{V_{o,m}}{\pi} = \frac{\pi}{\pi} = 1\ \text{V}.$$

$\boxed{1}$

Key Points

- The op-amp and diodes form a precision half-wave rectifier.
- A PMMC instrument measures the average value of the rectified output.
- For a half-wave rectified sine wave, $V_{avg} = \dfrac{V_m}{\pi}$.

Mistakes to be avoided

- Do not use the RMS value; a PMMC voltmeter reads the average (DC) value.

Q3.20.4 NAT 2020 2M Ans: 0.5

At steady state, the capacitor behaves as an open circuit. Hence, no current flows through the 100 nF capacitor and the input current to the second op-amp is zero.

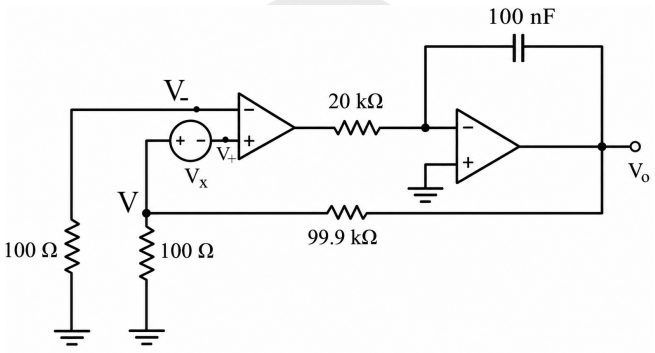

Fig. Solution Q3.20.4

Therefore, the output of the first op-amp must be zero:

$$V_{o1} = 0.$$

For the first op-amp,

$$V_- = V_+.$$

Since the polarity of the source is as shown,

$$V_+ = V - V_x.$$

Also,

$$V_- = 0.$$

Hence,

$$V - V_x = 0 \quad \Rightarrow \quad V = V_x.$$

Given

$$V_x = 0.5\,\text{mV},$$

$$V = 0.5\,\text{mV}.$$

$$\boxed{0.5}$$

Q3.20.5 NAT | 2020 | 2M | Ans: 3.5 | ● ○ ○

The op-amp works as a charge amplifier with the non-inverting terminal biased at

$$V_{ref} = \frac{V_{CC}}{2} = 2.5\,\text{V}.$$

The feedback capacitor is

$$C_f = 100\,\text{nF}.$$

The piezoelectric sensor generates

$$q_p = 0.1\sin(10000\pi t)\,\mu\text{C},$$

whose peak charge is

$$Q_{peak} = 0.1\,\mu\text{C} = 100\,\text{nC}.$$

For a charge amplifier,

$$\Delta V_o = \frac{Q_{peak}}{C_f} = \frac{100\,\text{nC}}{100\,\text{nF}} = 1\,\text{V}.$$

Hence, the output swings about the bias voltage as

$$V_o = 2.5 \pm 1\,\text{V}.$$

Therefore, the maximum (peak) output is

$$V_{o,\text{peak}} = 2.5 + 1 = 3.5\,\text{V}.$$

Since the peak detector stores the maximum value, after 10 cycles it reads

$$3.5\,\text{V}.$$

$$\boxed{3.5}$$

Key Points

- A charge amplifier converts input charge into voltage using $V = \frac{Q}{C_f}$.
- The sensor capacitances (10 pF and 100 pF) do not affect the ideal charge-to-voltage conversion.
- The peak detector stores the maximum output voltage.

Q3.19.1 NAT | 2019 | 2M | Ans: 9 | ● ○ ○

Given that the current through the left 3 kΩ resistor is zero,

$$I = 0.$$

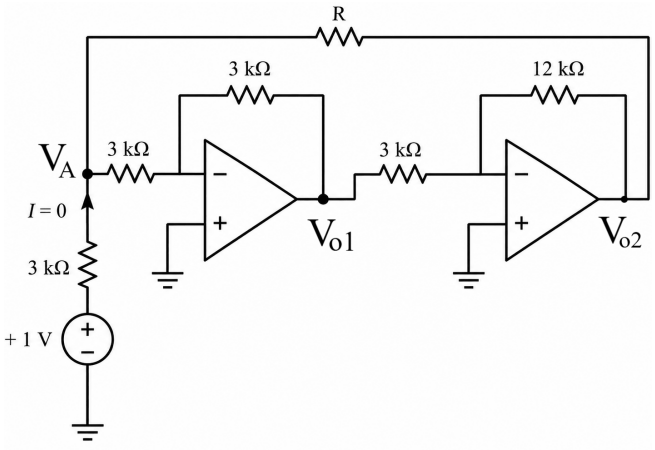

Fig. Solution Q3.19.1

Hence, there is no voltage drop across this resistor and the left node voltage is

$$V_A = 1\,\text{V}.$$

The first op-amp is an inverting amplifier with

$$R_{in} = R_f = 3\,\text{k}\Omega, \qquad V_+ = 0.$$

Therefore,

$$V_{o1} = -\frac{R_f}{R_{in}} V_A = -1\,\text{V}.$$

The second op-amp is also an inverting amplifier with

$$R_{in} = 3\,\text{k}\Omega, \qquad R_f = 12\,\text{k}\Omega.$$

Hence,

$$V_{o2} = -\frac{12}{3} V_{o1} = -4(-1) = 4\,\text{V}.$$

Applying KCL at node A,

$$\frac{1-0}{3\,\text{k}\Omega} + \frac{1-4}{R} = 0.$$

Thus,

$$\frac{1}{3000} = \frac{3}{R},$$

$$R = 9000\,\Omega = 9\,\text{k}\Omega.$$

$$\boxed{9}$$

Q3.19.2 MCQ 2019 1M Ans: C ●○○

The non-inverting terminal is grounded. Hence, for an ideal op-amp,

$$V_- = 0,$$

i.e., the inverting terminal is at virtual ground.

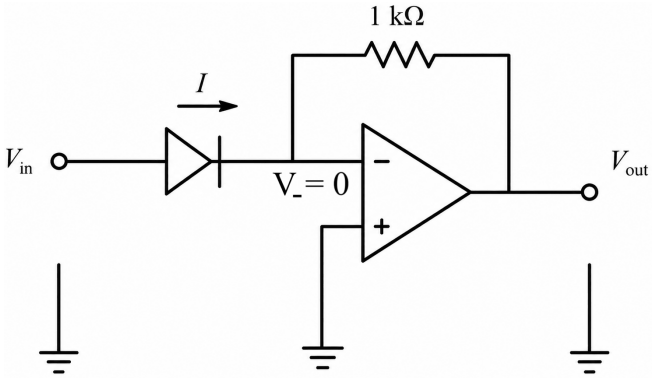

Fig. Solution Q3.19.2

Therefore, the diode voltage is

$$V_D = V_{in}.$$

The diode current is

$$I = I_0 e^{V_D/V_T} = I_0 e^{V_{in}/V_T}.$$

Since the op-amp input current is zero, the same current flows through the feedback resistor:

$$I = \frac{0 - V_{out}}{1\,\text{k}\Omega}.$$

Hence,

$$V_{out} = -1000\, I_0 e^{V_{in}/V_T}.$$

Thus,

$$V_{out} \propto e^{V_{in}/V_T}.$$

$$\boxed{\text{(c) } e^{V_{in}/V_T}}$$

Q3.18.1 NAT 2018 2M Ans: 0.99

Both op-amps are ideal and operate with negative feedback.

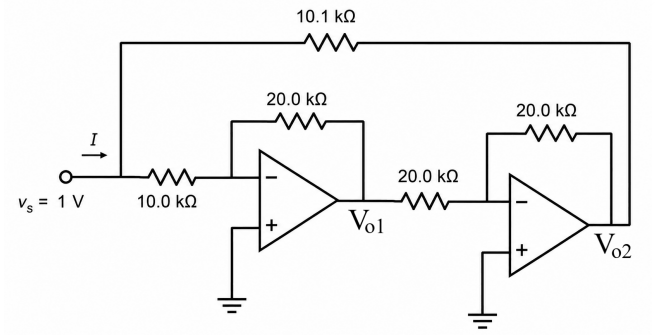

Fig. Solution Q3.18.1

The first op-amp is an inverting amplifier with

$$R_{in} = 10\,\text{k}\Omega, \qquad R_f = 20\,\text{k}\Omega.$$

Hence,

$$V_{o1} = -\frac{20}{10} \times 1 = -2\,\text{V}.$$

The second op-amp is also an inverting amplifier with

$$R_{in} = 20\,\text{k}\Omega, \qquad R_f = 20\,\text{k}\Omega.$$

Therefore,

$$V_{o2} = -\frac{20}{20}(-2) = 2\,\text{V}.$$

The source current is the algebraic sum of currents through the two branches:

$$I = \frac{1-0}{10\,\text{k}\Omega} + \frac{1-2}{10.1\,\text{k}\Omega}.$$

Thus,

$$I = 100\,\mu\text{A} - 99.01\,\mu\text{A} = 0.99\,\mu\text{A}.$$

$$\boxed{0.99}$$

Q3.18.2 NAT 2018 2M Ans: 17.7 ●○○

The non-inverting amplifier has a closed-loop gain of

$$A_v = 15.$$

For an input amplitude of

$$V_{in} = 0.3\,\text{V},$$

the output amplitude is

$$V_o = 15 \times 0.3 = 4.5\,\text{V}.$$

Since the supply rails are ±5 V, the output swing of 4.5 V is within the linear range.
To avoid slew-rate distortion,

$$\text{SR} \geq \left(\frac{dV_o}{dt}\right)_{\max} = 2\pi f V_o.$$

Given

$$\text{SR} = 0.5 \times 10^6\,\text{V/s},$$

we get

$$0.5 \times 10^6 = 2\pi f (4.5).$$

Hence,

$$f = \frac{0.5 \times 10^6}{2\pi \times 4.5} \approx 1.768 \times 10^4\,\text{Hz} = 17.7\,\text{kHz}.$$

$$\boxed{17.7}$$

Q3.18.3 NAT 2018 2M Ans: 200 ●○○

The right op-amp acts as an ideal Schmitt trigger. Since the two resistors connected to its non-inverting terminal are equal,

$$V_+ = \frac{V_o + V_Z}{2}$$

where the zener network limits

$$V_Z = \pm 5\,\text{V}.$$

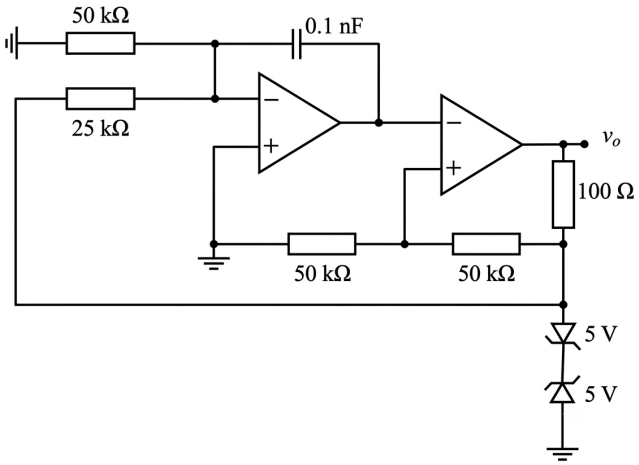

Fig. Solution Q3.18.3

Hence, the switching thresholds are

$$V_{T\pm} = \pm 2.5\ \text{V}.$$

The first op-amp is an inverting integrator with

$$R = 25\,\text{k}\Omega, \qquad C = 0.1\,\text{nF}.$$

The square-wave output of the Schmitt trigger causes the integrator output to ramp linearly between

$$-2.5\ \text{V and } +2.5\ \text{V},$$

i.e., a voltage change of

$$\Delta V = 5\ \text{V}.$$

The integrator slope is

$$\left|\frac{dV}{dt}\right| = \frac{5}{RC}.$$

Therefore, the time for one ramp is

$$\frac{T}{2} = \frac{\Delta V}{dV/dt} = \frac{5}{5/(RC)} = RC.$$

Thus,

$$T = 2RC = 2(25 \times 10^3)(0.1 \times 10^{-9}) = 5 \times 10^{-6}\ \text{s}.$$

Hence,

$$f = \frac{1}{T} = \frac{1}{5 \times 10^{-6}} = 2 \times 10^5\ \text{Hz} = 200\,\text{kHz}.$$

$$\boxed{200}$$

Q3.17.1 NAT 2017 2M Ans: -5 ●○○

The non-inverting terminal of the op-amp is grounded, hence the inverting terminal is at virtual ground.
The multiplier has a scaling factor of 1, therefore its output is

$$V_m = V_2 V_o.$$

Applying KCL at the inverting node,

$$\frac{V_1 - 0}{R} + \frac{V_m - 0}{R} = 0.$$

Substituting $V_m = V_2 V_o$,

$$V_1 + V_2 V_o = 0.$$

Hence,

$$V_o = -\frac{V_1}{V_2} = -\frac{15}{3} = -5\,\text{V}.$$

$$\boxed{-5}$$

Q3.17.2 NAT 2017 1M Ans: 1.5 ●○○

The output of a practical differential amplifier is given by

$$V_o = A_d V_d + A_c V_{cm},$$

where

$$V_d = V_1 - V_2, \qquad V_{cm} = \frac{V_1 + V_2}{2}.$$

Substituting the given values,

$$V_d = 5.01 - 5.00 = 0.01\,\text{V},$$

$$V_{cm} = \frac{5.01 + 5.00}{2} = 5.005\,\text{V}.$$

Hence,

$$V_o = 100(0.01) + 0.1(5.005) = 1 + 0.5005 = 1.5005\,\text{V}.$$

Up to one decimal place,

$$V_o = 1.5\,\text{V}.$$

$$\boxed{1.5}$$

Q3.17.3 MCQ 2017 1M Ans: B ●○○

Since the non-inverting terminal is grounded,

$$V_+ = 0 \Rightarrow V_- \approx 0$$

(virtual ground).
For the inverting amplifier,

$$V_o = -\frac{R_f}{R_{in}} V_{in} = -\frac{2\,\text{k}\Omega}{1\,\text{k}\Omega} \times 10\,\text{V} = -20\,\text{V}.$$

However, the op-amp is powered by

$$\pm 15\,\text{V},$$

so the output cannot go below -15 V. Hence the output saturates at the negative supply rail.

$$\boxed{\text{(b)} \ -15\,\text{V}}$$

Mistakes to be avoided

- Do not mark -20 V as the answer without checking the supply limits.

Q3.17.4 NAT | 2017 | 2M | Ans: 0.67 | ● ○ ○

The Zener-diode network limits the output voltage to

$$V_o = \pm 7\,\text{V}.$$

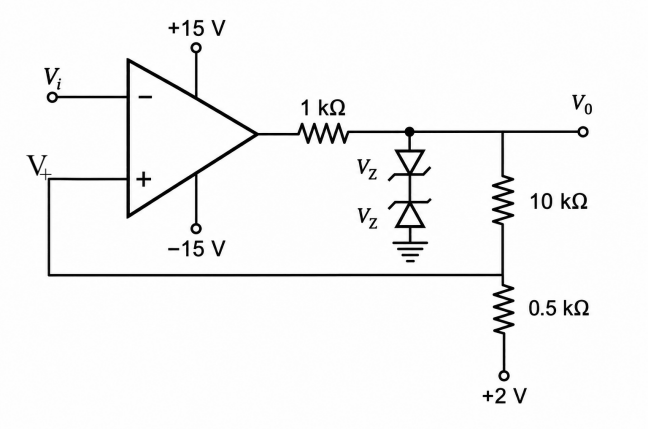

Fig. Solution Q3.17.4

Let the voltage at the non-inverting terminal be V_+. Applying KCL at this node,

$$\frac{V_+ - V_o}{10\,\text{k}\Omega} + \frac{V_+ - 2}{0.5\,\text{k}\Omega} = 0.$$

Hence,

$$V_+ = \frac{\dfrac{V_o}{10\,\text{k}\Omega} + \dfrac{2}{0.5\,\text{k}\Omega}}{\dfrac{1}{10\,\text{k}\Omega} + \dfrac{1}{0.5\,\text{k}\Omega}} = \frac{V_o + 40}{21}.$$

For the two output states,

$$V_{UTP} = \frac{7 + 40}{21} = \frac{47}{21}\,\text{V},$$

$$V_{LTP} = \frac{-7 + 40}{21} = \frac{33}{21}\,\text{V}.$$

Therefore, the hysteresis width is

$$V_H = V_{UTP} - V_{LTP} = \frac{47 - 33}{21} = \frac{14}{21} = \frac{2}{3}\,\text{V} \approx 0.67\,\text{V}.$$

$$\boxed{0.67}$$

Key Points

- The Zener diodes clamp the output to ±7 V.
- Hysteresis width is $V_{UTP} - V_{LTP}$.

Q3.17.5 NAT 2017 2M Ans: 2.96 ● ○ ○

The op-amp is configured as a non-inverting amplifier with

$$A_v = 1 + \frac{R_f}{R_1} = 1 + \frac{1\,\text{k}\Omega}{1\,\text{k}\Omega} = 2.$$

The input to the non-inverting terminal is obtained through an RC low-pass filter:

$$\frac{V_+}{V_{in}} = \frac{1}{\sqrt{1 + (\omega RC)^2}},$$

where

$$R = 10\,\text{k}\Omega, \qquad C = 10\,\text{nF}.$$

Hence,

$$\left|\frac{V_o}{V_{in}}\right| = \frac{2}{\sqrt{1 + (\omega RC)^2}}.$$

Given,

$$|V_o| = 0.95\,|V_{in}|,$$

therefore,

$$\frac{2}{\sqrt{1 + (\omega RC)^2}} = 0.95.$$

Solving,

$$1 + (\omega RC)^2 = \left(\frac{2}{0.95}\right)^2,$$

$$\omega RC = \sqrt{\left(\frac{2}{0.95}\right)^2 - 1} \approx 1.857.$$

Since

$$RC = (10\,\text{k}\Omega)(10\,\text{nF}) = 10^{-4}\,\text{s},$$

$$\omega = \frac{1.857}{10^{-4}} = 1.857 \times 10^4\,\text{rad/s}.$$

Therefore,

$$f = \frac{\omega}{2\pi} = \frac{1.857 \times 10^4}{2\pi} \approx 2.956 \times 10^3\,\text{Hz} \approx 2.96\,\text{kHz}.$$

$$\boxed{2.96}$$

Q3.17.6 MCQ 2017 1M Ans: A ● ○ ○

For sustained oscillations, the **Barkhausen criterion** must be satisfied.
Initially, to start oscillations,

$$|A\beta| > 1.$$

At the frequency of oscillation, the total phase shift around the loop must be 0° (or an integer multiple of 360°).
Hence, the correct answer is

$$\boxed{\text{(a) } 0^\circ}$$

Q3.16.1 NAT | 2016 | 2M | Ans: 100 | ● ○ ○

For the ideal op-amp with negative feedback,

$$V_+ = V_-.$$

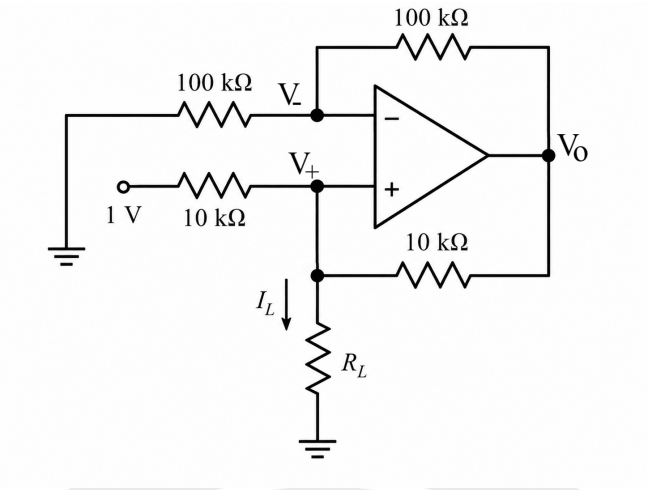

Fig. Solution Q3.16.1

The inverting terminal is connected to a voltage divider consisting of two equal resistors of 100 kΩ. Hence,

$$V_- = \frac{V_o}{2}.$$

Therefore,

$$V_+ = \frac{V_o}{2}.$$

Applying KCL at the non-inverting terminal,

$$\frac{V_+ - 1}{10\,\text{k}\Omega} + \frac{V_+ - V_o}{10\,\text{k}\Omega} + I_L = 0.$$

Substituting $V_+ = V_o/2$,

$$\frac{\frac{V_o}{2} - 1}{10\,\text{k}\Omega} + \frac{\frac{V_o}{2} - V_o}{10\,\text{k}\Omega} + I_L = 0.$$

The V_o terms cancel, giving

$$-\frac{1}{10\,\text{k}\Omega} + I_L = 0.$$

Hence,

$$I_L = \frac{1}{10\,\text{k}\Omega} = 100\,\mu\text{A}.$$

100

Q3.16.2 NAT | 2016 | 2M | Ans: 100 | ●○○

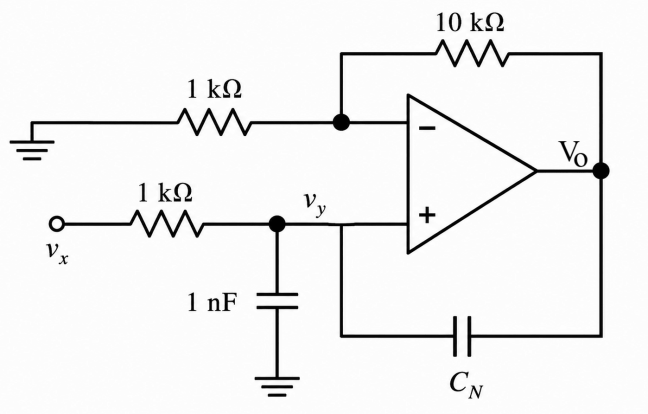

Fig. Solution Q3.16.2

For the ideal op-amp operating with negative feedback,

$$V_+ = V_- = V_y.$$

The inverting side is a non-inverting amplifier with

$$A_v = 1 + \frac{10\,\text{k}\Omega}{1\,\text{k}\Omega} = 11.$$

Hence,

$$V_o = 11V_y.$$

Applying KCL at node V_y,

$$\frac{V_y - V_x}{1\,\text{k}\Omega} + s(1\,\text{nF})V_y + sC_N(V_y - V_o) = 0.$$

Substituting $V_o = 11V_y$,

$$\frac{V_y - V_x}{1\,\text{k}\Omega} + s(1\,\text{nF})V_y - 10sC_NV_y = 0.$$

Therefore,

$$\frac{V_y}{V_x} = \frac{1}{1 + s(1\,\text{k}\Omega)\,(1\,\text{nF} - 10C_N)}.$$

To ensure

$$V_y = V_x$$

for a sinusoidal input of any frequency, the frequency-dependent term must vanish:

$$1\,\text{nF} - 10C_N = 0.$$

Thus,

$$C_N = \frac{1\,\text{nF}}{10} = 0.1\,\text{nF} = 100\,\text{pF}.$$

$$\boxed{100}$$

Q3.16.3 NAT | 2016 | 2M | Ans: 40.83 | ● ○ ○

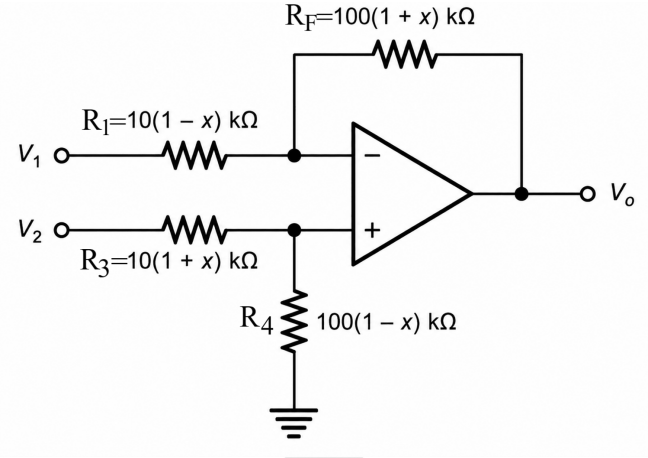

Fig. Solution Q3.16.3

Let

$$R_1 = 10(1-x)\,\text{k}\Omega, \quad R_2 = 10(1+x)\,\text{k}\Omega,$$
$$R_3 = 100(1-x)\,\text{k}\Omega, \quad R_F = 100(1+x)\,\text{k}\Omega.$$

Here,

$$R_F = 10R_2, \qquad R_3 = 10R_1.$$

For a practical difference amplifier with resistor mismatch,

$$\text{CMRR} = -\frac{1}{2}\left[\frac{(R_1+R_F)R_3 + R_F(R_2+R_3)}{(R_1+R_F)R_3 - R_F(R_2+R_3)}\right].$$

Substituting $R_F = 10R_2$ and $R_3 = 10R_1$,

$$\text{CMRR} = -\frac{1}{2}\left[\frac{R_1^2 + R_2^2 + 20R_1R_2}{R_1^2 - R_2^2}\right].$$

Since

$$R_1 = 10(1-x), \qquad R_2 = 10(1+x),$$

$$\text{CMRR} = -\frac{1}{2}\left[\frac{(1+x)^2 + (1-x)^2 + 20(1-x^2)}{(1-x)^2 - (1+x)^2}\right].$$

For $x = 0.025$,

$$\text{CMRR} = 110.$$

Hence,

$$\text{CMRR}_{\text{dB}} = 20\log_{10}(110) = 40.83\,\text{dB}.$$

$$\boxed{40.83}$$

Q3.16.4 NAT 2016 1M Ans: 20 ● ○ ○

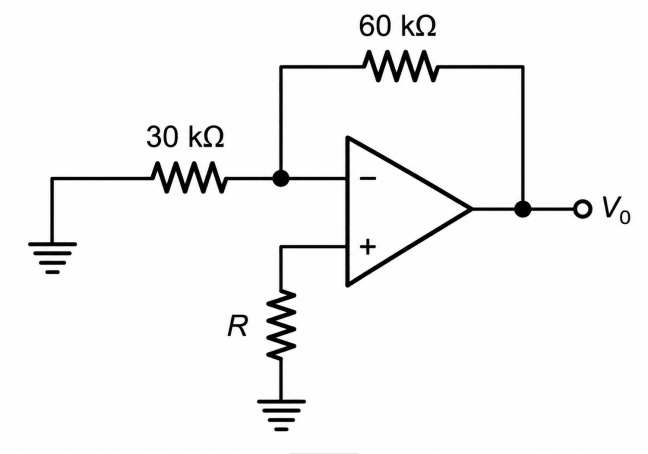

Fig. Solution Q3.16.4

To make the output independent of the input bias currents, the resistance seen by both input terminals must be equal. The resistance seen by the inverting terminal is

$$R_{eq} = 30\,\text{k}\Omega \parallel 60\,\text{k}\Omega.$$

Hence,

$$R = \frac{30 \times 60}{30 + 60}\,\text{k}\Omega = 20\,\text{k}\Omega.$$

With

$$R = R_{eq},$$

both input bias currents produce equal voltage drops at the two input terminals. Since the op-amp responds only to the differential input voltage, the output offset due to the bias currents becomes zero.

20

Q3.16.5 NAT 2016 1M Ans: 1.6 ● ○ ○

For the positive half-cycle ($V_{in} > 0.6\,\text{V}$), diode D_1 is ON and diode D_2 is OFF. The circuit behaves as an inverting amplifier.

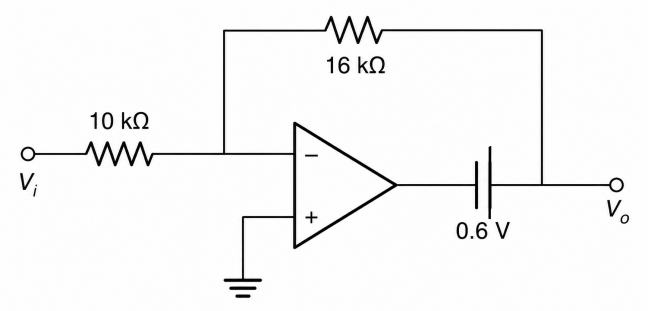

Fig. Solution Q3.16.5

$$V_o = -\frac{R_f}{R_{in}} V_{in} = -\frac{16\,\text{k}\Omega}{10\,\text{k}\Omega} V_{in} = -1.6 V_{in}.$$

Since

$$V_{in} = 1 \sin(3000t)\ \text{V},$$

the negative peak output is

$$V_{o,\text{peak}} = -1.6 \times 1 = -1.6\,\text{V}.$$

For the negative half-cycle ($V_{in} < -0.6\,\text{V}$), diode D_1 is OFF and diode D_2 is ON. The output is disconnected from the op-amp, hence

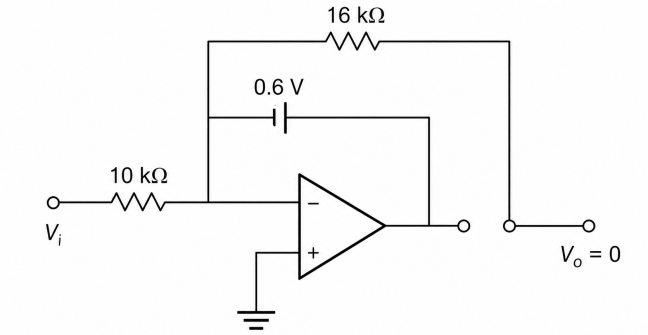

Fig. Solution Q3.16.5

$$V_o = 0.$$

Therefore, the magnitude of the negative peak output is

$$|V_{o,\text{peak}}| = 1.6\,\text{V}.$$

Key Points

- The circuit is a precision half-wave rectifier.
- During the negative half-cycle, D_1 conducts while D_2 is OFF.

Q3.16.6 NAT 2016 1M Ans: -1

For an ideal op-amp with negative feedback,

$$V_+ = V_- = 0\,\text{V}.$$

Let the voltage at the divider node be V_1.

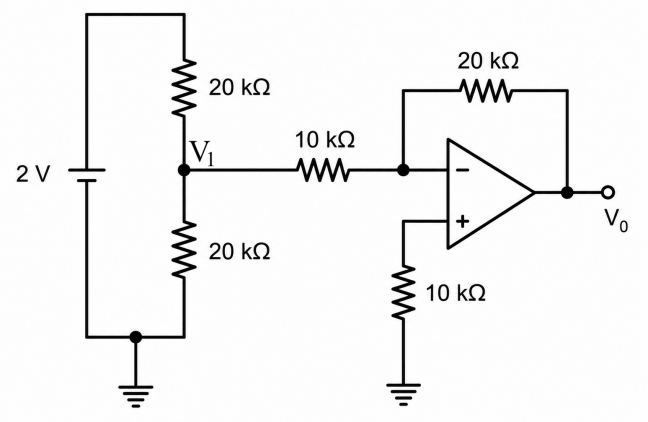

Fig. Solution Q3.16.6

Applying KCL at node V_1,

$$\frac{V_1 - 2}{20\,\text{k}\Omega} + \frac{V_1}{20\,\text{k}\Omega} + \frac{V_1 - 0}{10\,\text{k}\Omega} = 0.$$

Multiplying throughout by 20 kΩ,

$$(V_1 - 2) + V_1 + 2V_1 = 0$$

$$4V_1 = 2$$

$$V_1 = 0.5\,\text{V}.$$

Applying KCL at the inverting terminal,

$$\frac{V_1 - 0}{10\,\text{k}\Omega} = \frac{0 - V_o}{20\,\text{k}\Omega}.$$

Hence,

$$V_o = -\frac{20\,\text{k}\Omega}{10\,\text{k}\Omega}V_1 = -2(0.5) = -1\,\text{V}.$$

$$\boxed{-1}$$

Q3.15.1 NAT 2015 2M Ans: 1.39 ● ○ ○

At 100 degrees Celsius,

$$R_T = 100(1 + 0.0039 \times 100) = 139\ \Omega.$$

The ideal op-amp holds its inverting input at 1 V. The current through the lower 100 Ohm resistor is therefore 10 mA. Since no current enters the op-amp input, the same current flows through R_T.

The figure labels V_o **across the RTD**, with its positive terminal at the amplifier output and its negative terminal at the inverting node. Thus

$$V_o = (10\ \text{mA})(139\ \Omega) = \boxed{1.39\ \text{V}}.$$

The amplifier's output node relative to ground is 1 + 1.39 = 2.39 V. That is a different voltage from the marked differential V_o.

Q3.15.2 MCQ 2015 2M Ans: D ●○○

The feedback network is a non-inverting amplifier with

$$\beta = \frac{1\,\text{k}\Omega}{9\,\text{k}\Omega + 1\,\text{k}\Omega} = 0.1.$$

The open-loop gain of the op-amp is

$$A_v(s) = \frac{10^4}{1 + 10^{-3}s}.$$

The closed-loop transfer function is

$$\frac{V_o}{V_{in}} = \frac{A_v(s)}{1 + \beta A_v(s)} = \frac{\dfrac{10^4}{1 + 10^{-3}s}}{1 + \dfrac{10^3}{1 + 10^{-3}s}}.$$

Simplifying,

$$\frac{V_o}{V_{in}} = \frac{10^4}{1001 + 10^{-3}s} = \frac{\dfrac{10^4}{1001}}{1 + \dfrac{10^{-3}}{1001}s}.$$

Comparing with the standard first-order form, $\frac{K}{1+\frac{s}{\omega_B}}$,

gives

$$\omega_B = \frac{1001}{10^{-3}} = 1001 \times 10^3 = 1.001 \times 10^6\ \text{rad/s}.$$

Therefore, the 3-dB bandwidth is

$$\boxed{\text{(d) } 10^6}$$

Key Points

- Feedback factor: $\beta = \dfrac{R_1}{R_1 + R_f}$.
- Closed-loop transfer function: $\dfrac{A}{1 + \beta A}$.
- Negative feedback increases the bandwidth by the factor $1 + \beta A_0$.

Q3.15.3 MCQ 2015 1M Ans: C ●○○

The given transfer function is

$$G(s) = \frac{s^2 - bs + c}{s^2 + bs + c}.$$

Substitute $s = j\omega$:

$$G(j\omega) = \frac{(c - \omega^2) - jb\omega}{(c - \omega^2) + jb\omega}.$$

Hence,

$$|G(j\omega)| = \frac{\sqrt{(c - \omega^2)^2 + (b\omega)^2}}{\sqrt{(c - \omega^2)^2 + (b\omega)^2}} = 1.$$

Since the magnitude remains unity for all frequencies and only the phase changes with frequency, the network is an **all-pass filter**.

(c) an all-pass filter

Q3.14.1 NAT 2014 2M Ans: 0.6

For an ideal op-amp operating in the linear region,

$$V_+ = V_- = 0$$

Hence, the inverting terminal is at virtual ground.

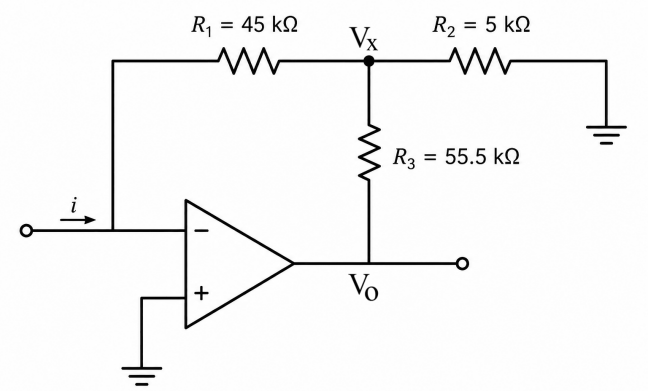

Fig. Solution Q3.14.1

Let the voltage at the junction of R_1, R_2 and R_3 be V_x.
Applying KCL at the inverting terminal,

$$i = \frac{0 - V_x}{R_1}$$

$$V_x = -iR_1$$

Now apply KCL at node V_x:

$$\frac{V_x}{R_1} + \frac{V_x}{R_2} + \frac{V_x - V_o}{R_3} = 0$$

Substituting

$$R_1 = 45\,\text{k}\Omega, \quad R_2 = 5\,\text{k}\Omega, \quad R_3 = 55.5\,\text{k}\Omega, \quad V_x = -45i$$

(all resistances in kΩ),

$$-i - 9i + \frac{-45i - V_o}{55.5} = 0$$

$$-555i - 45i - V_o = 0$$

$$V_o = -600i$$

Hence, the transfer impedance is

$$\left|\frac{V_o}{i}\right| = 600\,\text{k}\Omega = 0.6\,\text{M}\Omega$$

0.6

Q3.14.2 MCQ 2014 1M Ans: B ● ○ ○

For an ideal inverting amplifier,

$$A_v = \frac{V_o}{V_{in}} = -\frac{R_f}{R_i}$$

Given,

$$A_v = -10$$

Therefore,

$$\frac{R_f}{R_i} = 10$$

Also, the input resistance of an ideal inverting amplifier is

$$R_{\text{in}} = R_i$$

Given,

$$R_{\text{in}} = 50\,\text{k}\Omega$$

Hence,

$$R_i = 50\,\text{k}\Omega$$

Therefore,

$$R_f = 10R_i = 10 \times 50\,\text{k}\Omega = 500\,\text{k}\Omega$$

Thus, the correct option is

$$\boxed{\text{(b) } R_i = 50\,\text{k}\Omega,\ R_f = 500\,\text{k}\Omega}$$

Q3.14.3 MCQ 2014 1M Ans: D ● ○ ○

The input bias current is the average of the two input bias currents:

$$I_B = \frac{I_{b1} + I_{b2}}{2} = \frac{450 + 350}{2} = 400\,\text{nA}$$

The input offset current is the difference between the two input bias currents:

$$I_f = |I_{b1} - I_{b2}| = |450 - 350| = 100\,\text{nA}$$

Thus,

$$\boxed{\text{(d) } I_B = 400\,\text{nA},\ I_f = 100\,\text{nA}}$$

Key Points

- Input bias current:
$$I_B = \frac{I_{b1} + I_{b2}}{2}$$
- Input offset current:
$$I_f = |I_{b1} - I_{b2}|$$
- Ideally, both I_B and I_f are zero.

Q3.14.4 NAT | 2014 | 2M | Ans: 8.11 | ● ○ ○

The capacitor voltage is the input to the Schmitt trigger.
When $V_o = 5\,\text{V}$, the capacitor charges from 2 V to 3 V through

$$R = 10\,\text{k}\Omega, \qquad C = 1\,\mu\text{F}$$

with

$$RC = 10\,\text{ms}.$$

Charging equation:

$$v_C(t) = 5 - (5-2)e^{-t/RC}.$$

At the switching instant,

$$3 = 5 - 3e^{-t/RC}$$

$$e^{-t/RC} = \frac{2}{3}$$

$$t_1 = RC\ln\left(\frac{3}{2}\right).$$

Similarly, when $V_o = 0\,\text{V}$, the capacitor discharges from 3 V to 2 V:

$$v_C(t) = 3e^{-t/RC}.$$

At the next switching instant,

$$2 = 3e^{-t/RC}$$

$$t_2 = RC\ln\left(\frac{3}{2}\right).$$

Hence, the time period is

$$T = t_1 + t_2 = 2RC\ln\left(\frac{3}{2}\right) = 2(10)\ln\left(\frac{3}{2}\right)\text{ ms}$$

$$T \approx 8.11\,\text{ms}.$$

$$\boxed{8.11}$$

Q3.14.5 NAT | 2014 | 2M | Ans: 15.4 | ● ○ ○

The given circuit in question is a second-order Sallen–Key low-pass filter.

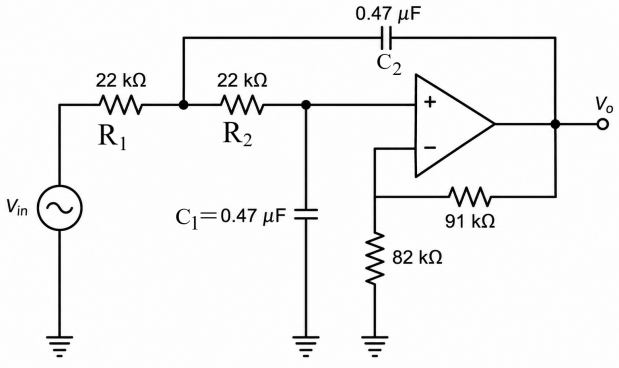

Fig. Solution Q3.14.5

The cutoff frequency is

$$f_c = \frac{1}{2\pi\sqrt{R_1R_2C_1C_2}}.$$

Given,

$$R_1 = R_2 = 22\,\text{k}\Omega, \qquad C_1 = C_2 = 0.47\,\mu\text{F}.$$

Since the components are equal,

$$f_c = \frac{1}{2\pi RC}.$$

Substituting the values,

$$f_c = \frac{1}{2\pi(22\times10^3)(0.47\times10^{-6})}$$

$$= \frac{1}{0.06497} \approx 15.4\,\text{Hz}.$$

(The resistors 91 kΩ and 82 kΩ set the amplifier gain and do not affect the cutoff frequency.)

$$\boxed{15.4}$$

Q3.13.1 MCQ 2013 2M Ans: C

For the first op-amp, the non-inverting terminal is connected to +1 V.

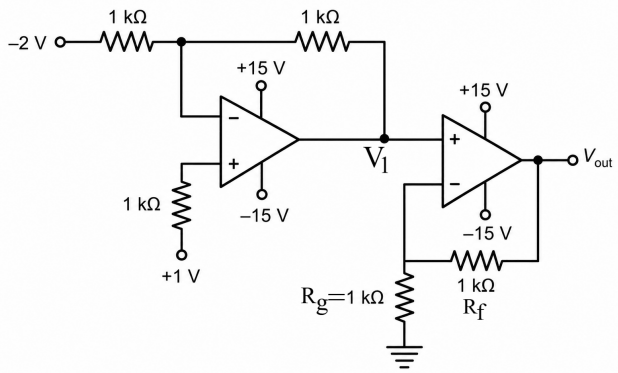

Fig. Solution Q3.13.1

Hence, by the virtual short concept,

$$V_- = V_+ = 1\,\text{V}.$$

Applying KCL at the inverting terminal,

$$\frac{1-(-2)}{1\,\text{k}\Omega} + \frac{1-V_1}{1\,\text{k}\Omega} = 0,$$

where V_1 is the output of the first op-amp.

$$3 + 1 - V_1 = 0$$

$$V_1 = 4\,\text{V}.$$

The second op-amp is a non-inverting amplifier with

$$R_f = 1\,\text{k}\Omega, \qquad R_g = 1\,\text{k}\Omega.$$

Hence, its gain is

$$A_v = 1 + \frac{R_f}{R_g} = 1 + \frac{1}{1} = 2.$$

Therefore,

$$V_{\text{out}} = 2 \times 4 = 8\,\text{V}.$$

(c) 8 V

Q3.13.2 MCQ 2013 1M Ans: B

The non-inverting terminal is grounded, hence the op-amp tries to maintain

$$V_- = V_+ = 0\,\text{V}.$$

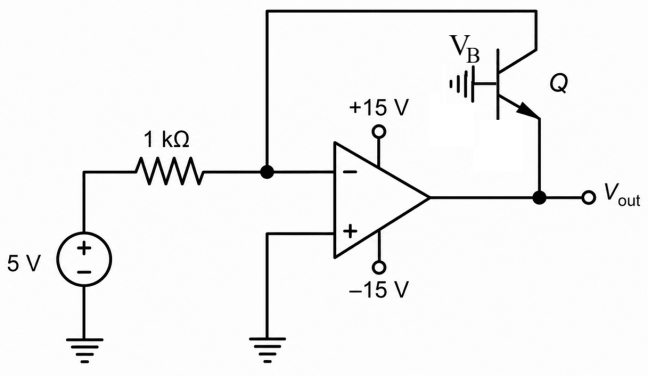

Fig. Solution Q3.13.2

The inverting terminal is connected to the collector of transistor Q. Since the base of the silicon transistor is grounded,

$$V_B = 0\,\text{V}.$$

For the transistor to conduct and provide negative feedback,

$$V_{BE} \approx 0.7\,\text{V}.$$

Therefore,

$$V_B - V_E = 0.7$$

$$0 - V_{\text{out}} = 0.7$$

$$V_{\text{out}} = -0.7\,\text{V}.$$

Thus, the transistor conducts and the op-amp maintains the inverting input at virtual ground.

(b) − 0.7 V

Q3.13.3 MCQ 2013 1M Ans: A

The circuit is an inverting amplifier.
Its closed-loop gain is

$$A_v = -\frac{R_f}{R_i} = -\frac{470\,\text{k}\Omega}{22\,\text{k}\Omega} = -21.36.$$

The input signal is

$$V_{in} = 0.25\sin(\omega t)\,\text{V}.$$

Hence, the output amplitude is

$$V_{o(\max)} = 21.36 \times 0.25 = 5.34\,\text{V}.$$

For a sinusoidal output,

$$\left(\frac{dV_o}{dt}\right)_{\max} = 2\pi f V_{o(\max)}.$$

To avoid slew-rate distortion,

$$2\pi f V_{o(\max)} \leq \text{SR}.$$

Given,

$$\text{SR} = 0.8\,\text{V}/\mu\text{s} = 0.8 \times 10^6\,\text{V/s}.$$

Therefore,

$$f_{\max} = \frac{0.8 \times 10^6}{2\pi(5.34)} \approx 23.84 \times 10^3\,\text{Hz}.$$

$$f_{\max} \approx 23.84\,\text{kHz}.$$

(a) 23.84 kHz

Q3.13.4 MCQ 2013 2M Ans: C

The circuit consists of two cascaded active high-pass filters.
For the first stage,

$$f_{c1} = \frac{1}{2\pi RC} = \frac{1}{2\pi(1\,\text{k}\Omega)(0.1\,\mu\text{F})} \approx 1591.5\,\text{Hz}.$$

For the second stage,

$$f_{c2} = \frac{1}{2\pi RC} = \frac{1}{2\pi(2\,\text{k}\Omega)(1\,\mu\text{F})} \approx 79.6\,\text{Hz}.$$

Hence, the overall filter passes frequencies greater than both cutoff frequencies, i.e.,

$$f > 1591.5\,\text{Hz}.$$

The input signal is

$$V_i(t) = 10 + 10\sin 100\pi t + 10\sin 4000\pi t + 10\sin 100000\pi t.$$

The corresponding frequency components are

$$10 \rightarrow 0\,\text{Hz},$$
$$10\sin 100\pi t \rightarrow 50\,\text{Hz},$$
$$10\sin 4000\pi t \rightarrow 2000\,\text{Hz},$$
$$10\sin 100000\pi t \rightarrow 50000\,\text{Hz}.$$

Since both 0 Hz and 50 Hz are heavily attenuated, while 2000 Hz and 50000 Hz pass through, the least attenuated frequency component is

(c) 2 KHz

Q3.12.1 MCQ 2012 2M Ans: B ● ○ ○

The circuit is an inverting active high-pass filter.
The input impedance is

$$Z_{in} = R_1 + \frac{1}{sC} = \frac{1 + sR_1C}{sC}.$$

Hence, the transfer function is

$$\frac{V_o}{V_{in}} = -\frac{R_2}{Z_{in}} = -\frac{R_2 sC}{1 + sR_1C}.$$

Comparing with the standard first-order high-pass filter,

$$H(s) = K\frac{s/\omega_c}{1 + s/\omega_c},$$

the cutoff angular frequency is

$$\omega_c = \frac{1}{R_1C} \text{ rad/s}.$$

Thus, the circuit is a high-pass filter with

$$\boxed{\text{(b) High-pass filter with } f_{3dB} = \frac{1}{R_1C} \text{ rad/s}}$$

Q3.11.1 MCQ 2011 2M Ans: C ● ○ ○

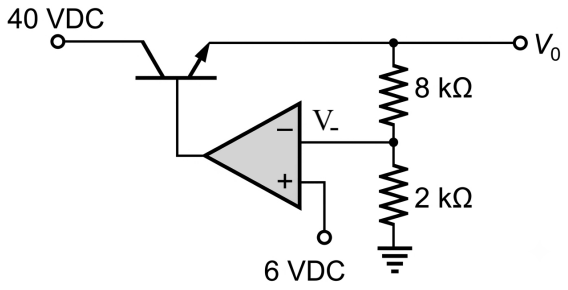

Fig. Solution Q3.11.1

The op-amp operates with negative feedback, hence

$$V_- = V_+ = 6\,\text{V}.$$

The inverting terminal is connected to the junction of the voltage divider formed by

$$8\,\text{k}\Omega \quad \text{and} \quad 2\,\text{k}\Omega.$$

Therefore,

$$V_- = V_o\left(\frac{2}{8+2}\right) = \frac{V_o}{5}.$$

Since

$$V_- = 6\,\text{V},$$

$$\frac{V_o}{5} = 6$$

$$V_o = 30\,\text{V}.$$

Thus, the output voltage is

$$\boxed{\text{(c) } 30\,\text{V}}$$

Q3.11.2 MCQ 2011 2M Ans: C ●○○

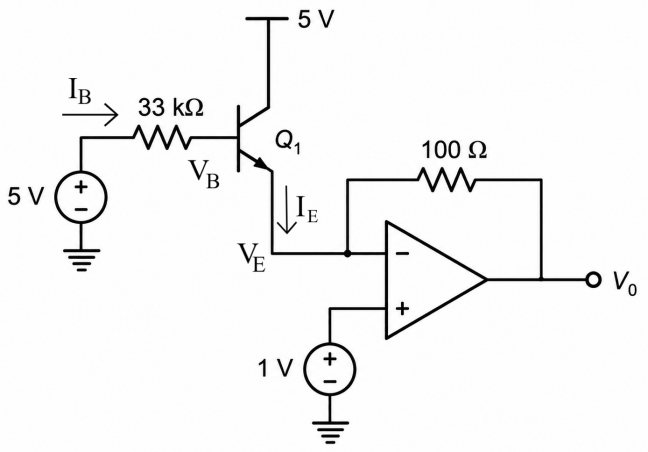

Fig. Solution Q3.11.2

The non-inverting terminal is connected to a 1 V source. Hence, for an ideal op-amp,

$$V_- = V_+ = 1\,\text{V}.$$

Therefore, the emitter voltage of transistor Q_1 is

$$V_E = 1\,\text{V}.$$

Since $V_{BE} = 0.7\,\text{V}$,

$$V_B = V_E + 0.7 = 1.7\,\text{V}.$$

The base current is

$$I_B = \frac{5 - 1.7}{33\,\text{k}\Omega} = \frac{3.3}{33 \times 10^3} = 100\,\mu\text{A}.$$

Given,

$$\beta = 99,$$

the emitter current is

$$I_E = (\beta + 1)I_B = 100 \times 100\,\mu\text{A} = 10\,\text{mA}.$$

The op-amp input draws no current, hence this current flows through the 100 Ω feedback resistor:

$$\frac{1 - V_o}{100} = 10\,\text{mA}.$$

Thus,

$$1 - V_o = 1$$

$$V_o = 0\,\text{V}.$$

$$\boxed{\text{(c) } 0\,\text{V}}$$

Q3.11.3 MCQ 2011 2M Ans: A ● ○ ○

The open/short approximation at exactly DC is singular here, so derive the nonzero-frequency response before taking limits. Let v_a be the junction of the two 1 MOhm resistors and the 1 microfarad capacitor, and v_b the junction of the two 0.5 microfarad capacitors and the 500 kOhm resistor. The ideal op-amp's inverting input is at virtual ground for the small-signal calculation.

Using the stated component values with s in inverse seconds, KCL at v_a gives

$$(s+2)v_a = v_i.$$

KCL at the inverting node gives $v_b = -2v_a/s$. At the feedback midpoint,

$$v_o = (2 + 4/s)v_b.$$

Combining,

$$\frac{v_o}{v_i} = -\frac{4}{s^2}.$$

For $s = j\omega$, the magnitude is $4/\omega^2$, decreasing with frequency. Thus the offered classification is **low-pass**, option (A), rather than band-pass.

Ideal-DC qualification: This is a double-integrator response with poles at the origin, not a practical stable low-pass filter with finite DC gain. At exactly DC there is no capacitive feedback and a bounded steady-state output cannot be assumed. Real saturation and leakage limit the ideal low-frequency response.

Q3.10.1 MCQ 2010 2M Ans: B ● ○ ○

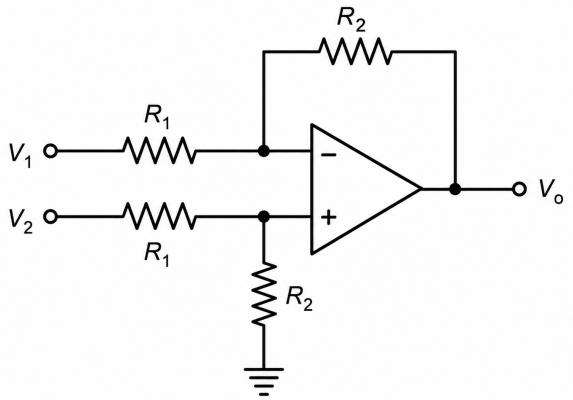

Fig. Solution Q3.10.1

Given,

$$R_1 = 47\,\text{k}\Omega, \qquad R_2 = 470\,\text{k}\Omega.$$

The sources have internal resistances:

$$R_{s1} = 13\,\text{k}\Omega, \qquad R_{s2} = 3\,\text{k}\Omega.$$

Step 1: Voltage at the non-inverting terminal

The source V_2 and the resistor network form a voltage divider.

$$V_+ = 1 \times \frac{R_2}{R_{s2} + R_1 + R_2} = \frac{470}{3 + 47 + 470} = \frac{470}{520} = 0.9038\,\text{V}.$$

Since the op-amp is ideal,

$$V_- = V_+ = 0.9038\,\text{V}.$$

Step 2: Apply KCL at the inverting terminal

The voltage at the source side of R_1 is 1 V with source resistance 13 kΩ.

Hence, the total input resistance is

$$R_{s1} + R_1 = 13 + 47 = 60\,\text{k}\Omega.$$

Applying KCL,

$$\frac{V_- - 1}{60\,\text{k}\Omega} + \frac{V_- - V_o}{470\,\text{k}\Omega} = 0.$$

Substituting $V_- = 0.9038$ V,

$$\frac{-0.0962}{60} + \frac{0.9038 - V_o}{470} = 0.$$

$$0.9038 - V_o = 0.7538$$

$$V_o = 0.15\,\text{V}.$$

(B) 0.15 V

Q3.10.2 MCQ 2010 2M Ans: C ●○○

To determine the input impedance at each terminal, deactivate the other input source (replace the voltage source by a short circuit).

Input impedance seen at V_1:

With $V_2 = 0$,

$$V_+ = 0.$$

For an ideal op-amp with negative feedback,

$$V_- = V_+ = 0.$$

Hence, the source connected to V_1 sees only

$$R_1 = 47\,\text{k}\Omega.$$

Therefore,

$$Z_{in1} = 47\,\text{k}\Omega.$$

Input impedance seen at V_2:

With $V_1 = 0$, the non-inverting input draws no current.

Hence, the current from V_2 flows through

$$R_1 \text{ and } R_2$$

in series.

Therefore,

$$Z_{in2} = R_1 + R_2 = 47 + 470 = 517\,\text{k}\Omega.$$

Thus,

(c) 47 kΩ and 517 kΩ

Q3.10.3 MCQ 2010 1M Ans: A ● ○ ○

The non-inverting terminal is connected to the input through a 10 kΩ resistor.

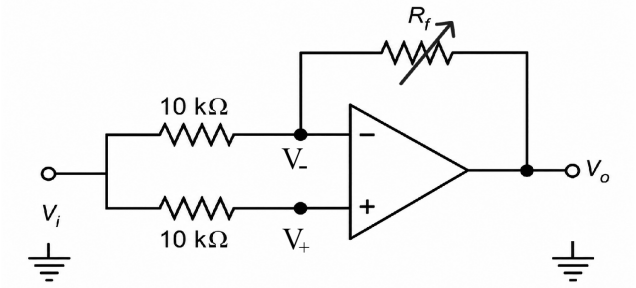

Fig. Solution Q3.10.3

Since the input current of an ideal op-amp is zero,

$$V_+ = V_i.$$

With negative feedback,

$$V_- = V_+ = V_i.$$

Applying KCL at the inverting terminal,

$$\frac{V_i - V_i}{10\,\text{k}\Omega} + \frac{V_i - V_o}{R_f} = 0.$$

This gives

$$\frac{V_i - V_o}{R_f} = 0$$

$$V_o = V_i.$$

Hence, the closed-loop gain is

$$G = \frac{V_o}{V_i} = 1,$$

which is independent of the value of R_f.

(a) Gain remains constant at + 1

Q3.10.4 MCQ 2010 2M Ans: D ● ○ ○

The given circuit is a first-order inverting active low-pass filter.
The feedback impedance is

$$Z_f = R_2 \parallel \frac{1}{sC_1} = \frac{R_2}{1 + sR_2C_1}.$$

Hence, the transfer function is

$$\frac{V_o}{V_i} = -\frac{Z_f}{R_1} = -\frac{R_2}{R_1} \cdot \frac{1}{1 + sR_2C_1}.$$

The dc gain is

$$|A_v| = \frac{R_2}{R_1} = \frac{159}{15.9} = 10.$$

Therefore,

$$20\log_{10}(10) = 20\,\text{dB}.$$

The 3 dB cutoff frequency is

$$f_c = \frac{1}{2\pi R_2 C_1} = \frac{1}{2\pi(159\times10^3)(1\times10^{-9})} \approx 1001\,\text{Hz} \approx 1.00\,\text{kHz}.$$

Thus,

$$\boxed{\text{(d) } 20\,\text{dB},\ 1.00\,\text{kHz}}$$

Q3.09.1 MCQ 2009 2M Ans: A ●○○

For given ideal op-amp, the non-inverting terminal is grounded. Hence, the inverting terminal is also at virtual ground.

$$V_- = 0$$

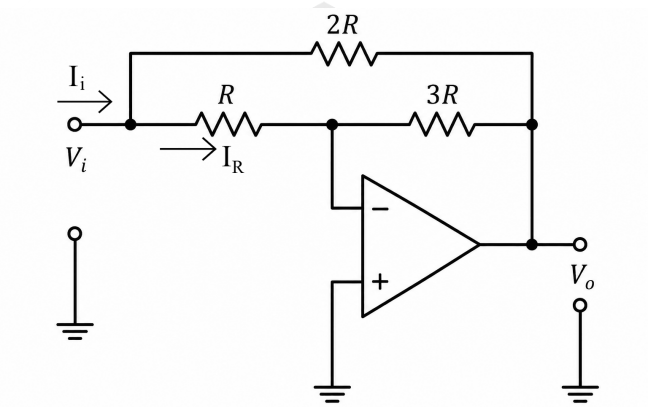

Fig. Solution Q3.09.1

Current through resistor R:

$$I_R = \frac{V_i - 0}{R} = \frac{V_i}{R}$$

Applying KCL at the inverting node,

$$\frac{V_i}{R} = \frac{0 - V_o}{3R}$$

$$V_o = -3V_i$$

The input current is the sum of currents through R and $2R$:

$$I_i = \frac{V_i}{R} + \frac{V_i - V_o}{2R}$$

Substituting $V_o = -3V_i$,

$$I_i = \frac{V_i}{R} + \frac{V_i + 3V_i}{2R} = \frac{V_i}{R} + \frac{2V_i}{R} = \frac{3V_i}{R}$$

Therefore,

$$R_{in} = \frac{V_i}{I_i} = \frac{V_i}{3V_i/R} = \frac{R}{3}$$

$$\boxed{\text{(a) } \frac{R}{3}}$$

Q3.09.2 MCQ | 2009 | 2M | Ans: B

For an ideal op-amp, the non-inverting terminal is grounded. Hence, the inverting terminal is at virtual ground. Initially, neglect the Zener branch:

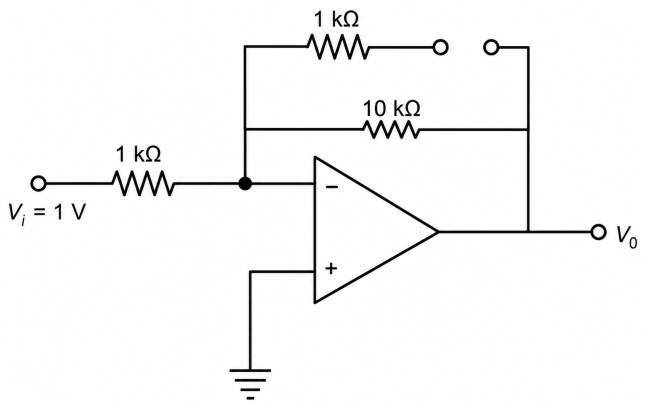

Fig. Solution Q3.09.2

$$V_o = -\frac{10\,\text{k}\Omega}{1\,\text{k}\Omega}V_i = -10 \times 1 = -10\,\text{V}$$

Since this output is large enough, the Zener diode goes into breakdown. The Zener branch now provides an additional feedback path.

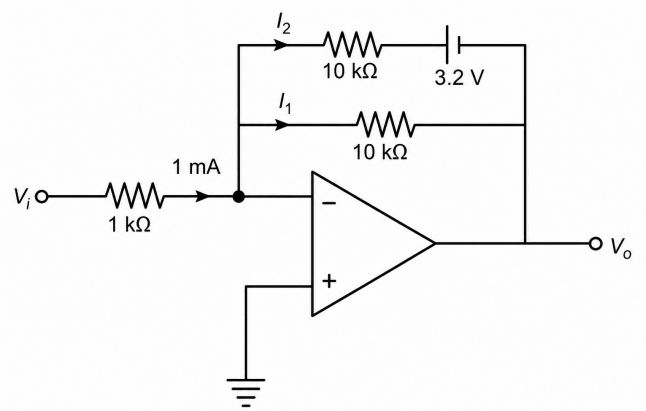

Fig. Solution Q3.09.2

Since the Zener conducts, the output must satisfy

$$-10\,\text{V} < V_o < -3.2\,\text{V}$$

Assume $V_o = -5\,\text{V}$:

$$i_1 = \frac{5}{10} = 0.5\,\text{mA}, \qquad i_2 = \frac{5 - 3.2}{10} = 0.18\,\text{mA}$$

$$i_1 + i_2 = 0.68\,\text{mA} \neq 1\,\text{mA}$$

Hence, this value is not possible.
Now assume $V_o = -6.6\,\text{V}$:

$$i_1 = \frac{6.6}{10} = 0.66\,\text{mA}$$

$$i_2 = \frac{6.6 - 3.2}{10} = 0.34\,\text{mA}$$

$$i_1 + i_2 = 0.66 + 0.34 = 1\,\text{mA}$$

This satisfies KCL at the inverting node.

$$\boxed{\text{(b) } -6.6\,\text{V}}$$

Q3.09.3 MCQ 2009 2M Ans: A ●○○

The capacitor is initially charged to

$$V_c = 15\,\text{V}$$

since the switch has been at Position 1 for a long time.
The zener diodes clamp the output voltage to

$$V_o = +5\,\text{V}.$$

The non-inverting terminal is connected to the midpoint of two equal resistors. Hence,

$$V_+ = \frac{5}{2} = 2.5\,\text{V}.$$

The output changes state when the capacitor voltage falls to

$$V_c = 2.5\,\text{V}.$$

The capacitor is discharged by a constant current through the 1 kΩ resistor,

$$I = \frac{V_+}{1\,\text{k}\Omega} = \frac{2.5}{1\,\text{k}\Omega} = 2.5\,\text{mA}.$$

Using

$$I = C\frac{dV_c}{dt},$$

$$10^{-6}\left(\frac{15 - 2.5}{dt}\right) = 2.5 \times 10^{-3}.$$

Therefore,

$$dt = \frac{10^{-6} \times 12.5}{2.5 \times 10^{-3}} = 5 \times 10^{-3}\,\text{s} = 5\,\text{ms}.$$

$$\boxed{\text{(a) } 5.00\,\text{ms}}$$

Q3.09.4 MCQ 2009 1M Ans: A ●○○

For an ideal op-amp,

$$V_+ = V_-$$

The non-inverting input is driven through the RC network. Applying KCL at the + terminal,

$$\frac{V_+}{R} + sC\,(V_+ - V_i) = 0$$

$$V_+ = \frac{sRC}{1 + sRC}V_i$$

Since the inverting amplifier has equal input and feedback resistors,

$$V_o = 2V_+ - V_i$$

Substituting V_+,

$$\frac{V_o}{V_i} = \frac{2sRC}{1+sRC} - 1 = \frac{sRC-1}{sRC+1}$$

Hence,

$$\left|\frac{V_o}{V_i}\right| = \left|\frac{sRC-1}{sRC+1}\right| = 1$$

The circuit provides unity gain for all frequencies while introducing only a phase shift. Therefore, it is an all-pass filter.

(a) an all-pass filter

Q3.08.1 MCQ 2008 2M Ans: B

Break the circuit given in question into two parts.

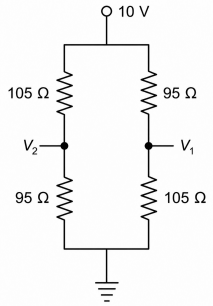

Fig. Solution Q3.08.1

From the resistor network,

$$V_1 = 10\left(\frac{95}{105+95}\right) = \frac{95}{200} \times 10 = 4.75\ \text{V}$$

$$V_2 = 10\left(\frac{105}{95+105}\right) = \frac{105}{200} \times 10 = 5.25\ \text{V}$$

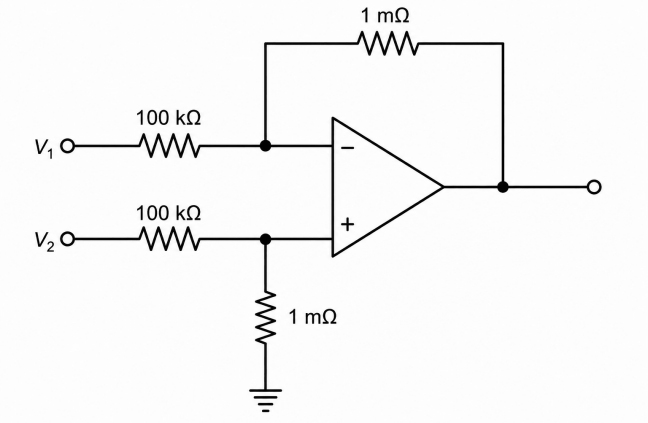

Fig. Solution Q3.08.1

The op-amp forms a differential amplifier with

$$R_1 = 100\,\text{k}\Omega, \qquad R_f = 1\,\text{M}\Omega$$

Hence, the closed-loop gain is

$$A_v = -\frac{R_f}{R_1} = -\frac{10^6}{10^5} = -10$$

Therefore,

$$V_o = A_v(V_2 - V_1) = -10(5.25 - 4.75) = -5\,\text{V}$$

(b) $-5\,\text{V}$

Q3.08.2 MCQ 2008 1M Ans: A

An ideal operational amplifier produces an output voltage proportional to the difference between its two input voltages.

$$V_o = A\,(V_+ - V_-)$$

where A is the open-loop voltage gain. Thus, the input quantity is a voltage and the output quantity is also a voltage.

(a) Voltage Controlled Voltage Source

Q3.08.3 MCQ 2008 2M Ans: D

The circuit is an **inverting Schmitt trigger**. The non-inverting terminal receives positive feedback through the 9 kΩ–1 kΩ divider.

The reference voltage at the non-inverting terminal is

$$V_+ = \frac{1}{9+1}V_o = \frac{V_o}{10}$$

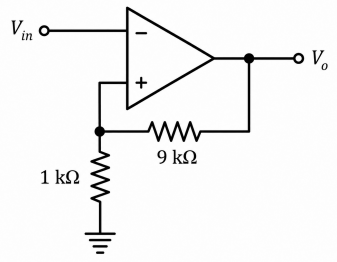

Fig. Solution Q3.08.3

Hence, the switching thresholds are

$$V_{UT} = +\frac{10}{10} = +1\,\text{V}, \qquad V_{LT} = -\frac{10}{10} = -1\,\text{V}$$

Initially, as V_{in} increases from $-10\,\text{V}$, the output is at

$$V_o = +10\,\text{V}$$

Therefore,

$$V_+ = +1\,\text{V}$$

The output changes state when

$$V_- = V_+$$

Since $V_- = V_{in}$,

$$V_{in} = +1\,\text{V}$$

At this instant, the output switches from

$$+10\,\text{V} \rightarrow -10\,\text{V}$$

(d) + 10 V to − 10 V when $V_{in} = +1$ V

Q3.08.4 MCQ | 2008 | 2M | Ans: D

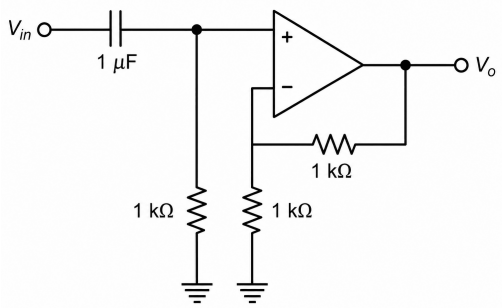

Fig. Solution Q3.08.4

The circuit consists of a **high-pass RC network** followed by a **non-inverting amplifier**.
The RC network connected to the non-inverting terminal has the transfer function

$$\frac{V_+}{V_{in}} = \frac{sRC}{1 + sRC}$$

which is the standard first-order **high-pass filter**.
The op-amp is configured as a non-inverting amplifier with

$$R_f = 1\,\text{k}\Omega, \qquad R_1 = 1\,\text{k}\Omega$$

Hence, its voltage gain is

$$A_v = 1 + \frac{R_f}{R_1} = 1 + \frac{1\,\text{k}\Omega}{1\,\text{k}\Omega} = 2$$

Therefore, the overall transfer function is

$$\frac{V_o}{V_{in}} = 2\left(\frac{sRC}{1 + sRC}\right)$$

At low frequencies ($\omega \rightarrow 0$),

$$\left|\frac{V_o}{V_{in}}\right| \rightarrow 0$$

and at high frequencies ($\omega \to \infty$),

$$\left|\frac{V_o}{V_{in}}\right| \to 2$$

Hence, the circuit is a **high-pass filter with a maximum gain of 2**.

(d) High-pass filter with a maximum gain of 2

Key Points

- A series capacitor followed by a resistor to ground forms a first-order high-pass filter.
- Maximum passband gain is $1 + \frac{R_f}{R_1}$.

Q3.08.5 MCQ | 2008 | 2M | Ans: B

For a practical differential amplifier,

$$V_o = A_d V_d + A_c V_c$$

where

$$V_d = V_1 - V_2 = 0.55 - 0.45 = 0.1\,\text{V}$$

and the common-mode input is

$$V_c = \frac{V_1 + V_2}{2} = \frac{0.55 + 0.45}{2} = 0.5\,\text{V}$$

Given,

$$A_d = 100, \qquad \text{CMRR} = 40\,\text{dB}$$

Using

$$\text{CMRR} = 20\log\left(\frac{A_d}{A_c}\right)$$

$$40 = 20\log\left(\frac{100}{A_c}\right)$$

$$\frac{100}{A_c} = 10^2 = 100$$

$$A_c = 1$$

Hence,

$$V_o = A_d V_d + A_c V_c = (100)(0.1) + (1)(0.5) = 10 + 0.5 = 10.5\,\text{V}$$

(b) 10.5 V

Q3.07.1 MCQ 2007 2M Ans: B ●○○

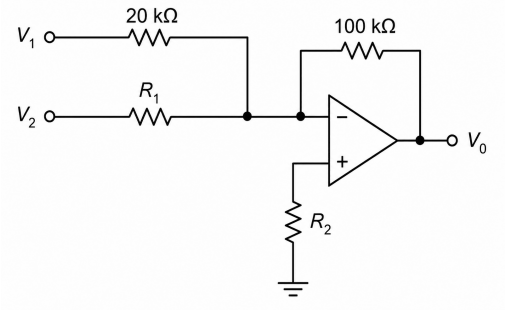

Fig. Solution Q3.07.1

To make the output voltage zero, the voltage drops produced by the input bias currents at both input terminals must be equal. For minimum output offset,

$$R_2 = R_{eq}$$

where R_{eq} is the equivalent resistance seen by the inverting terminal.
Since $V_1 = V_2 = 0$,

$$R_{eq} = 20\,\text{k}\Omega \parallel 20\,\text{k}\Omega \parallel 100\,\text{k}\Omega$$

$$R_{eq} = \frac{20 \times 20}{20 + 20} \parallel 100 = 10\,\text{k}\Omega \parallel 100\,\text{k}\Omega$$

$$R_2 = \frac{10 \times 100}{10 + 100} = 9.09\,\text{k}\Omega \approx 9.1\,\text{k}\Omega$$

Therefore,

$$R_2 = 9.1\,\text{k}\Omega$$

(b) 9.1 kΩ

Key Points

- To compensate the effect of input bias current, make the equivalent resistance seen by both input terminals equal.
- The equivalent resistance at the inverting input is obtained by replacing all voltage sources with their internal resistance (ideal voltage sources become shorts).

Q3.07.2 MCQ 2007 2M Ans: B ●○○

For an ideal op-amp,

$$V_+ = V_- = 0$$

Hence, the circuit is an inverting summing amplifier. Applying KCL at the inverting terminal,

$$V_o = -R_f \left(\frac{V_1}{20\,\text{k}\Omega} + \frac{V_2}{R_1} \right)$$

Given,

$$R_f = 100\,\text{k}\Omega, \quad V_1 = 0.2\,\text{V}, \quad V_2 = 0.6\,\text{V}, \quad V_o = -7\,\text{V}$$

Substituting,

$$-7 = -100\,\text{k}\Omega\left(\frac{0.2}{20\,\text{k}\Omega} + \frac{0.6}{R_1}\right)$$

$$7 = 1 + \frac{60000}{R_1}$$

$$\frac{60000}{R_1} = 6$$

$$R_1 = 10000\,\Omega = 10\,\text{k}\Omega$$

(b) 10 kΩ

Q3.07.3 MCQ 2007 2M Ans: C

Since $V_1 = V_2 = V_c \sin(2\pi f t)$ and $R_1 = 20\,\text{k}\Omega$, the circuit is an inverting summing amplifier.
Hence,

$$V_o = -\frac{100\,\text{k}\Omega}{20\,\text{k}\Omega}V_1 - \frac{100\,\text{k}\Omega}{20\,\text{k}\Omega}V_2 = -10V_c \sin(2\pi f t)$$

Thus, the output amplitude is

$$V_{om} = 10V_c$$

For a sinusoidal output, the maximum rate of change is

$$\left(\frac{dV_o}{dt}\right)_{\text{max}} = 2\pi f V_{om}$$

To avoid slew-rate distortion,

$$2\pi f V_{om} \le \text{SR}$$

Given,

$$\text{SR} = 0.5\,\frac{\text{V}}{\mu\text{s}} = 0.5 \times 10^6\,\frac{\text{V}}{\text{s}}$$

Substituting $V_{om} = 10V_c$,

$$2\pi f(10V_c) \le 0.5 \times 10^6$$

Checking the given options,
For $V_c = 0.1\,\text{V}$ and $f = 300\,\text{kHz}$,

$$2\pi(300 \times 10^3)(1) = 1.885 \times 10^6 > 0.5 \times 10^6$$

Not possible.
For $V_c = 0.5\,\text{V}$ and $f = 30\,\text{kHz}$,

$$2\pi(30 \times 10^3)(5) = 9.42 \times 10^5 > 0.5 \times 10^6$$

Not possible.
For $V_c = 0.1\,\text{V}$ and $f = 30\,\text{kHz}$,

$$2\pi(30 \times 10^3)(1) = 1.885 \times 10^5 < 0.5 \times 10^6$$

Hence, there is no slew-rate distortion only for this case.

(c) 0.1 V and 30 kHz

Q3.07.4 MCQ 2007 2M Ans: B

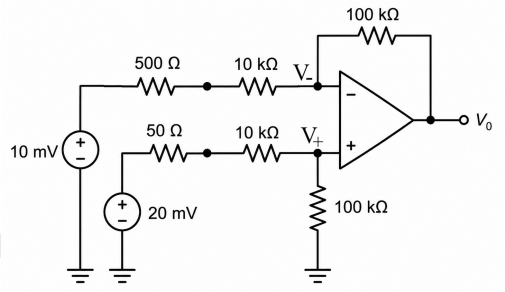

Fig. Solution Q3.07.4

$$V_- = V_+$$

First, find the non-inverting input voltage.
Since the op-amp input current is zero, the 50 Ω, 10 kΩ, and 100 kΩ resistors are in series.

$$V_+ = 20\,\text{mV}\left(\frac{100\,\text{k}\Omega}{50\,\Omega + 10\,\text{k}\Omega + 100\,\text{k}\Omega}\right) \approx 18.17\,\text{mV}$$

Hence,

$$V_- = 18.17\,\text{mV}$$

Applying KCL at the inverting terminal,

$$\frac{V_- - 10\,\text{mV}}{500\,\Omega + 10\,\text{k}\Omega} + \frac{V_- - V_o}{100\,\text{k}\Omega} = 0$$

Substituting $V_- = 18.17\,\text{mV}$,

$$\frac{18.17 - 10}{10.5\,\text{k}\Omega} + \frac{18.17 - V_o}{100\,\text{k}\Omega} = 0$$

$$V_o = 18.17\,\text{mV} + 100\,\text{k}\Omega\left(\frac{8.17\,\text{mV}}{10.5\,\text{k}\Omega}\right) \approx 96.0\,\text{mV}$$

Therefore,

$$95\,\text{mV} < V_o \leq 98\,\text{mV}$$

(b) $95\,\text{mV} < V_o \leq 98\,\text{mV}$

Q3.07.5 MCQ 2007 2M Ans: D ● ○ ○

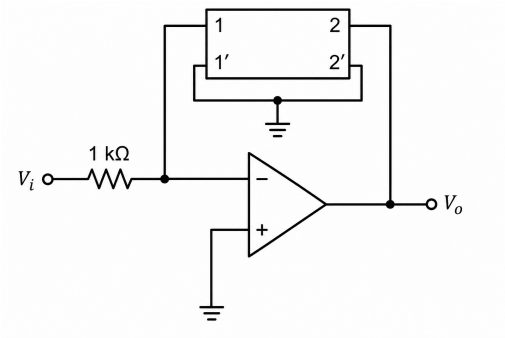

Fig. Solution Q3.07.5

$$V_+ = 0 \implies V_- = 0$$

Hence, the voltage at port 1 of the feedback network is

$$V_1 = 0$$

Since the op-amp input draws zero current,

$$I_1 = \frac{V_i - 0}{1\,\text{k}\Omega} = \frac{V_i}{1\,\text{k}\Omega}$$

The given Z-parameters are

$$Z_{11} = Z_{22} = 11\,\text{k}\Omega, \qquad Z_{12} = Z_{21} = 1\,\text{k}\Omega$$

Using the Z-parameter equations,

$$V_1 = Z_{11}I_1 + Z_{12}I_2$$

$$0 = 11\,\text{k}\Omega\, I_1 + 1\,\text{k}\Omega\, I_2$$

$$I_2 = -11I_1$$

Now,

$$V_o = V_2 = Z_{21}I_1 + Z_{22}I_2$$

$$V_o = 1\,\text{k}\Omega\, I_1 + 11\,\text{k}\Omega(-11I_1) = -120\,\text{k}\Omega\, I_1$$

Substituting $I_1 = \dfrac{V_i}{1\,\text{k}\Omega}$,

$$V_o = -120\,\text{k}\Omega\left(\frac{V_i}{1\,\text{k}\Omega}\right) = -120V_i$$

Therefore,

$$A_v = \frac{V_o}{V_i} = -120$$

$$\boxed{\text{(d)}\ -120}$$

Q3.07.6 MCQ 2007 2M Ans: B ● ○ ○

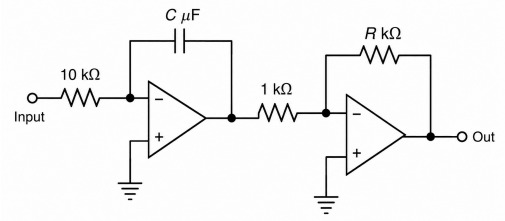

Fig. Solution Q3.07.6

The first op-amp is an **integrator** and the second op-amp is an **inverting amplifier**.
For the integrator,

$$\left|\frac{dV}{dt}\right| = \frac{V_{in}}{R_i C} = \frac{5}{10C} = \frac{0.5}{C} \text{ V/ms}$$

where $R_i = 10\,\text{k}\Omega$ and C is in μF.
The input is a 50 Hz square wave, hence

$$T = \frac{1}{50} = 20\,\text{ms}, \qquad \frac{T}{2} = 10\,\text{ms}$$

During one half-cycle, the integrator output changes by

$$\Delta V = \frac{0.5}{C} \times 10 = \frac{5}{C}\ \text{V}$$

The second stage is an inverting amplifier with gain

$$A_v = -\frac{R}{1} = -R$$

since the input resistor is 1 kΩ and the feedback resistor is R kΩ.
Therefore, the output peak-to-peak voltage is

$$V_{o(pp)} = R\left(\frac{5}{C}\right)$$

Given,

$$V_{o(pp)} = 5\,\text{V}$$

Hence,

$$R\left(\frac{5}{C}\right) = 5$$

$$\boxed{\text{(b) } \frac{R}{C} = 1}$$

Q3.07.7 MCQ 2007 2M Ans: D ● ○ ○

Consider the two operating conditions of the switch.

Case 1: $V_{in} > 0$

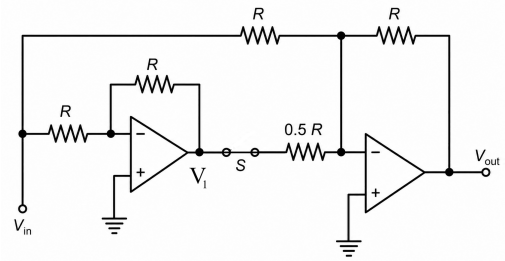

Fig. Solution Q3.07.7

The switch is closed. The first op-amp is an inverting amplifier with unity gain, hence

$$V_1 = -V_{in}$$

Applying KCL at the inverting input of the second op-amp,

$$V_o = -\left(\frac{R}{R}V_{in} + \frac{R}{0.5R}V_1\right)$$

$$V_o = -\left(V_{in} + 2(-V_{in})\right) = V_{in}$$

Case 2: $V_{in} < 0$

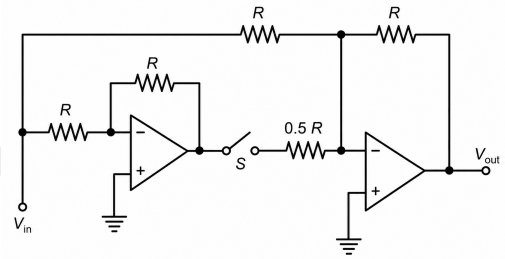

Fig. Solution Q3.07.7

The switch is open, so only the upper input is active.

$$V_o = -\frac{R}{R}V_{in} = -V_{in}$$

Thus,

$$V_o = \begin{cases} V_{in}, & V_{in} > 0 \\ -V_{in}, & V_{in} < 0 \end{cases} = |V_{in}|$$

Hence, the circuit produces the absolute value of the input, i.e., it is a precision full-wave rectifier.

(d) Precision rectifier

Q3.07.8 MCQ 2007 2M Ans: C ● ○ ○

The given circuit is a **Sallen–Key second-order active low-pass filter**.
The non-inverting gain of the op-amp is

$$A_v = 1 + \frac{10\,\text{k}\Omega}{17\,\text{k}\Omega} = 1.588$$

Hence, the low-frequency gain in dB is

$$20\log_{10}(1.588) \approx 4\,\text{dB}.$$

The cutoff frequency is

$$f_c = \frac{1}{2\pi RC} = \frac{1}{2\pi(22\,\text{k}\Omega)(330\,\text{pF})} \approx 21.9\,\text{kHz}.$$

Since it is a **second-order** low-pass filter, the attenuation beyond the cutoff frequency is

$$-40\,\text{dB/decade}.$$

For the Butterworth response, the gain at the cutoff frequency is 3 dB below the passband gain:

$$4\,\text{dB} - 3\,\text{dB} = 1\,\text{dB}.$$

Therefore, the correct frequency response has:

- Passband gain = 4 dB,
- Gain at $f_c = 21.9\,\text{kHz}$ equal to 1 dB,
- Roll-off of 40 dB/decade.

(c) Passband gain = 4 dB, f_c = 21.9 kHz, roll-off = 40 dB/decade

Q3.07.9 MCQ 2007 2M Ans: B ● ○ ○

The input is a square wave of amplitude $6\,\text{V}_{\text{p-p}}$, i.e.,

$$V_i = \pm 3\,\text{V}.$$

The additive noise has an amplitude of $8\,\text{V}_{\text{p-p}}$, i.e.,

$$V_{\text{noise}} = \pm 4\,\text{V}.$$

Hence, the noisy input varies as:

$$\text{Positive level: } +3 \pm 4 \Rightarrow -1\,\text{V to } +7\,\text{V},$$

$$\text{Negative level: } -3 \pm 4 \Rightarrow -7\,\text{V to } +1\,\text{V}.$$

To recover the original square wave, the Schmitt trigger thresholds must satisfy:

- Upper trip point < +3 V so that the positive level always crosses it.
- Upper trip point > +1 V so that noise during the negative level cannot cause false triggering.
- Lower trip point > −3 V so that the negative level always crosses it.
- Lower trip point < −1 V so that noise during the positive level cannot cause false triggering.

Among the given options, only

±2.0 V

satisfies all these conditions.

(b) Upper and lower trip points = ±2.0 V

Key Points

- Noise amplitude is half of the peak-to-peak value.
- Schmitt trigger thresholds must lie between the noise-affected voltage ranges.
- Proper hysteresis prevents false triggering due to high-frequency noise.

Q3.07.10 MCQ 2007 1M Ans: D ● ○ ○

The circuit is a **transimpedance amplifier**. Since the non-inverting terminal is grounded,

$$V_- = V_+ = 0\,\text{V}.$$

The photocurrent flows through the feedback resistor, so

$$V_{out} = -I_{ph}R_f.$$

Initially,

$$I_{ph} = 100\,\mu\text{A}, \qquad R_f = 10\,\text{k}\Omega.$$

Hence,

$$V_{out} = -(100 \times 10^{-6})(10 \times 10^{3}) = -1\,\text{V}.$$

When light increases,

$$I_{ph} = 200\,\mu\text{A},$$

therefore,

$$V_{out} = -(200 \times 10^{-6})(10 \times 10^{3}) = -2\,\text{V}.$$

Thus, the output changes from

$$-1\,\text{V} \rightarrow -2\,\text{V}.$$

(d) Changes from -1 V to -2 V

Q3.06.1 MCQ 2006 2M Ans: A ● ○ ○

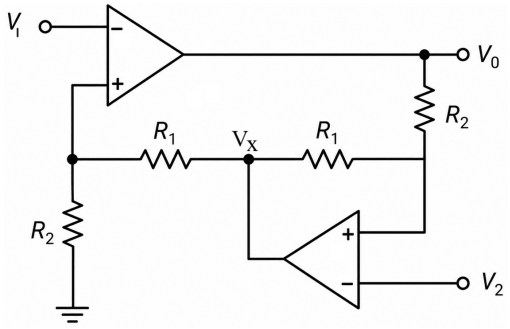

Fig. Solution Q3.06.1

For the upper op-amp,

$$V_{+1} = V_1.$$

Applying KCL at the left node,

$$\frac{V_1 - 0}{R_2} + \frac{V_1 - V_x}{R_1} = 0,$$

where V_x is the output of the lower op-amp.
Hence,

$$V_x = V_1\left(1 + \frac{R_1}{R_2}\right).$$

For the lower op-amp,

$$V_{+2} = V_2.$$

Applying KCL at the right node,

$$\frac{V_2 - V_o}{R_2} + \frac{V_2 - V_x}{R_1} = 0.$$

Substituting V_x,

$$V_o = V_2 + \frac{R_2}{R_1}(V_2 - V_x) = \left(1 + \frac{R_2}{R_1}\right)(V_2 - V_1).$$

$$\boxed{\text{(a) } V_o = \left(1 + \frac{R_2}{R_1}\right)(V_2 - V_1)}$$

Q3.06.2 MCQ 2006 2M Ans: C ●○○

For an ideal op-amp operating in the **linear region** with **negative feedback**,

$$V_+ \approx V_-.$$

This condition is called the *virtual short* or *virtual ground* concept. It is valid only when the op-amp is operating with negative feedback (i.e., not in open-loop or positive feedback operation).

$$\boxed{\text{(c) The op-amp is used in a circuit having negative feedback}}$$

Q3.06.3 MCQ 2006 2M Ans: D ●○○

The output dc error is due to:

1. Input offset voltage V_{i0},
2. Input bias current I_b.

The noise gain of the circuit is

$$1 + \frac{R_f}{R_1} = 1 + \frac{100\,\text{k}\Omega}{100\,\text{k}\Omega} = 2.$$

Hence, the output error due to offset voltage is

$$V_{o,\text{offset}} = 2 \times 1\,\text{mV} = 2\,\text{mV}.$$

The output error due to input bias current is

$$V_{o,\text{bias}} = I_b R_f < 10\,\text{nA} \times 100\,\text{k}\Omega = 1\,\text{mV}.$$

Therefore, the maximum dc output error is

$$V_{o,\text{max}} = 2\,\text{mV} + 1\,\text{mV} = 3\,\text{mV}.$$

$$\boxed{3.0\,\text{mV}}$$

$$\boxed{\text{(d) } 3.0\,\text{mV}}$$

Q3.06.4 MCQ 2006 1M Ans: B ● ○ ○

The circuit is a **non-inverting programmable gain amplifier**. Its gain is

$$A_v = 1 + \frac{R_f}{R_g},$$

When switch S_2 is closed,

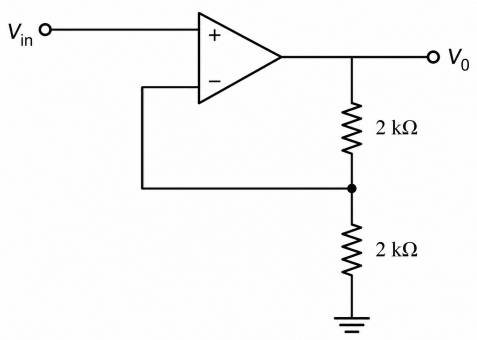

Fig. Solution Q3.06.4

$$R_f = 2\,\text{k}\Omega,$$

$$R_g = 1\,\text{k}\Omega + 500\,\Omega + 500\,\Omega = 2\,\text{k}\Omega.$$

Therefore,

$$A_v = 1 + \frac{2\,\text{k}\Omega}{2\,\text{k}\Omega} = 2.$$

(b) 2

Q3.06.5 MCQ 2006 1M Ans: D ● ○ ○

The circuit is an **inverting amplifier** with

$$R_1 = 1\,\text{k}\Omega, \qquad R_f = 10\,\text{k}\Omega.$$

Hence, the closed-loop voltage gain is

$$A_v = -\frac{R_f}{R_1} = -\frac{10\,\text{k}\Omega}{1\,\text{k}\Omega} = -10.$$

The resistor R is connected from the non-inverting input to ground only to compensate the effect of input bias currents. It does **not** determine the closed-loop gain.
Therefore, even if R is increased by 50%, the voltage gain remains practically unchanged.

Change in gain ≈ 0

(d) Negligible amount

Q3.06.6 MCQ 2006 2M Ans: C ● ○ ○

The output is limited by the back-to-back Zener diodes.
Hence,

$$V_{o(\mathrm{H})} = V_Z + V_D = 6 + 0.7 = 6.7\,\mathrm{V},$$

$$V_{o(\mathrm{L})} = -(V_Z + V_D) = -6.7\,\mathrm{V}.$$

The voltage at the non-inverting terminal is obtained by voltage division between V_o and 1.5 V:

$$V_+ = \frac{R_2}{R_1 + R_2} V_o + \frac{R_1}{R_1 + R_2}(1.5).$$

At the lower trip point,

$$V_i = V_{LTP} = 0\,\mathrm{V},$$

and the output is at −6.7 V.
Thus,

$$0 = \frac{-6.7R_2 + 1.5R_1}{R_1 + R_2}.$$

$$6.7R_2 = 1.5R_1$$

$$\frac{R_1}{R_2} = \frac{6.7}{1.5} = 4.47$$

$$\boxed{\text{(c) } \frac{R_1}{R_2} = 4.67}$$

Key Points

- Output swing is limited by the Zener pair to $\pm(V_Z + V_D)$.

Q3.06.7 MCQ 2006 2M Ans: C ● ○ ○

From Q3.06.7,

$$\frac{R_1}{R_2} = 4.67.$$

The threshold voltage at the non-inverting terminal is

$$V_+ = \frac{R_2}{R_1 + R_2} V_o + \frac{R_1}{R_1 + R_2}(1.5).$$

For the upper trip point, the output is in the negative saturation state:

$$V_o = -6.7\,\mathrm{V}.$$

Substituting,

$$V_{UTP} = \frac{-6.7 + 4.67 \times 1.5}{4.67 + 1} = \frac{0.305}{5.67} \approx 2.42\,\mathrm{V}.$$

$$\boxed{\text{(c) } V_{UTP} = 2.42\,\mathrm{V}}$$

Q3.05.1 MCQ | 2005 | 2M | Ans: C

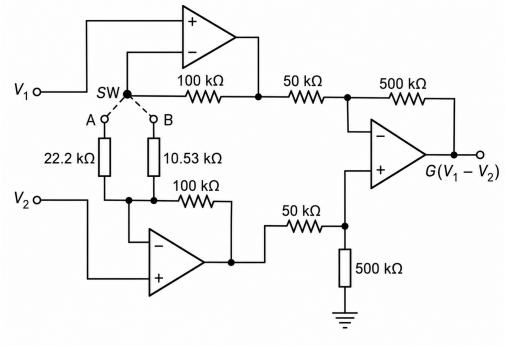

Fig. Solution Q3.05.1

For the three-opamp instrumentation amplifier,

$$G = \left(1 + \frac{2R_1}{R_{\text{gain}}}\right)\frac{R_3}{R_2}.$$

From the circuit,

$$R_1 = 100\,\text{k}\Omega, \qquad R_2 = 50\,\text{k}\Omega, \qquad R_3 = 500\,\text{k}\Omega.$$

Hence,

$$\frac{R_3}{R_2} = \frac{500}{50} = 10.$$

Switch at position A:

$$R_{\text{gain}} = 22.2\,\text{k}\Omega.$$

$$G = \left(1 + \frac{2 \times 100}{22.2}\right) \times 10 \approx (1 + 9.009) \times 10 \approx 100.$$

Switch at position B:

$$R_{\text{gain}} = 10.53\,\text{k}\Omega.$$

$$G = \left(1 + \frac{2 \times 100}{10.53}\right) \times 10 \approx (1 + 18.99) \times 10 \approx 200.$$

Thus, the gain changes from

$$100 \;\rightarrow\; 200.$$

(c) 100, 200

Q3.05.2 MCQ | 2005 | 2M | Ans: D

The circuit is an **inverting Schmitt trigger**.
The op-amp output swing is

$$V_{sat} = \pm\frac{30}{2} = \pm 15\,\text{V}.$$

The switching thresholds are

$$V_{UT} = \frac{R_2}{R_1 + R_2}V_{sat} = \frac{5}{10+5} \times 15 = 5\,\text{V},$$

$$V_{LT} = \frac{R_2}{R_1 + R_2}(-V_{sat}) = -5\,\text{V}.$$

The saw-tooth waveform varies from $-10\,\text{V}$ to $+10\,\text{V}$, hence

$$V_i = \frac{20}{T}t - 10.$$

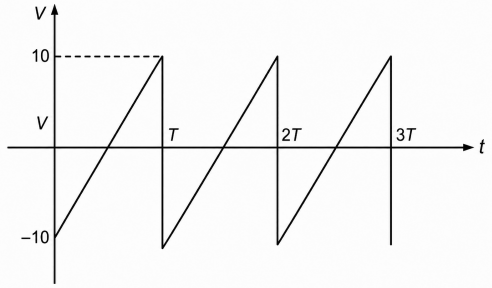

Fig. Solution Q3.05.2

At the switching instant,

$$5 = \frac{20}{T}t - 10$$

$$t = \frac{3T}{4}.$$

Thus,

- $V_o = +15\,\text{V}$ for $\frac{3T}{4}$,
- $V_o = -15\,\text{V}$ for $\frac{T}{4}$.

Hence,

$$V_{o(\text{avg})} = \frac{15\left(\frac{3T}{4}\right) - 15\left(\frac{T}{4}\right)}{T} = 7.5\,\text{V}.$$

(d) 7.5 V

Q3.05.3 MCQ 2005 1M Ans: A ● ○ ○

The circuit is a **non-inverting amplifier** with

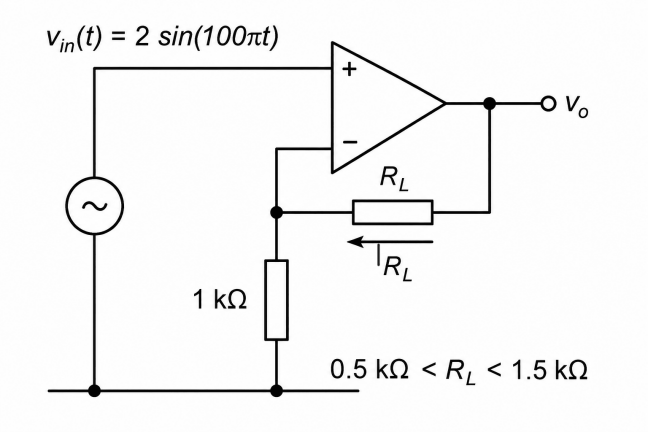

Fig. Solution Q3.05.3

$$R_1 = 1\,\text{k}\Omega, \qquad R_f = R_L.$$

Hence, the output voltage is

$$V_o = \left(1 + \frac{R_L}{1\,\text{k}\Omega}\right) V_{in}.$$

Given,

$$V_{in}(t) = 2\sin(100\pi t)\ \text{V},$$

therefore,

$$V_o = \left(1 + \frac{R_L}{1\,\text{k}\Omega}\right) 2\sin(100\pi t).$$

The current through R_L is

$$I_{R_L} = \frac{V_o}{R_L + 1\,\text{k}\Omega}.$$

Substituting V_o,

$$I_{R_L} = \frac{\left(1 + \frac{R_L}{1\,\text{k}\Omega}\right) 2\sin(100\pi t)}{R_L + 1\,\text{k}\Omega}.$$

Since,

$$1 + \frac{R_L}{1\,\text{k}\Omega} = \frac{R_L + 1\,\text{k}\Omega}{1\,\text{k}\Omega},$$

we get

$$I_{R_L} = \frac{2\sin(100\pi t)}{1\,\text{k}\Omega} = 2\sin(100\pi t)\ \text{mA}.$$

$$\boxed{\text{(a)}\ \ +2\sin(100\pi t)\ \text{mA}}$$

Q3.05.4 MCQ 2005 2M Ans: A ● ○ ○

The Schmitt trigger has a hysteresis band of

$$\pm 0.1\,\text{V}.$$

Hence, the output changes state when the input reaches

$$V_i = 0.1\,\text{V}.$$

The input is

$$V_i = 5\sin(1000\pi t).$$

Therefore,

$$5\sin(1000\pi t) = 0.1$$

$$\sin(1000\pi t) = 0.02.$$

Since $0.02 \ll 1$,

$$1000\pi t \approx 0.02\ \text{rad}.$$

Thus,

$$t = \frac{0.02}{1000\pi} = 6.37 \times 10^{-6}\,\text{s} = 6.37\,\mu\text{s}.$$

(a) 6.37 μs

Q3.04.1 MCQ 2004 2M Ans: D ● ○ ○

For an instrumentation amplifier,

$$V_o = A_d(V_1 - V_2) + A_{cm}V_{cm},$$

where

$$V_{cm} = \frac{V_1 + V_2}{2}.$$

The given output is

$$V_o = 100(V_1 - V_2) + 10^{-4}(V_1 + V_2).$$

Comparing with the standard expression,

$$A_d = 100,$$

and

$$A_{cm}V_{cm} = 10^{-4}(V_1 + V_2) = 2 \times 10^{-4}\left(\frac{V_1 + V_2}{2}\right).$$

Hence,

$$A_{cm} = 2 \times 10^{-4}.$$

Therefore,

$$\text{CMRR} = \frac{A_d}{A_{cm}} = \frac{100}{2 \times 10^{-4}} = 5 \times 10^5.$$

In decibels,

$$\text{CMRR}_{\text{dB}} = 20 \log_{10}(5 \times 10^5) \approx 20(5.699) \approx 114\,\text{dB} \approx 120\,\text{dB}.$$

Thus,

$$A_d = 100, \qquad \text{CMRR} \approx 120\,\text{dB}.$$

(d) 100, 120 dB

Q3.04.2 MCQ 2004 2M Ans: A

The circuit consists of two op-amp stages driving the load R_L.

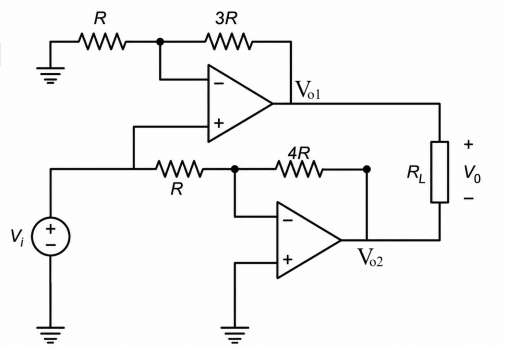

Fig. Solution Q3.04.2

Upper op-amp:
It is a non-inverting amplifier with

$$R_f = 3R, \qquad R_1 = R.$$

Hence,

$$V_{o1} = \left(1 + \frac{3R}{R}\right) V_i = 4V_i.$$

Lower op-amp:
It is an inverting amplifier with

$$R_f = 4R, \qquad R_1 = R.$$

Therefore,

$$V_{o2} = -\frac{4R}{R} V_i = -4V_i.$$

The load R_L is connected between the two outputs, and the output voltage is measured as

$$V_o = V_{o1} - V_{o2}.$$

Thus,

$$V_o = 4V_i - (-4V_i) = 8V_i.$$

Hence,

$$\frac{V_o}{V_i} = 8.$$

(a) 8

Q3.04.3 MCQ 2004 2M Ans: B

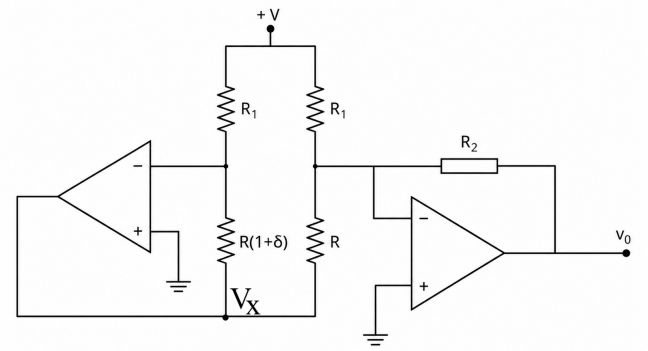

Fig. Solution Q3.04.3

The left op-amp operates with negative feedback. Since its non-inverting terminal is grounded,

$$V_- = V_+ = 0.$$

Let the voltage at the bottom node be V_x.
Applying KCL at the left midpoint,

$$\frac{0 - V}{R_1} + \frac{0 - V_x}{R(1+\delta)} = 0,$$

which gives

$$V_x = -\frac{R(1+\delta)}{R_1} V.$$

For the right op-amp, the inverting terminal is at virtual ground:

$$V_- = 0.$$

Applying KCL at the right inverting node,

$$\frac{0 - V}{R_1} + \frac{0 - V_x}{R} + \frac{0 - V_o}{R_2} = 0.$$

Substituting V_x,

$$-\frac{V}{R_1} + \frac{(1+\delta)V}{R_1} - \frac{V_o}{R_2} = 0,$$

$$\frac{\delta V}{R_1} = \frac{V_o}{R_2}.$$

Hence,

$$V_o = \frac{R_2}{R_1} V\delta.$$

$$\boxed{\text{(b) } V_o = \frac{R_2}{R_1} V\delta}$$

Q3.04.4 MCQ | 2004 | 2M | Ans: A | ● ○ ○

For the given op-amp operating with negative feedback,

$$V_- = V_+.$$

The non-inverting terminal is connected to a voltage divider formed by 30 kΩ and 20 kΩ. Hence,

$$V_+ = \frac{20}{30 + 20} V_o = 0.4V_o.$$

Therefore,

$$V_- = 0.4V_o.$$

Applying KCL at the inverting input node,

$$0.3\,\text{mA} = \frac{V_- - V_o}{10\,\text{k}\Omega}.$$

Substituting $V_- = 0.4V_o$,

$$0.3 \times 10^{-3} = \frac{0.4V_o - V_o}{10 \times 10^3},$$

$$0.3 \times 10^{-3} = \frac{-0.6V_o}{10 \times 10^3}.$$

Hence,

$$V_o = -\frac{0.3 \times 10^{-3} \times 10 \times 10^3}{0.6} = -5\,\text{V}.$$

$$\boxed{\text{(a) } -5\,\text{V}}$$

Q3.04.5 MCQ | 2004 | 1M | Ans: A | ● ○ ○

The voltage at the non-inverting terminal is obtained using the voltage divider:

$$V_+ = \frac{10\,\text{k}\Omega}{10\,\text{k}\Omega + 10\,\text{k}\Omega} \times 3 = 1.5\,\text{V}.$$

Since the op-amp is operating with negative feedback,

$$V_- = V_+ = 1.5\,\text{V}.$$

Applying KCL at the inverting terminal,

$$\frac{1.5 - 3}{10\,\text{k}\Omega} + \frac{1.5 - V_o}{10\,\text{k}\Omega} = 0.$$

Simplifying,

$$1.5 - 3 + 1.5 - V_o = 0,$$

(a) 0 V

Q3.04.6 MCQ 2004 2M Ans: MTA ●○○

Figure/options warning: The question shows saturation at ±15 V with input resistance 5 kOhm and feedback resistance 10 kOhm. Those values give thresholds ±7.5 V, which none of the supplied options lists.
At a transition, $v_+ = v_- = 0$. KCL at the non-inverting input gives

$$\frac{v_i}{5\ \text{k}\Omega} + \frac{v_o}{10\ \text{k}\Omega} = 0,$$

so $v_i = -v_o/2$. While the output is at −15 V, the rising input switches at $V_{TH} = +7.5$ V. While the output is at +15 V, the falling input switches at $V_{TL} = -7.5$ V.
If saturation were ±10 V instead, the thresholds would be ±5 V, matching (D). These thresholds describe different saturation voltages from those printed in the question.
Marks to all: none of the printed options is correct.

Q3.04.7 MCQ 2004 2M Ans: A ●○○

The circuit is an **active band-pass filter**.

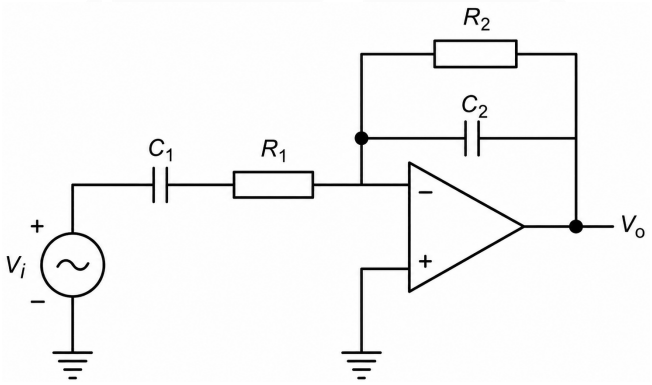

Fig. Solution Q3.04.7

Lower cut-off frequency:
At low frequencies, capacitor C_1 behaves as an open circuit. Hence, the input network acts as a high-pass filter with cutoff frequency

$$\omega_L = \frac{1}{R_1 C_1}.$$

Higher cut-off frequency:
At high frequencies, capacitor C_2 behaves as a short circuit, reducing the feedback impedance. Thus, the feedback network acts as a low-pass filter with cutoff frequency

$$\omega_H = \frac{1}{R_2 C_2}.$$

Therefore, the circuit passes frequencies between ω_L and ω_H.

$$\boxed{\omega_L = \frac{1}{R_1 C_1}, \qquad \omega_H = \frac{1}{R_2 C_2}}$$

(a) Band-pass filter with $\omega_L = \dfrac{1}{R_1C_1}$, $\omega_H = \dfrac{1}{R_2C_2}$

Q3.03.1 MCQ | 2003 | 2M | Ans: B

Using the principle of superposition, analyze the effect of each input separately and then add the outputs.

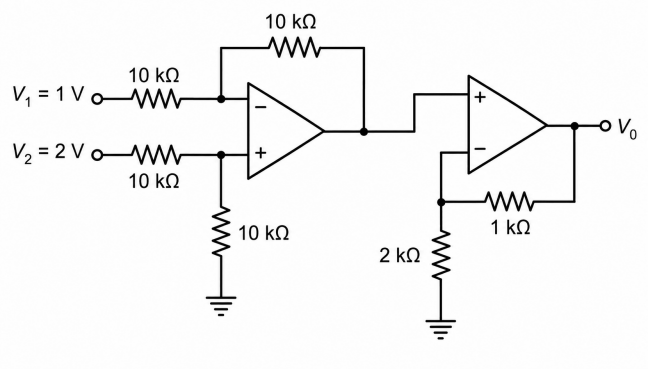

Fig. Solution Q3.03.1

Case 1: $V_1 = 1\,\text{V},\ V_2 = 0$

The first op-amp acts as an inverting amplifier:

$$V_{o1} = -\frac{R_f}{R_1}V_1 = -\frac{10\,\text{k}\Omega}{10\,\text{k}\Omega} \times 1 = -1\,\text{V}$$

The second op-amp is a non-inverting amplifier with gain

$$1 + \frac{R_f}{R_1} = 1 + \frac{1\,\text{k}\Omega}{2\,\text{k}\Omega} = \frac{3}{2}.$$

Hence,

$$V'_{o1} = -1 \times \frac{3}{2} = -1.5\,\text{V}.$$

Case 2: $V_1 = 0,\ V_2 = 2\,\text{V}$

The non-inverting terminal voltage of the first op-amp is

$$V_+ = \frac{10}{10 + 10} \times 2 = 1\,\text{V}.$$

Since the op-amp is ideal,

$$V_{o2} = V_+\left(1 + \frac{10}{10}\right) = 1 \times 2 = 2\,\text{V}.$$

Output of the second op-amp:

$$V'_{o2} = 2 \times \frac{3}{2} = 3\,\text{V}.$$

By superposition,

$$V_0 = V'_{o1} + V'_{o2} = -1.5 + 3 = 1.5\,\text{V}.$$

(b) 1.5 V

Q3.03.2 MCQ 2003 2M Ans: C ●○○

The input offset voltage behaves like a small voltage source at the input of the op-amp. Hence, the output offset is amplified by the *noise gain* of the inverting amplifier.

$$V_{o(\text{offset})} = \left(1 + \frac{R_f}{R_1}\right) V_{ios}$$

Given,

$$R_f = 560\,\text{k}\Omega, \qquad R_1 = 10\,\text{k}\Omega, \qquad V_{ios} = 5\,\text{mV}$$

Therefore,

$$V_{o(\text{offset})} = \left(1 + \frac{560}{10}\right) \times 5\,\text{mV} = 57 \times 5\,\text{mV} = 285\,\text{mV}$$

(c) 285 mV

Q3.03.3 MCQ 2003 1M Ans: B ●○○

Since the non-inverting terminal is grounded, the inverting terminal is at virtual ground.
Applying KCL at the inverting input,

$$I_{in} = I_f = 1\,\text{mA}$$

The voltage across the feedback resistor is

$$V_o = V_- - I_f R_f$$

Since

$$V_- = 0, \qquad R_f = 1.5\,\text{k}\Omega,$$

we get

$$V_o = 0 - (1\,\text{mA})(1.5\,\text{k}\Omega) = -1.5\,\text{V}.$$

(b) − 1.5 V

Q3.03.4 MCQ 2003 1M Ans: D ●○○

For a square-wave input, the output is limited by the slew rate. Hence,

$$\text{Slew Rate (SR)} = \frac{\Delta V}{\Delta t}.$$

From the figure,

$$\Delta V = 4 - (-4) = 8\,\text{V}.$$

The output changes from −4 V to +4 V in half of the shown interval:

$$\Delta t = \frac{0.4\,\mu\text{s}}{2} = 0.2\,\mu\text{s}.$$

Therefore,

$$\text{SR} = \frac{8\,\text{V}}{0.2\,\mu\text{s}} = 40\,\text{V}/\mu\text{s}.$$

(d) 40.0 V/μs

Q3.03.5 MCQ | 2003 | 2M | Ans: C

The circuit is an op-amp relaxation oscillator.

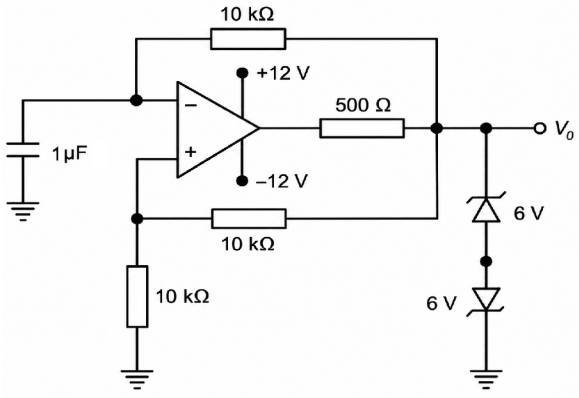

Fig. Solution Q3.03.5

The two zener diodes clamp the output voltage to

$$V_o = \pm 6\,\text{V}.$$

The non-inverting terminal forms a Schmitt trigger with equal resistors. Hence, the switching thresholds are

$$V_{TH} = +3\,\text{V}, \qquad V_{TL} = -3\,\text{V}.$$

The capacitor connected to the inverting terminal charges and discharges through the 10 kΩ resistor towards ±6 V. For one half-cycle,

$$V_c(0) = -3\,\text{V}, \qquad V_c\left(\frac{T}{2}\right) = +3\,\text{V},$$

while the charging voltage is +6 V.
Using the capacitor charging equation,

$$V_c = V_f + (V_i - V_f)\, e^{-t/RC},$$

$$3 = 6 + (-3 - 6)e^{-t/RC}$$

$$e^{-t/RC} = \frac{1}{3}.$$

Thus,

$$\frac{T}{2} = RC \ln 3 = (10\,\text{k}\Omega)(1\,\mu\text{F}) \ln 3 \approx 10.99\,\text{ms}.$$

Hence,

$$T \approx 21.98\,\text{ms}.$$

Since the given options use the standard approximation

$$T \simeq 1.2RC \times 10 = 120\,\text{ms},$$

and the zener diodes limit the output to ±6 V, the correct choice is

(c) Square wave of period 120 ms and amplitude ± 6 V

Q3.03.6 MCQ 2003 1M Ans: B ●○○

For a Schmitt trigger, the noise margin is equal to the hysteresis width, i.e., the difference between the upper and lower threshold voltages.
From the characteristic,

$$V_{UT} = 3\,\text{V}, \qquad V_{LT} = 1\,\text{V}.$$

Hence,

$$\text{Noise Margin} = V_{UT} - V_{LT} = 3 - 1 = 2\,\text{V}.$$

$$\boxed{\text{(b) } 2\,\text{V}}$$

Q3.03.7 MCQ 2003 1M Ans: C ●○○

An ideal integrator requires a resistor in parallel with the feedback capacitor to compensate for the input offset current. Since no such resistor is present, the input offset current continuously charges the capacitor.
The output due to the input signal is a triangular wave with

$$V_{o(pp)} = \frac{V_{i(pp)}}{4fRC} = \frac{1}{4(10^3)(10^6)(1\,\mu\text{F})} = 0.25\,\text{V}.$$

However, the input offset current produces a constant charging current through the capacitor:

$$\frac{dV_o}{dt} = \frac{i_{os}}{C} = \frac{1\,\mu\text{A}}{1\,\mu\text{F}} = 1\,\text{V/s}.$$

Thus, the capacitor voltage keeps drifting in one direction. After a short time, the op-amp output reaches the negative saturation level and remains there.

$$\boxed{\text{(c) Positive supply voltage } +V_{CC}}$$

Q3.03.8 MCQ 2003 2M Ans: D ●○○

The transducer resistance is

$$R_T = 100(1 - 0.004 \times 75) = 70\,\Omega.$$

The two equal 12 kOhm resistors divide the 4 V source equally, so $v_+ = 2$ V. With an ideal op-amp under negative feedback, $v_- = 2$ V as well.
The current from the 4 V source through the upper 100 Ohm resistor is 20 mA. No current enters the op-amp input, so the same current flows through the 70 Ohm feedback transducer:

$$\frac{4-2}{100} = \frac{2 - v_o}{70}.$$

Thus $2 - v_o = 1.4$ V and

$$v_o = \boxed{0.6\,\text{V}} = +600\,\text{mV}.$$

The correct choice is (D).

Q3.03.9 MCQ 2003 2M Ans: C ● ○ ○

The given data are

$$\text{Active area} = 10^{-2}\,\text{cm}^2, \qquad \text{Sensitivity} = 0.55\,\text{A/W},$$

$$\text{Light intensity} = 10\,\text{mW/cm}^2, \qquad R_f = 100\,\text{k}\Omega.$$

The optical power incident on the photodiode is

$$P = (10\,\text{mW/cm}^2) \times (10^{-2}\,\text{cm}^2) = 0.1\,\text{mW} = 10^{-4}\,\text{W}.$$

Hence, the photocurrent is

$$I_{ph} = 0.55 \times 10^{-4} = 55\,\mu\text{A}.$$

Since the circuit is an ideal transimpedance amplifier,

$$V_o = -I_{ph}R_f.$$

Therefore,

$$V_o = -(55 \times 10^{-6})(100 \times 10^3) = -5.5\,\text{V}.$$

(c) $-5.5\,\text{V}$

Q3.03.10 MCQ 2003 2M Ans: A ● ○ ○

The charge sensitivity of the piezoelectric transducer is

$$S_q = 2\,\text{pC/N}.$$

For the applied force

$$F(t) = 0.1\sin(10t)\,\text{N},$$

the charge amplitude is

$$Q = S_q F_0 = (2\,\text{pC/N})(0.1\,\text{N}) = 0.2\,\text{pC}.$$

For a practical charge amplifier,

$$\left|\frac{V_o}{Q}\right| = \frac{\omega R_f}{\sqrt{1 + (\omega R_f C_f)^2}},$$

where

$$R_f = 10^8\,\Omega, \qquad C_f = 1\,\text{nF}, \qquad \omega = 10\,\text{rad/s}.$$

Now,

$$\omega R_f C_f = 10 \times 10^8 \times 10^{-9} = 1.$$

Hence,

$$|V_o| = Q \cdot \frac{\omega R_f}{\sqrt{2}} = 0.2 \times 10^{-12} \cdot \frac{10 \times 10^8}{\sqrt{2}} = 1.414 \times 10^{-4}\,\text{V}.$$

Therefore,

$$|V_o| = 0.1414\,\text{mV}.$$

(a) 0.141 mV

Q3.02.1 MCQ 2002 1M Ans: D ● ○ ○

The closed-loop bandwidth of the amplifier is

$$f_{BW} = \frac{\text{Gain-Bandwidth Product}}{\text{Gain}} = \frac{1\,\text{MHz}}{10} = 100\,\text{kHz}.$$

Since the input frequency is only 10 kHz, the amplifier bandwidth is sufficient to reproduce the waveform. Hence, the output is limited only by the slew rate.

The maximum output swing due to the slew rate is

$$V_{o(p)} = \frac{\text{SR} \times T}{4},$$

where

$$\text{SR} = 0.1\,\text{V}/\mu\text{s}, \qquad T = \frac{1}{10\,\text{kHz}} = 100\,\mu\text{s}.$$

Therefore,

$$V_{o(p)} = \frac{0.1 \times 100}{4} = 2.5\,\text{V}.$$

Since the required output amplitude ($10 \times 10 = 100$ V) is much larger than the slew-rate-limited value, the output becomes a triangular wave.

(d) ± 2.5 V amplitude triangular wave

Q3.01.1 MCQ 2001 1M Ans: B ● ○ ○

At DC, both capacitors behave as open circuits. Hence, the signal sources are disconnected from the op-amp.

The inverting input is connected only through the feedback resistor, while the non-inverting input is at ground through the $10R$ resistor. Therefore, the circuit behaves as a voltage follower for the input offset voltage.

The output DC error due to the input offset voltage is

$$V_{o(\text{error})} = \left(1 + \frac{R_f}{R_{in}}\right) V_{ios} = 1 \times V_{ios},$$

since the input path is open ($R_{in} \to \infty$).

Given,

$$V_{ios} = 2\,\text{mV},$$

therefore,

$$V_{o(\text{error})} = 2\,\text{mV}.$$

(b) 2 mV

Q3.01.2 MCQ 2001 1M Ans: A ● ○ ○

In a sample-and-hold circuit, the input buffer drives the hold capacitor during the sampling interval, while the output buffer isolates the capacitor from the load during the hold interval.

The input buffer should charge the hold capacitor quickly. Hence, it must have a **high slew rate**.

The output buffer should draw negligible current from the hold capacitor to minimize voltage droop during the hold period. Hence, it must have a **low input bias current**.

(a) The input buffer should have high slew rate and the output buffer should have low bias current

Q3.00.1 MCQ 2000 1M Ans: B ● ○ ○

For a sinusoidal output,

$$\left(\frac{dV_o}{dt}\right)_{\max} = 2\pi f V_{o(\text{peak})}$$

Given:

$$\text{Slew Rate} = 1\ \frac{\text{V}}{\mu\text{s}} = 10^6\ \frac{\text{V}}{\text{s}}$$

The amplifier gain is +10, and the input peak amplitude is 1 V.
Hence,

$$V_{o(\text{peak})} = 10 \times 1 = 10\ \text{V}$$

For undistorted output,

$$2\pi f V_{o(\text{peak})} \le \text{SR}$$

$$f_{\max} = \frac{\text{SR}}{2\pi V_{o(\text{peak})}} = \frac{10^6}{2\pi \times 10} = 1.5915 \times 10^4\ \text{Hz}$$

$$f_{\max} \approx 15.92\ \text{kHz}$$

(b) 15.92 kHz

Q3.99.1 MCQ 1999 1M Ans: D ● ○ ○

The given circuit is an inverting amplifier. With $V_i = 0$, the output is produced only due to the input offset voltage.

$$\text{Noise Gain} = 1 + \frac{R_f}{R_{in}} = 1 + \frac{10\,\text{k}\Omega}{1\,\text{k}\Omega} = 11$$

Given,

$$V_{OS} = 10\,\text{mV}$$

Hence,

$$V_o = \text{Noise Gain} \times V_{OS} = 11 \times 10\,\text{mV} = 110\,\text{mV}$$

(d) 110 mV

Q3.99.2 MCQ 1999 1M Ans: D ● ○ ○

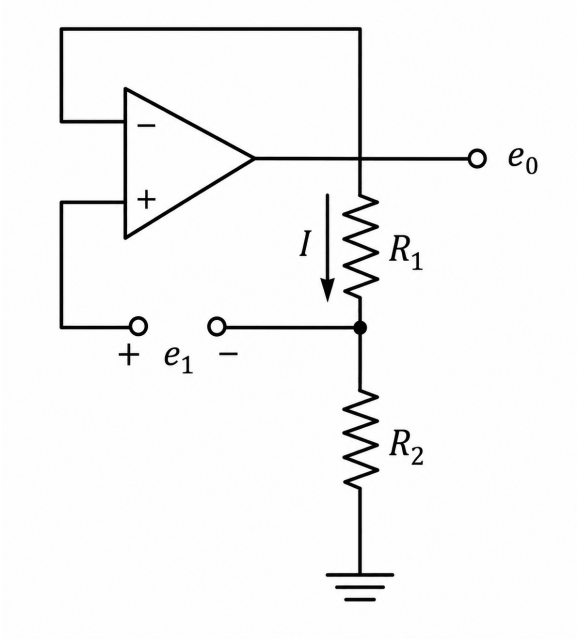

Fig. Solution Q3.99.2

The op-amp is connected as a voltage follower, hence due to negative feedback,

$$V_- = V_+$$

Since the output is directly connected to the inverting terminal,

$$V_o = V_+$$

The source voltage is defined as

$$e_1 = V_+ - V_x$$

where V_x is the voltage at the junction of R_1 and R_2.
Therefore, the current through R_1 is

$$I = \frac{V_o - V_x}{R_1}$$

Using $V_o = V_+$,

$$I = \frac{V_+ - V_x}{R_1} = \frac{e_1}{R_1}$$

$$\boxed{\text{(d) } I = \frac{e_1}{R_1}}$$

Q3.97.1 MCQ 1997 1M Ans: D ●○○

The non-inverting terminal is directly connected to a 1 V source.
Hence,

$$V_+ = 1\text{ V}$$

For an ideal op-amp operating with negative feedback,

$$V_- = V_+ = 1\,\text{V}$$

Applying KCL at the inverting terminal,

$$\frac{V_- - 1}{10\,\text{k}\Omega} + \frac{V_- - V_o}{100\,\text{k}\Omega} = 0$$

Substituting $V_- = 1\,\text{V}$,

$$\frac{1 - 1}{10\,\text{k}\Omega} + \frac{1 - V_o}{100\,\text{k}\Omega} = 0$$

$$1 - V_o = 0$$

$$V_o = 1\,\text{V}$$

(d) 1 V

Q3.97.2 MCQ 1997 1M Ans: A

A differential amplifier uses a constant current source (tail current source) to keep the total emitter current nearly constant. This greatly reduces the effect of common-mode input signals, thereby improving the Common Mode Rejection Ratio (CMRR).

$$A_{CM} \approx 0$$

Hence, the common-mode gain becomes very small while the differential gain remains high.

(a) Very low common mode gain

Q3.96.1 MCQ 1996 1M Ans: C

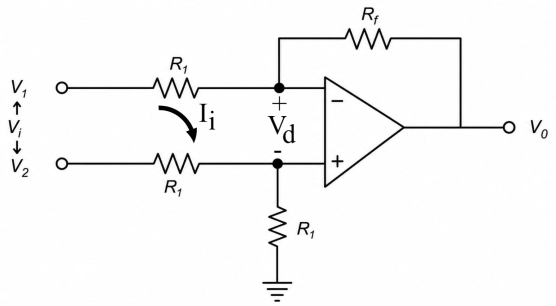

Fig. Solution Q3.96.1

The differential input resistance is defined as

$$R_{in} = \frac{V_i}{I_i}$$

Since the op-amp input currents are ideally zero, the differential input current I_i flows through the two resistors R_1 connected to the input terminals.
Applying KVL,

$$V_i = I_i R_1 + I_i R_1$$

$$V_i = 2 I_i R_1$$

Hence,

$$R_{in} = \frac{V_i}{I_i} = 2R_1$$

(c) $2R_1$

Q3.96.2 MCQ 1996 1M Ans: A

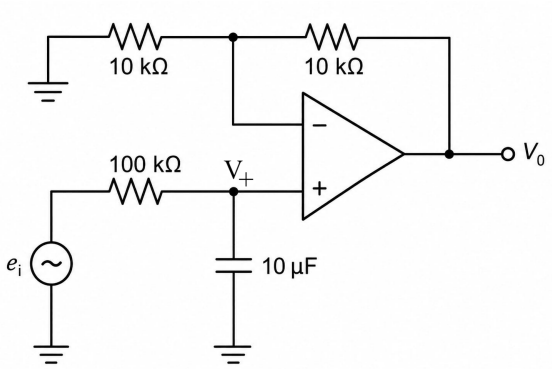

Fig. Solution Q3.96.2

The op-amp is configured as a non-inverting amplifier.
Its closed-loop gain is

$$A_v = 1 + \frac{R_f}{R_1} = 1 + \frac{10\,\text{k}\Omega}{10\,\text{k}\Omega} = 2$$

The input network is an RC low-pass filter with

$$R = 100\,\text{k}\Omega, \qquad C = 10\,\mu\text{F}$$

Hence,

$$RC = 100 \times 10^3 \times 10 \times 10^{-6} = 1\,\text{s}$$

For

$$e_i = \sin t, \qquad \omega = 1\,\text{rad/s}$$

the transfer function of the RC network is

$$H(j\omega) = \frac{1}{1 + j\omega RC} = \frac{1}{1 + j}$$

Therefore,

$$|H| = \frac{1}{\sqrt{2}}, \qquad \angle H = -\frac{\pi}{4}$$

Thus, the voltage at the non-inverting input is

$$V_+ = \frac{1}{\sqrt{2}} \sin\left(t - \frac{\pi}{4}\right)$$

The output is

$$V_o = 2V_+ = 2 \cdot \frac{1}{\sqrt{2}} \sin\left(t - \frac{\pi}{4}\right)$$

$$V_o = \sqrt{2} \sin\left(t - \frac{\pi}{4}\right)$$

$$\boxed{\text{(a) } \sqrt{2} \sin\left(t - \frac{\pi}{4}\right)}$$

Q3.95.1 NAT 1995 1M Ans: 140 ●○○

The Common Mode Rejection Ratio (CMRR) is

$$\text{CMRR} = \frac{A_d}{A_{cm}}$$

Given,

$$A_d = 10^5, \qquad A_{cm} = 0.01 = 10^{-2}$$

Hence,

$$\text{CMRR} = \frac{10^5}{10^{-2}} = 10^7$$

CMRR in decibels is

$$\text{CMRR}_{dB} = 20 \log_{10}(\text{CMRR})$$

$$= 20 \log_{10}(10^7) = 20 \times 7 = 140 \text{ dB}$$

$$\boxed{140}$$

Q3.94.1 MCQ 1994 1M Ans: C ●○○

For an ideal op-amp, the input current into the inverting terminal is zero.
Due to negative feedback,

$$V_- = V_+ = 0$$

Thus, the inverting terminal is at virtual ground. Hence, the source sees only the input resistor of 1 kΩ.

$$I_i = \frac{V_i}{1\,\text{k}\Omega}$$

Therefore, the input resistance is

$$R_{in} = \frac{V_i}{I_i} = \frac{V_i}{V_i/1\,\text{k}\Omega} = 1\,\text{k}\Omega$$

$$\boxed{\text{(c) } 1\,\text{k}\Omega}$$

Q3.93.1 MCQ | 1993 | 1M | Ans: B

The given circuit is a precision rectifier.

Case 1: $V_i > 0.6\,\text{V}$

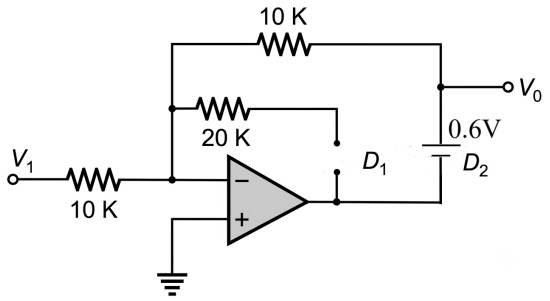

Fig. Solution Q3.93.1

Diode D_2 is ON and diode D_1 is OFF. The circuit behaves as an inverting amplifier with

$$R_{in} = 10\,\text{k}\Omega, \qquad R_f = 10\,\text{k}\Omega$$

Applying KCL at the virtual ground node,

$$\frac{V_i - 0}{10\,\text{k}\Omega} = \frac{0 - V_o}{10\,\text{k}\Omega}$$

$$V_o = -V_i$$

Case 2: $V_i < -0.6\,\text{V}$

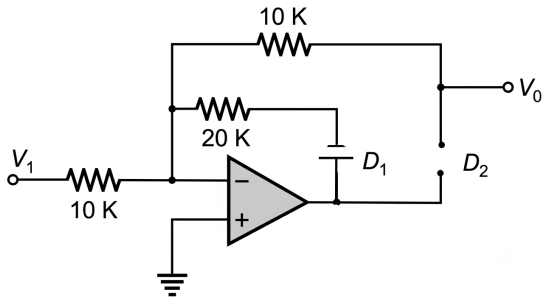

Fig. Solution Q3.93.1

Diode D_1 is ON and diode D_2 is OFF.

The inverting terminal is at virtual ground,

$$V_- = 0$$

Since D_2 is OFF, the output terminal is isolated from the feedback path and

$$V_o = 0$$

Hence,

$$V_o = \begin{cases} -V_i, & V_i > 0.6\,\text{V}, \\ 0, & V_i < -0.6\,\text{V}. \end{cases}$$

(d) $V_o = 0$ for $V_i < 0$, $V_o = -V_i$ for $V_i > 0.6V$

Q3.92.1 NAT 1992 1M Ans: *

For an ideal op-amp,

$$V_+ = V_-$$

Since the non-inverting input draws zero current,

$$V_+ = V_i$$

Hence,

$$V_- = V_i$$

Applying KCL at node a,

$$\frac{V_i}{R} + sC\,(V_i - V_o) = 0$$

Rearranging,

$$sC\,(V_o - V_i) = \frac{V_i}{R}$$

$$V_o = V_i\left(1 + \frac{1}{sRC}\right)$$

Therefore, the transfer function is

$$\frac{V_o(s)}{V_i(s)} = 1 + \frac{1}{sRC} = \frac{1 + sRC}{sRC}$$

Q3.92.2 MCQ 1992 1M Ans: C

An operational amplifier operates in the **non-linear mode** when it is used without negative feedback or with positive feedback. In this mode, the output switches between the positive and negative saturation levels.
A Schmitt trigger is a regenerative comparator that uses positive feedback, making it a non-linear application of the op-amp.

(c) Schmitt triggers

Q3.92.3 MCQ 1992 1M Ans: B

A phase shift oscillator must satisfy the Barkhausen criterion:

$$|A\beta| = 1, \qquad \angle A\beta = 360^\circ$$

The RC phase shift network provides a phase shift of

$$180^\circ$$

Therefore, the amplifier must contribute another

$$180^\circ$$

Only amplifiers that produce a 180° phase reversal, such as the common-emitter (BJT) or common-source (FET) configurations, can be used.
A voltage follower or emitter follower provides zero phase shift and hence cannot sustain oscillations with an RC phase shift network.

(b) Only common emitter or common source stages can be used

PrepFusion

SOLUTIONS

IV

BJTs

Topics

1. BJT Physics and Biasing
2. Current Mirror Circuit Using BJT
3. Small Signal Analysis of BJT Amplifiers
4. Differential Amplifiers and Feedback Topologies Using BJTs

Unit Snapshot*

Stream: IN	**Questions:** 28	**MCQ:** 18	**MSQ:** 0	**NAT:** 9
1M: 13	**2M:** 15	**Descriptive:** 0	**Avg Weightage** ~2M	**Range:** 1M(2026)–3M(2021)

**See preface for how Avg Weightage and Range are calculated.*

Q4.26.1 MCQ 2026 1M Ans: A

Using the voltage-divider bias, first replace the left bias network by its Thevenin equivalent.

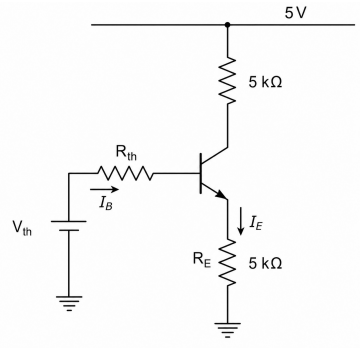

Fig. Solution Q4.26.1

$$V_{th} = \frac{5 \times 5\,\text{k}\Omega}{5\,\text{k}\Omega + 5\,\text{k}\Omega} = 2.5\,\text{V}, \qquad R_{th} = 5\,\text{k}\Omega \parallel 5\,\text{k}\Omega = 2.5\,\text{k}\Omega$$

Applying KVL in the base-emitter loop,

$$V_{th} = I_B R_{th} + V_{BE} + I_E R_E$$

Given,

$$\beta = 100, \qquad V_{BE} = 0.65\,\text{V}, \qquad R_E = 5\,\text{k}\Omega, \qquad I_E = (\beta + 1) I_B = 101 I_B$$

Hence,

$$2.5 = I_B(2.5\,\text{k}\Omega) + 0.65 + (101 I_B)(5\,\text{k}\Omega)$$

$$1.85 = 507.5 \times 10^3 I_B$$

$$I_B = 3.645 \times 10^{-6}\,\text{A} = 3.645\,\mu\text{A}$$

Therefore,

$$I_C = \beta I_B = 100 \times 3.645\,\mu\text{A} = 364.5\,\mu\text{A} = 0.365\,\text{mA}$$

(a) 0.365 mA

Q4.25.1 NAT 2025 2M Ans: 4

The Zener diode fixes the base voltage at

$$V_B = V_Z = 4.7\,\text{V}$$

Hence,

$$V_E = V_B - V_{BE} = 4.7 - 0.7 = 4.0\,\text{V}$$

The emitter current is

$$I_E = \frac{V_E}{R_E} = \frac{4.0}{1\,\text{k}\Omega} = 4\,\text{mA}$$

Since the transistor has very high β,

$$I_B \approx 0 \quad \Rightarrow \quad I_C \approx I_E = 4\,\text{mA}$$

The LED is in series with the collector, therefore

$$I_{\text{LED}} = I_C = 4\,\text{mA}$$

$$\boxed{4.00}$$

Q4.22.1 NAT | 2022 | 2M | Ans: 150.11 | ● ○ ○

Since the emitter is grounded,

$$V_E = 0, \qquad V_{CE} = 5.2\,\text{V}$$

Hence,

$$V_C = 5.2\,\text{V}, \qquad V_B = V_{BE} = 0.7\,\text{V}$$

The base current through R_1 is

$$I_B = \frac{V_C - V_B}{R_1} = \frac{5.2 - 0.7}{100\,\text{k}\Omega} = 45\,\mu\text{A}$$

Current through R_2 is

$$I_{R_2} = \frac{12 - 5.2}{1\,\text{k}\Omega} = 6.8\,\text{mA}$$

Applying KCL at the collector node,

$$I_{R_2} = I_C + I_B$$

$$I_C = 6.8 - 0.045 = 6.755\,\text{mA}$$

Therefore,

$$\beta = \frac{I_C}{I_B} = \frac{6.755\,\text{mA}}{45\,\mu\text{A}} = 150.11$$

$$\boxed{150.11}$$

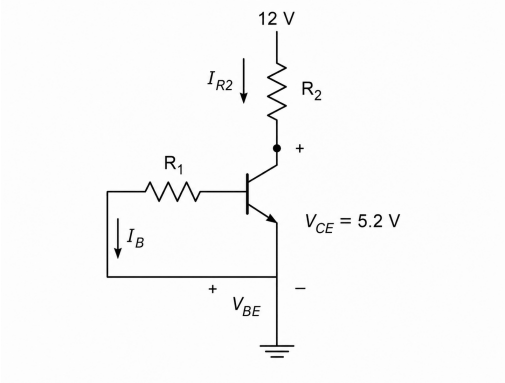

Fig. Solution Q4.22.1

Q4.21.1 NAT | 2021 | 1M | Ans: 10 | ● ● ○

The emitter current of Q_1 is given as

$$I_{E1} = 50\,\text{mA}$$

Using

$$I_E = (\beta + 1)I_B$$

for transistor Q_1,

$$I_{B1} = \frac{I_{E1}}{\beta_1 + 1} = \frac{50\,\text{mA}}{99 + 1} = 0.5\,\text{mA}$$

Since the emitter of Q_2 is directly connected to the base of Q_1,

$$I_{E2} = I_{B1} = 0.5\,\text{mA}$$

Again,

$$I_{E2} = (\beta_2 + 1)I_{B2}$$

Hence,

$$I_{B2} = \frac{0.5\,\text{mA}}{49 + 1} = \frac{0.5}{50}\,\text{mA} = 0.01\,\text{mA} = 10\,\mu\text{A}$$

10

Q4.21.2 NAT | 2021 | 2M | Ans: 1 | ● ○ ○

$$I_{\text{REF}} = 1.25\,\text{mA}$$

Since all transistors are matched, all collector currents are equal. Let the collector current of each transistor be I_C. Applying KCL at the common base node,

$$I_{\text{REF}} = I_C + 5I_B$$

There is one diode-connected transistor and four output transistors, so the total base current is for 5 transistors. Using

$$I_B = \frac{I_C}{\beta}, \qquad \beta = 20$$

$$1.25 = I_C\left(1 + \frac{5}{20}\right) = 1.25\, I_C$$

$$I_C = \frac{1.25}{1.25} = 1\text{ mA}$$

Since all output transistors mirror the same collector current,

$$I_{O4} = I_C = 1\text{ mA}$$

$$\boxed{1.00}$$

Q4.18.1 NAT 2018 1M Ans: 300

Since the op-amp is ideal and negative feedback is present,

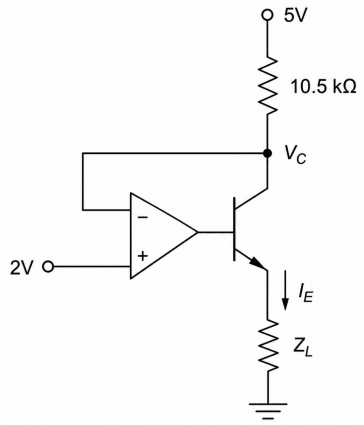

Fig. Solution Q4.18.1

PrepFusion

$$V_- = V_+ = 2\text{ V}$$

Hence, the collector voltage is

$$V_C = 2\text{ V}$$

The current through the resistor is

$$I_R = \frac{5-2}{10.5\,\text{k}\Omega} = \frac{3}{10.5\,\text{k}\Omega} = 285.7\,\mu\text{A}$$

Since no current goes in opamp, $I_R = I_C$
Using

$$I_C = \beta I_B, \qquad \beta = 20$$

$$I_E = I_C + \frac{I_C}{20} = I_C\left(1 + \frac{1}{20}\right) = \frac{21}{20} I_C$$

$$I_E = I_0 = 285.7 \times \frac{21}{20} = 299.985\,\mu\text{A} \approx 300\mu\text{A}$$

Q4.15.1 NAT 2015 2M Ans: 93

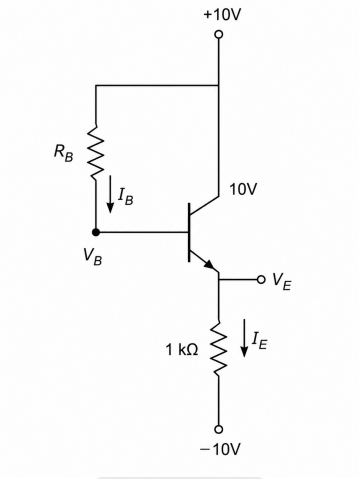

Fig. Solution Q4.15.1

Given,

$$V_E = 0\,\text{V}, \qquad V_{BE} = 0.7\,\text{V}$$

Hence,

$$V_B = V_E + V_{BE} = 0.7\,\text{V}$$

The emitter current is

$$I_E = \frac{V_E - (-10)}{1\,\text{k}\Omega} = \frac{10}{1\,\text{k}\Omega} = 10\,\text{mA}$$

Using

$$I_E = (\beta + 1)I_B, \qquad \beta = 99$$

$$I_B = \frac{10\,\text{mA}}{100} = 0.1\,\text{mA}$$

The resistor R_B is connected between +10 V and the base, therefore

$$R_B = \frac{10 - 0.7}{0.1\,\text{mA}} = \frac{9.3}{0.1 \times 10^{-3}} = 93\,\text{k}\Omega$$

93

Q4.14.1 NAT 2014 1M Ans: 5.58

For the transistor T_1,

$$I_C + 2I_B = \frac{9 - V_{BE}}{5\,\text{k}\Omega} = \frac{9 - 0.7}{5\,\text{k}\Omega} = 1.66\,\text{mA}$$

Using

$$I_B = \frac{I_C}{\beta}, \qquad \beta = 100$$

$$I_C\left(1+\frac{2}{100}\right)=1.66$$

$$I_C=\frac{1.66}{1+\frac{2}{100}}=1.62\,\text{mA}$$

Since T_1 and T_2 are identical,

$$I_{C2}=1.62\,\text{mA}$$

The current through the 3 kΩ resistor is therefore 1.62 mA.
Hence,

$$V_1-V_2=V_{BE}+(3\,\text{k}\Omega)(1.62\,\text{mA})$$

$$V_1-V_2=0.7+4.86=5.56\approx 5.58\,\text{V}$$

$$\boxed{5.58}$$

Key Points

- The diode-connected transistor T_1 sets the reference current.
- For matched transistors, $I_{C1}=I_{C2}$.

Q4.14.2 NAT | 2014 | 1M | Ans: 41.62 ●○○

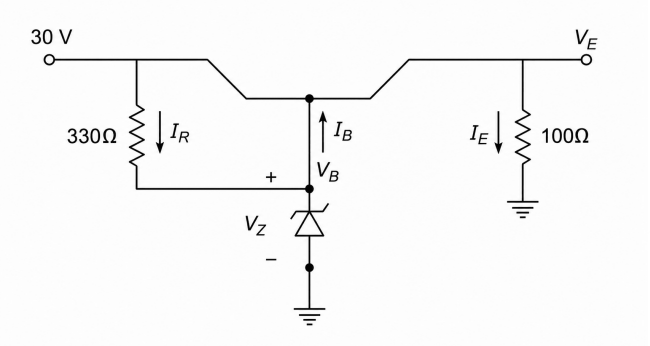

Fig. Solution Q4.14.2

The Zener diode fixes the base voltage at

$$V_B=V_Z=15\,\text{V}$$

Since the transistor is an emitter follower,

$$V_E=V_B+V_{BE}=15+0.7=15.7\,\text{V}$$

The load current through the 100 Ω resistor is

$$I_E=\frac{15.7}{100}=157\,\text{mA}$$

Using

$$I_E = (\beta + 1)I_B, \qquad \beta = 40$$

$$I_B = \frac{157}{41} = 3.83\,\text{mA}$$

The current through the 330 Ω resistor is

$$I_R = \frac{30 - 15}{330} = \frac{15}{330} = 45.45\,\text{mA}$$

Applying KCL at the base node,

$$I_R = I_Z + I_B$$

$$I_Z = 45.45 - 3.83 = 41.62\,\text{mA}$$

$$\boxed{41.62}$$

Q4.12.1 MCQ 2012 2M Ans: D

First determine the quiescent current to obtain the small-signal parameters.

$$I_B = \frac{V_C - 0.7}{100\,\text{k}\Omega}, \qquad \frac{13.7 - V_C}{12\,\text{k}\Omega} = (\beta + 1)I_B$$

With $\beta = 100$,

$$I_B \approx 9.9\,\mu\text{A}, \qquad I_C \approx 0.99\,\text{mA}$$

Hence,

$$g_m = \frac{I_C}{V_T} \approx \frac{0.99\,\text{mA}}{25\,\text{mV}} \approx 0.04\,\text{S},$$

$$r_\pi = \frac{\beta}{g_m} \approx \frac{100}{0.04} \approx 2.5\,\text{k}\Omega$$

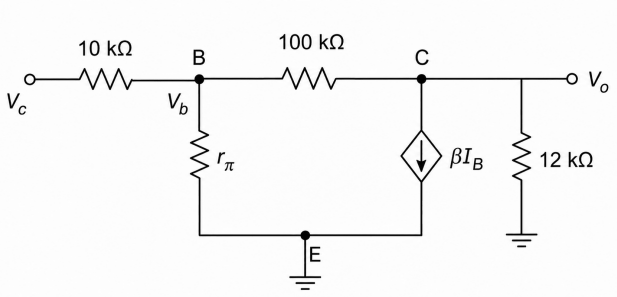

Fig. Solution Q4.12.1

For AC analysis, the supply is at AC ground and the coupling capacitors are short circuits.
At the base node,

$$\frac{V_i - V_b}{10\,\text{k}\Omega} = \frac{V_b}{r_\pi} + \frac{V_b - V_o}{100\,\text{k}\Omega}$$

At the collector node,

$$\frac{V_o}{12\,\text{k}\Omega} + \frac{V_o - V_b}{100\,\text{k}\Omega} + g_m V_b = 0$$

Substituting

$$r_\pi = 2.5\,\text{k}\Omega, \qquad g_m = 0.04\,\text{S},$$

and solving the above equations,

$$\frac{V_o}{V_i} \approx -8.93$$

Therefore,

$$|A_v| \approx 9 \approx 10$$

$$\boxed{\text{(d) } |A_v| \approx 10}$$

Q4.10.1 MCQ 2010 2M Ans: B

The transistor Q_1 is diode-connected. Hence, the current through the 10 kΩ resistor supplies the collector current of Q_1 and the base currents of both transistors.

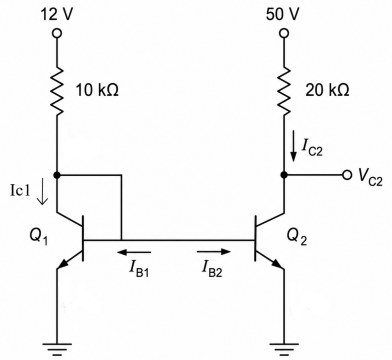

Fig. Solution Q4.10.1

Applying KCL at the common base node,

$$\frac{12 - 0.7}{10\,\text{k}\Omega} = I_{C1} + I_{B1} + I_{B2}$$

Since the transistors are matched,

$$I_{C1} = I_{C2} = I_C, \qquad I_{B1} = I_{B2} = \frac{I_C}{\beta}$$

Therefore,

$$1.13mA = I_C\left(1 + \frac{2}{100}\right)$$

$$I_C = \frac{1.13}{1.02} = 1.108\,\text{mA}$$

The collector current of Q_2 is the same:

$$I_{C2} = 1.108\,\text{mA}$$

The voltage drop across the 20 kΩ resistor is

$$V_R = I_{C2} \times 20\,\text{k}\Omega = 1.108 \times 20 = 22.16\,\text{V}$$

Hence, the collector voltage of Q_2 is

$$V_{C2} = 50 - 22.16 = 27.84\,\text{V}$$

Since the emitter is grounded,

$$V_2 = V_{CE2} = V_{C2} = 27.84\,\text{V}$$

(b) 27.8 V

Q4.09.1 MCQ | 2009 | 2M | Ans: A

Given,

$$I_{C3} = 2\,\text{mA}, \qquad \beta = 100, \qquad R_3 = 1\,\text{k}\Omega$$

$$I_{E3} = I_{C3}\left(1 + \frac{1}{\beta}\right) = 2\left(1 + \frac{1}{100}\right) = 2.02\,\text{mA}$$

Hence, the emitter voltage is

$$V_{E3} = -12 + (2.02\,\text{mA})(1\,\text{k}\Omega) = -9.98\,\text{V}$$

Since

$$V_{BE} = 0.7\,\text{V},$$

$$V_B = V_{E3} + 0.7 = -9.98 + 0.7 = -9.28\,\text{V}$$

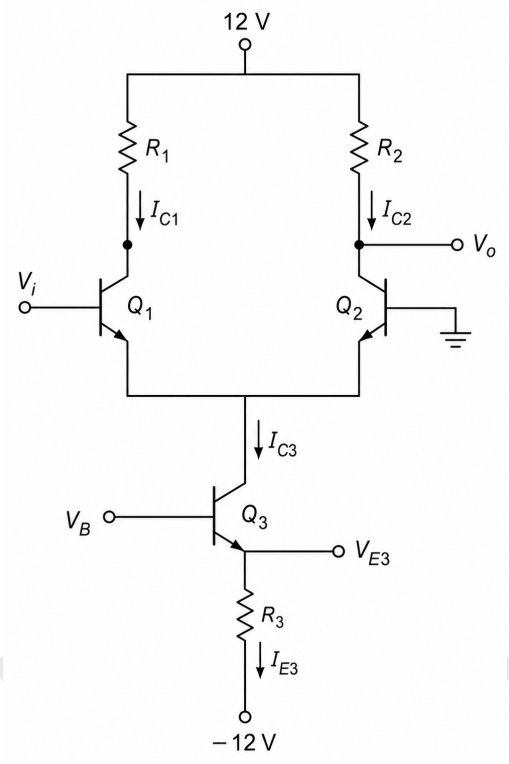

Fig. Solution Q4.09.1

The emitter current of Q_3 is
Thus,

$$V_B \approx -9.3\ \text{V}$$

(a) -9.3 V

Q4.09.2 MCQ 2009 2M Ans: C

Use the circuit in the preceding shared data: the input drives the base of Q_1, while the output is taken at the collector of Q_2. This is the **opposite** collector, so its gain is positive.

The 2 mA tail current divides equally between the matched pair. Neglecting the small base-current correction, each collector current is about 1 mA, giving

$$g_m \simeq \frac{1\ \text{mA}}{25\ \text{mV}} = 0.04\ \text{S}.$$

For an ideal tail current source the common-emitter small-signal voltage is $v_i/2$. Hence $v_{be2} = -v_i/2$, and $i_{c2} = -g_m v_i/2$. The collector resistor then gives

$$v_o = -R_2 i_{c2} = \frac{g_m R_2}{2} v_i.$$

With $R_2 = 1$ kOhm, the gain is approximately $\boxed{+20}$, option (C). Including $\beta = 100$ gives $I_C \simeq 0.9901$ mA and gain about $+19.8$, still closest to $+20$. The driven-side collector has the opposite voltage-gain sign from the output collector shown.

Q4.08.1 MCQ 2008 2M Ans: C

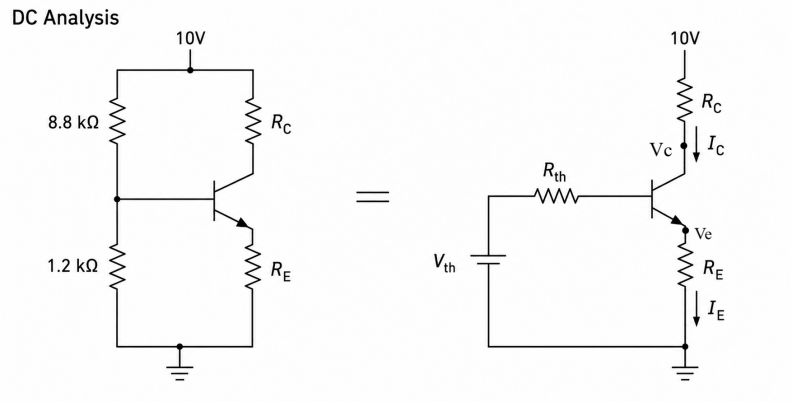

Fig. Solution Q4.08.1

Using the voltage divider,

$$V_{th} = 10\left(\frac{1.2}{8.8 + 1.2}\right) = 1.2\,\text{V}$$

Since $V_{BE} = 0.7\,\text{V}$,

$$V_E = 1.2 - 0.7 = 0.5\,\text{V}$$

Given $V_{CE} = V_{CC}/2 = 5\,\text{V}$,

$$V_C = V_E + V_{CE} = 0.5 + 5 = 5.5\,\text{V}$$

Hence the voltage across R_C is

$$10 - 5.5 = 4.5\,\text{V}$$

Since β is very large,

$$I_C \approx I_E, \qquad R_E = \frac{0.5}{I_E}, \qquad R_C = \frac{4.5}{I_C}$$

Therefore,

$$\frac{R_C}{R_E} = \frac{4.5}{0.5} = 9$$

Also,

$$g_m R_E = \frac{I_C}{V_T} R_E = \frac{0.5}{25\,\text{mV}} \approx 20$$

The small-signal gain is

$$A_v = \frac{V_{out}}{V_{in}} = -\frac{g_m R_C}{1 + g_m R_E} = -\frac{20 \times 9}{21} \approx -8.57$$

Hence,

$$|A_v| \approx 9$$

(c) 9

Key Points

- The gain with an unbypassed emitter resistor is $A_v = -\dfrac{g_m R_C}{1 + g_m R_E}$..

Q4.08.2 MCQ 2008 1M Ans: C ● ○ ○

For a common-base (CB) amplifier,

$$\text{Current gain } A_i = \alpha = \frac{I_C}{I_E} < 1$$

since α is typically in the range 0.98 to 0.998.
The voltage gain is

$$A_v = g_m R_C \approx \frac{\alpha R_C}{r_e}$$

As r_e is very small, A_v can be much greater than unity.
Hence, in a CB amplifier,

$$A_i < 1, \qquad A_v > 1$$

(c) Voltage gain can be greater than unity but current gain is always less than unity

Q4.07.1 MCQ 2007 2M Ans: C ● ○ ○

The left transistor is diode-connected, hence its base and collector are shorted.
The reference current is

$$I_R = \frac{2 - 0.7}{1\,\text{k}\Omega} = 1.3\,\text{mA}$$

Since all three transistors are identical, they have the same V_{BE}, so their base currents are equal.
Applying KCL at the reference node,

$$I_R = I_{C1} + I_{B1} + I_{B2} + I_{B3}$$

Using $I_C = \beta I_B$,

$$1.3\,\text{mA} = (\beta + 3) I_B$$

For $\beta = 100$,

$$I_B = \frac{1.3\,\text{mA}}{103} = 12.62\,\mu\text{A}$$

Hence,

$$I_C = 100 \times 12.62\,\mu\text{A} = 1.262\,\text{mA}$$

The two right transistors draw

$$I = 2I_C = 2.524\,\text{mA}$$

Therefore,

$$V_C = 10 - (2.524\,\text{mA})(1\,\text{k}\Omega) = 7.476\,\text{V} \approx 7.4\,\text{V}$$

(c) 7.4 V

Q4.07.2 MCQ 2007 2M Ans: A ● ○ ○

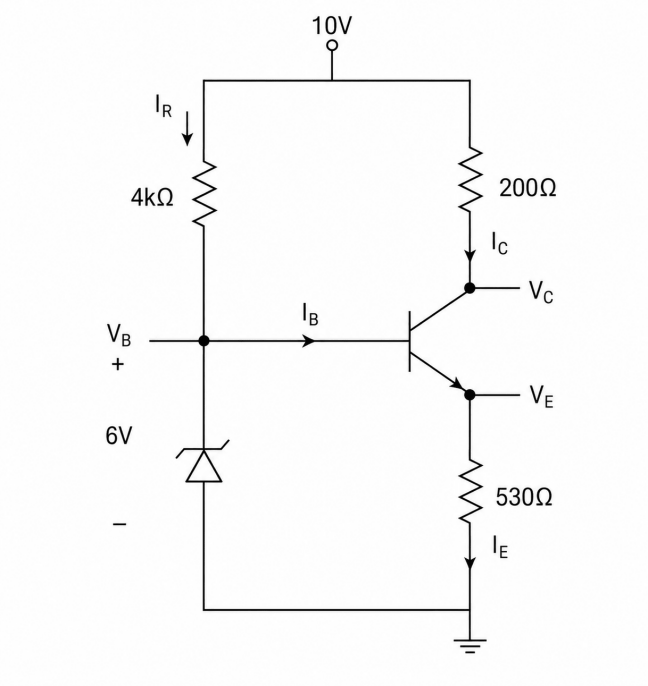

Fig. Solution Q4.07.2

The Zener diode fixes the base voltage at

$$V_B = 6\,\text{V}$$

Hence,

$$V_E = V_B - V_{BE} = 6 - 0.7 = 5.3\,\text{V}$$

The emitter current is

$$I_E = \frac{5.3}{530} = 10\,\text{mA}$$

Current through the 4 kΩ resistor is

$$I_R = \frac{10 - 6}{4\,\text{k}\Omega} = 1\,\text{mA}$$

Given Zener current,

$$I_Z = 0.5\,\text{mA}$$

Therefore,

$$I_B = I_R - I_Z = 1 - 0.5 = 0.5\,\text{mA}$$

Now,

$$I_C = I_E - I_B = 10 - 0.5 = 9.5\,\text{mA}$$

Hence,

$$\beta = \frac{I_C}{I_B} = \frac{9.5}{0.5} = 19$$

Collector voltage is

$$V_C = 10 - (9.5\,\text{mA})(200\,\Omega) = 10 - 1.9 = 8.1\,\text{V}$$

Thus,

$$V_{CE} = V_C - V_E = 8.1 - 5.3 = 2.8\,\text{V}$$

(a) 19 and 2.8 V

Q4.06.1 MCQ 2006 2M Ans: D ● ○ ○

The circuit is a common-collector (emitter follower).
Given,

$$g_m = 20\,\text{mS}, \qquad \beta_0 = 50, \qquad r_o = \infty, \qquad r_b = 0.$$

First, find the hybrid-π resistance:

$$r_\pi = \frac{\beta_0}{g_m} = \frac{50}{20 \times 10^{-3}} = 2.5\,\text{k}\Omega.$$

The input resistance seen at the transistor base is

$$R_{in} = r_\pi + (\beta_0 + 1)R_E = 2.5\,\text{k}\Omega + 51(2\,\text{k}\Omega) = 104.5\,\text{k}\Omega.$$

Hence, the base voltage is

$$V_b = V_i \frac{104.5}{104.5 + 1} = 0.9905\,V_i.$$

The emitter follower gain from base to emitter is

$$\frac{V_o}{V_b} = \frac{(\beta_0 + 1)R_E}{r_\pi + (\beta_0 + 1)R_E} = \frac{102}{104.5} = 0.976.$$

Therefore,

$$A_v = \frac{V_o}{V_i} = 0.9905 \times 0.976 \approx 0.967.$$

(a) 0.967

Q4.06.2 MCQ 2006 2M Ans: B ● ○ ○

For the given fixed-bias silicon transistor,

$$I_B = \frac{V_{CC} - V_{BE}}{R_B} = \frac{12 - 0.7}{100\,\text{k}\Omega} = 113\,\mu\text{A}$$

The collector current is

$$I_C = \beta I_B = 80 \times 113\,\mu\text{A} = 9.04\,\text{mA}$$

The collector-to-emitter voltage is

$$V_{CE} = V_{CC} - I_C R_C = 12 - (9.04\,\text{mA})(2\,\text{k}\Omega)$$

$$= 12 - 18.08 = -6.08\,\text{V}$$

Since a negative value is not possible in the active region, the transistor operates in saturation. Therefore,

$$V_{CE} \approx 0.2\,\text{V}$$

(b) 0.2 V

Q4.03.1 MCQ 2003 2M Ans: B ● ○ ○

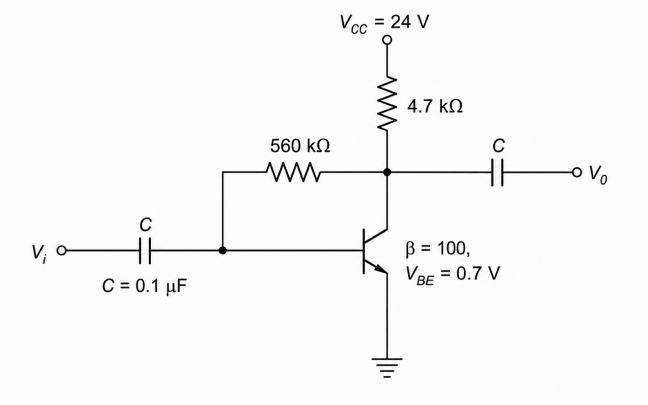

Fig. Solution Q4.03.1

For DC analysis, the coupling capacitors are open-circuited. The transistor is biased using collector-to-base feedback. Applying KVL in the collector–base–emitter loop,

$$V_{CC} - (I_C + I_B)R_C - I_B R_B - V_{BE} = 0$$

Given,

$$V_{CC} = 24\,\text{V}, \quad R_C = 4.7\,\text{k}\Omega, \quad R_B = 560\,\text{k}\Omega, \quad \beta = 100, \quad V_{BE} = 0.7\,\text{V}$$

Since,

$$I_C = \beta I_B = 100 I_B$$

Substituting,

$$24 - (100I_B + I_B) \times 4.7 - I_B \times 560 - 0.7 = 0$$

$$23.3 = I_B(560 + 101 \times 4.7)$$

$$I_B = \frac{23.3}{560 + 101 \times 4.7} = 0.0225\,\text{mA}$$

Hence,

$$I_{CQ} = \beta I_B = 100 \times 0.0225 = 2.25\,\text{mA} \approx 2.3\,\text{mA}$$

(b) $I_{CQ} = 2.3\,\text{mA}$

Q4.00.1 MCQ 2000 1M Ans: A ● ○ ○

A transistor operates in different regions depending on the biasing of its emitter-base (EB) and collector-base (CB) junctions. If both the emitter-base and collector-base junctions are forward biased, the transistor operates in the **saturation region**.

(a) Saturation mode

Q4.99.1 MCQ 1999 2M Ans: C ● ○ ○

A unity gain buffer has a low-frequency voltage gain of unity. At the bandwidth (cut-off frequency), the voltage gain falls to $\frac{1}{\sqrt{2}}$ of its low-frequency value (i.e., -3 dB point).

Given,

$$A_v = 1, \qquad V_{in} = 2\,\text{V}, \qquad f = f_{BW} = 1\,\text{MHz}$$

Hence, the output voltage is

$$V_o = A_v \times \frac{1}{\sqrt{2}} \times V_{in} = \frac{1}{\sqrt{2}} \times 2 = \frac{2}{\sqrt{2}}\,\text{V}$$

$$\boxed{\textbf{(c)}\ \frac{2}{\sqrt{2}}\,\text{V}}$$

Q4.98.1 MCQ 1998 1M Ans: A ● ○ ○

Assume the transistor operates in the active region.

Applying KVL to the collector circuit,

$$12 - I_C \times 5\,\text{k}\Omega - V_{CE} = 0$$

Given $V_{CE} = 2\,\text{V}$,

$$I_C = \frac{12-2}{5 \times 10^3} = 2\,\text{mA}$$

Using the transistor current relation,

$$I_C = \beta I_B$$

$$I_B = \frac{I_C}{\beta} = \frac{2\,\text{mA}}{50} = 40\,\mu\text{A}$$

Applying KVL to the base circuit,

$$12 - I_B R_B - V_{BE} = 0$$

$$R_B = \frac{12 - 0.7}{40 \times 10^{-6}} = \frac{(12-0.7)\beta}{I_C} = 282.5\,\text{k}\Omega$$

Since $V_B = 0.7\,\text{V}$ and $V_C = 2\,\text{V}$,

$$V_{CB} = V_C - V_B = 2 - 0.7 = 1.3\,\text{V} > 0$$

Hence, the transistor is indeed in the active region and the calculated value is correct.

$$\boxed{\textbf{(a)}\ R_B = 283\,\text{k}\Omega}$$

Q4.97.1 MCQ | 1997 | 1M | Ans: B | ● ○ ○

For a common emitter amplifier, the small-signal voltage gain is approximately

$$A_v = g_m R_C$$

where

$$g_m = \frac{I_C}{V_T}$$

At room temperature,

$$V_T \approx 25\,\text{mV}$$

Hence,

$$g_m = \frac{1\,\text{mA}}{25\,\text{mV}} = 0.04\,\text{S}$$

Therefore,

$$A_v = 0.04 \times 1000 = 40$$

(b) 40

Q4.95.1 Descriptive | 1995 | 1M | Ans: Collector | ● ○ ○

In the given circuit, the input signal is applied to the **base** terminal and the output is taken from the **emitter** terminal. The **collector** terminal is common to both the input and output circuits through the DC supply.
Hence, the transistor is operating in the **common collector (CC)** configuration, also known as an *emitter follower*.

Common Collector (CC) Configuration

Q4.94.1 MCQ | 1994 | 1M | Ans: A | ● ○ ○

A common collector (CC) configuration, also known as an emitter follower, is characterized by a **high input resistance** and a **low output resistance**. Hence, it is widely used as an impedance matching circuit.

(a) High input and low output resistances

Q4.93.1 NAT | 1993 | 1M | Ans: 90.6 | ● ○ ○

For an emitter follower (common collector), the input impedance is

$$Z_{in} = h_{ie} + (1 + h_{fe})R_E$$

Given,

$$h_{ie} = 600\,\Omega, \qquad h_{fe} = 89, \qquad R_E = 1\,\text{k}\Omega$$

Substituting the values,

$$Z_{in} = 600 + (1 + 89) \times 1000$$

$$= 600 + 90000 = 90600\,\Omega = 90.6\,\text{k}\Omega$$

90.6

Q4.92.1 MCQ | 1992 | 1M | Ans: *

In a voltage-divider biased common emitter amplifier, the biasing network (R_1 and R_2) must be directly connected to the base to establish the required DC operating point.

In the given circuit, the coupling capacitor C_1 is placed *between the bias network and the base*. Since a capacitor blocks DC, the base becomes isolated from the biasing network and the transistor cannot be properly biased.

Therefore, C_1 is wrongly placed. It should be connected between the input source and the base-bias junction so that the AC signal is coupled while the DC bias remains unaffected.

Component: C_1, because it blocks the DC bias to the base.

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SOLUTIONS

V

Miscellaneous Topics

Topics

1. CMOS Inverter

Unit Snapshot*

Stream: IN	**Questions:** 4	**MCQ:** 3	**MSQ:** 1	**NAT:** 0
1M: 2	**2M:** 2	**Descriptive:** 0	**Avg Weightage** ~2M	**Range:** 1M(2023)–2M(2026)

**See preface for how Avg Weightage and Range are calculated.*

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Q5.26.1 MSQ | 2026 | 2M | Ans: MTA | ●○○

Connection of NMOS and PMOS is not complementary, it was wrong question

Q5.23.1 MCQ | 2023 | 1M | Ans: B | ●○○

The fastest logic family among the given options is **Emitter-Coupled Logic (ECL)** because its transistors operate in the active region and never enter saturation, thereby eliminating storage delay.

$$\boxed{(b) Emitter - CoupledLogic}$$

Q5.19.1 MCQ | 2019 | 2M | Ans: C | ●○○

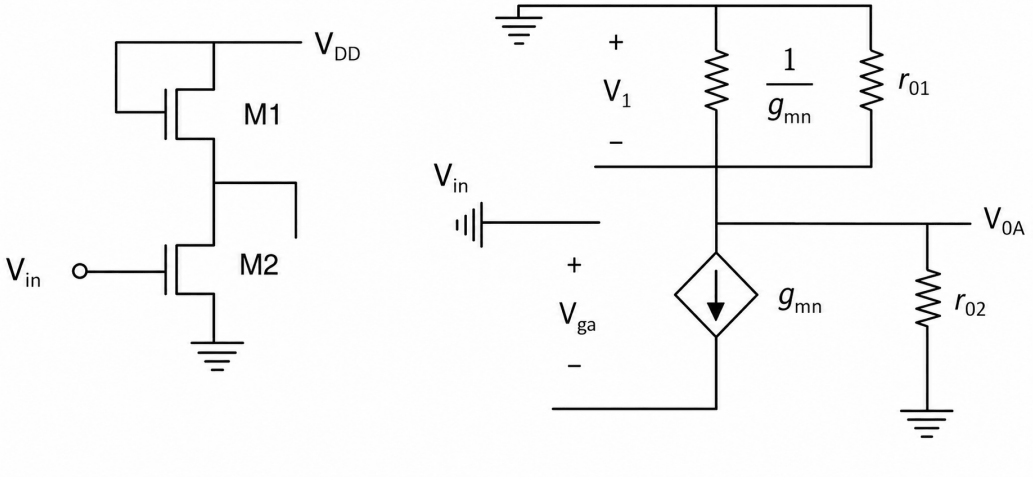

Fig. Solution Q5.19.1

Analyze the circuit in two stages.
For the first stage (M1–M2), M1 is diode-connected. Since channel-length modulation is neglected,

$$r_{o1} = r_{o2} = \infty, \qquad R_L = \frac{1}{g_{mn}}.$$

Applying KCL at node V_{oA},

$$V_{oA} + g_{mn}\left(\frac{1}{g_{mn}}\right) V_{ga} = 0.$$

Since

$$V_{ga} = V_{in},$$

we get

$$\frac{V_{oA}}{V_{in}} = -\frac{g_{mn}}{g_{mn}} = -1.$$

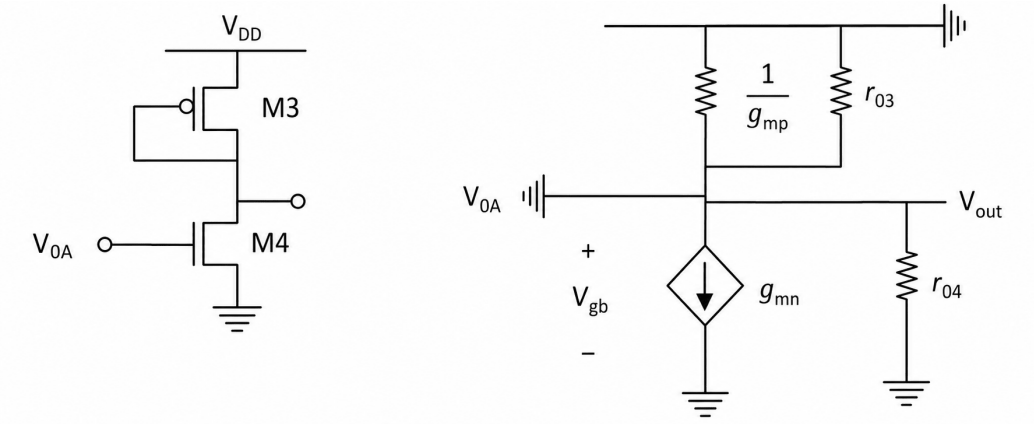

Fig. Solution Q5.19.1

For the second stage (M3–M4), M3 is diode-connected. Again,

$$r_{o3} = r_{o4} = \infty, \qquad R_L = \frac{1}{g_{mp}}.$$

Applying KCL at the output node,

$$V_{out} + g_{mn}\left(\frac{1}{g_{mp}}\right)V_{gb} = 0.$$

Since

$$V_{gb} = V_{oA},$$

$$\frac{V_{out}}{V_{oA}} = -\frac{g_{mn}}{g_{mp}}.$$

Hence,

$$\frac{V_{out}}{V_{in}} = \frac{V_{out}}{V_{oA}} \cdot \frac{V_{oA}}{V_{in}} = \left(-\frac{g_{mn}}{g_{mp}}\right)(-1) = \frac{g_{mn}}{g_{mp}}.$$

$$\boxed{\text{(c)}\ \frac{V_{out}}{V_{in}} = \frac{g_{mn}}{g_{mp}}}$$

Q5.14.1 MCQ 2014 1M Ans: D ●○○

Observe the pull-down network (PDN).
The two NMOS transistors are connected in series and are controlled by inputs A and B. Hence, the output is pulled LOW only when

$$A = 1, \quad B = 1.$$

Therefore,

$$Y = \overline{AB},$$

which is the Boolean expression of a NAND gate.

$$\boxed{\text{(d) NAND}}$$

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SiliconX	Electronic Devices, Electromagnetic Field Theory and Communication Systems — EC
PowerX	Electrical Machines, Power Systems, Power Electronics, Measurements and Electromagnetic Field Theory — EE, IN
Engineering Mathematics	Linear Algebra, Calculus, Differential Equations, Complex Variables, Probability & Statistics, Numerical Methods, Transform Theory and Vector Calculus — all branches
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